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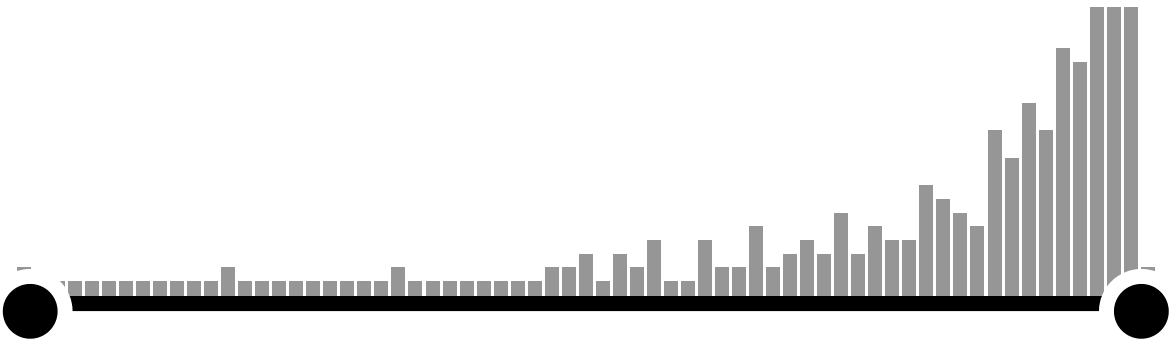
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	Document title	Authors	Source	Year	Citations
1	Article • *Open access* **Adaptive acoustic comfort: Evidence from an acoustic neutrality approach in naturally ventilated office buildings**	Yuksel Dicle S.,Aletta F., Kang J.	Applied Acoustics, 257, 111627	2027	0

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Acoustic comfort is commonly assessed using fixed *sound*-level criteria that assume stable exposure–response relationships. However, in naturally ventilated buildings, occupants' evaluations may vary with changing acoustic exposure and adaptive responses. Although such adaptive relationships are well established in *thermal* comfort research, they remain underexplored in acoustics. This study examines an adaptive perspective on acoustic comfort assessment through a field study conducted in six naturally ventilated or mixed-mode open-plan offices involving 72 participants and 480 observations. Repeated in situ perceptual evaluations, concurrent indoor and *outdoor sound*-level measurements, and self-reported adaptive behaviours were collected and analysed using mixed-effects modelling approaches. For level-based evaluations, an Acoustic Sensation Vote (ASV) of 0 ("neither loud nor quiet") emerged as the operational acoustic neutral reference, representing a threshold-like transition to dissatisfaction. Office-specific neutral levels ranged from 47.2 to 53.3 dB and tended to be higher in offices with higher indoor *sound*-level exposure. Adaptive actions were more likely at higher preceding-hour indoor L_{Aeq} (OR = 1.55 per 5 dB), and observations with reported adaptive behaviour were slightly more quiet-oriented after accounting for concurrent indoor *sound* level and office. *Outdoor* L_{Aeq} did not provide additional explanatory information for ASV; however, among neutral evaluations, each 1-dB increase in *outdoor* L_{Aeq} was associated with an approximately 0.16-dB increase in concurrent indoor *sound* level. Overall, the findings provide empirical support for a context-sensitive adaptive acoustic comfort perspective in which acoustic evaluations are considered in relation to indoor exposure, office context, behavioural responses and *outdoor* acoustic conditions. The findings should be interpreted within the scope of six London offices and a field-based design in which acoustic and contextual conditions could not be fully standardised across participants. For office design and assessment, the results suggest that fixed *sound*-level criteria can be complemented by consideration of office-specific acoustic exposure and occupants' opportunities for behavioural adjustment, particularly in naturally ventilated workplaces.

	Document title	Authors	Source	Year	Citations
2	Article • *Open access* **Evaluation of heat stress mitigation potentials of climate change adaptation measures in cities: Scenario analyses for technical- and nature-based solutions using 3D microclimate modelling**	Eingruber N., Zimmermann D., Korres W.,Lohnert U., Schneider K.	Climate Risk Management , 54, 100894	2026	0

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Urban areas are expected to warm significantly more than the global average according to climate projections. This will lead to many adverse effects such as increasing health risks and mortality rates, making climate change adaptation crucial for future urban planning. Identifying effective and suitable adaptation measures requires assessment of cooling potentials, societal acceptance and cost effectiveness based on *sound*-scientific analyses. High-resolution microclimate modelling and scenario analyses are a promising approach to evaluate *thermal* effects of climate adaptation measures. A physically-based 3D-gridded ENVI-met model was parameterized based on field measurements and remote sensing data to simulate the microclimate of a 16-ha urban high-density area in Cologne/Germany. The model was successfully validated using a densely-distributed NETATMO sensor network in the study area. We evaluated 92 adaptation scenarios for a 20-year extreme heat event to quantify heat mitigation potentials. Scenarios included technical- and nature-based solutions at buildings as well as in street corridors. Measures were simulated individually and in combinations, across varying idealized and more realistic implementation degrees. Results showed a significant mean air temperature reduction by -3.0 K up to -6.3 K as well as significant mean cooling effects on physiological equivalent temperature (PET) of -4.3 K up to -11.1 K in pedestrian height level, particularly for scenarios with high greenery coverage. Combinations of different nature-based and technical solutions performed better than single measures. Even partial implementations (e.g., 25% greened buildings) provided significant cooling effects in the entire quarter, while high-reflective surfaces and shading structures had limited impact on *thermal outdoor* comfort.

	Document title	Authors	Source	Year	Citations

Article • *Open access*

3 **Exploring interactions of thermal, acoustic, and aesthetic environments on thermal experience in parks**

Zhang Y.,Uchiyama Y., Sato M.

Scientific Reports, 16(1), 8564

2026

0

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The *thermal* comfort of urban parks is crucial for residents' well-being and urban livability. There is limited research on the integration of different environmental factors in urban parks, with most studies focusing on hot, sunny days in summer. This study explored the interactions of multi-sensory factors, including *thermal*, acoustic, and aesthetic, on *thermal* experience in parks across various weather types. The results reveal that *thermal* conditions directly dominate *thermal* experience. Acoustic comfort has a relatively weak positive effect on *thermal* comfort, and this effect disappears under strong heat-stress conditions. Natural *sounds* help increase *thermal* acceptability, whereas mechanically dominated *soundscapes* have the opposite effect. Aesthetic satisfaction also contributes to higher *thermal* acceptability. Compared with blue spaces, aesthetic satisfaction with green spaces exerts a more consistently positive influence on *thermal* experience. The negative effect of PET is significantly weaker than the positive effects of perceived acoustic comfort and aesthetic satisfaction on overall satisfaction. The revealed multi-sensory interactions yield practical recommendations for urban planners and designers in addressing climate change and enhancing urban livability.

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Article

4 **Soundscape diversity governs thermo-physiological and psychological states in a sexually dimorphic manner**

Tang Z.,Huang W., Zhang M.,Deng P.,Hong B.

Building and Environment, 301, 114742

2026

0

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Although *soundscapes* modulate *outdoor thermal* perception, sex differences in physiological and psychological mechanisms under *thermal* stress remain elusive. This study examined *thermo*-acoustic interactions across six acoustic conditions that varied in diversity (low/medium/high) and environmental context (nature-dominant versus anthropogenic-dominant) in male and female participants exposed to graded *thermal* stress. Physiological indicators (blood pressure, heart rate, and eardrum temperature), electroencephalographic (EEG) activity, and psychological metrics were continuously monitored alongside microclimatic parameters. Results demonstrate marked sexual dimorphism in *thermo*-acoustic perception, primarily within nature-dominant environments. Males exhibited cardiovascular attenuation under slight *thermal* stress, with systolic pressure declining 6.08 mmHg (medium diversity) and diastolic pressure 6.61 mmHg (high diversity) versus low diversity ($p < 0.05$), though physiological responsivity diminished as *thermal* stress escalated. Females conversely demonstrated intensified physiological reactions with increasing *thermal* load, including reduced heart rate under no *thermal* stress conditions in anthropogenic settings ($p < 0.05$). In nature-dominant *sound* environments, low *soundscape* diversity attenuated male δ power while potentiating α power, consistent with diminished somnolence and heightened relaxation; elevated female θ power, conversely, indexed enhanced positive affect. Psychologically, males demonstrated consistently higher sensitivity, while females exhibited a biphasic pattern of initial attenuation followed by enhancement. In anthropogenic-dominant environments, males displayed heightened physiological and psychological sensitivity to acoustic diversity, whereas females showed asynchronous patterns: physiological responses weakened then strengthened, while psychological responses progressively intensified with increasing *thermal* stress. Our findings inform sex-sensitive acoustic optimization frameworks for urban design, advancing evidence-based healthy built environment strategies.

Article

	Document title	Authors	Source	Year	Citations
5	**Critical analysis of smart materials in adaptive transparent systems for building façade**	Jaffar H.,Javed N.,Riaz A., ... Zhou C.,Zhang J.	Energy and Environment , 37(5)	2026	4

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Adaptive transparent system (ATS) is an aspiring and effective building system that can adapt itself through environmental changes in spontaneous and reversible ways. It proposes an automatic and real-time response to indoor and *outdoor* conditions by increasing building energy efficiency and user comfort. This research presents a critical analysis to provide a preference order of the smart materials being used in the adaptive transparent systems. The approach of using smart material technologies in the building façades is discussed as the promising research direction for the future. A comprehensive literature review is conducted to identify the smart materials used in adaptive transparent systems along with their properties. Seven types of smart materials that are aerogel (AG), phase change material (PCM), photovoltaic (PV), electrochromic (EC), *thermo*-chromic (TC), *thermo*-tropic (TT), and liquid crystal polymers (LCP) are selected based on evaluation criteria determined by the literature. The identified criteria are bulk density (BD), *thermal* conductivity (TCd), *sound* insolation (SI), fire retardation (FR), heat transfer coefficient (HTC), solar transmittance (ST), air purification (AP), annual energy saving (AES), cost saving (CS), ultraviolet (UV)/near infrared (NIR) control (UNC), and embodied carbon (EC). The identified seven smart materials are investigated with the help of two multi-criteria decision-making (MCDM) techniques named analytical hierarchical process (AHP) and technique for order preference by similarity to ideal solution (TOPSIS) for analysis and ranking of materials. Results of the analysis demonstrate that PV is the most optimum material for adaptive transparent systems while others are ranked according to their performance.

Article

	Document title	Authors	Source	Year	Citations
6	**Phase Change Microcapsules Loaded in Polyurethane Foam for Dual-Purpose Thermal and Acoustic Insulation**	Zhu H.,Tan J.,Qiang X., Yin Y.,Wang C.	ACS Applied Polymer Materials , 8(14)	2026	0

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Growing demand for lightweight and energy-efficient materials in building envelope systems has promoted the advancement of high-performance *thermal* insulation foams. Polyurethane foams (PUFs) are extensively adopted as porous insulation fillers owing to their low density, structural continuity, and suitability for scale construction deployment. However, insufficient skeleton strength and nonuniform pore size distribution limit the *thermal* insulation performance. In this study, n-tetradecane microcapsules were incorporated into a polyurethane raw material system, and phase change polyurethane composite foams (PCM-PUFs) were fabricated via a one-step foaming process. Microstructural characterization reveals that the composite foams exhibit a uniform closed-cell morphology, with the n-tetradecane microcapsules well dispersed and firmly embedded within the polyurethane cell walls; the average cell size is approximately 150 μm. At an optimal microcapsule loading of 15 wt % and a corresponding encapsulation efficiency of 89.7%, the resulting PCM_{15}-PUF foam demonstrates substantially enhanced resilience and *thermal* buffering capacity. The support factor reaches 5.45, indicating excellent load-bearing capability and comfortability. The *thermal* conductivity is reduced to 0.05 W/(m·K). Under *outdoor* solar exposure, the surface temperature of PCM_{15}-PUF is 6.2 °C lower than that of neat PUF. Furthermore, PCM_{15}-PUF achieves an average *sound* absorption coefficient (AAC) above 0.9 and a noise reduction coefficient (NRC) exceeding 0.7, confirming that the incorporation of microcapsules does not compromise the *sound* insulation performance. The developed composite foam holds strong promise for building applications, particularly for regulating temperature fluctuations in scenarios with significant indoor–*outdoor* temperature differences.

Article

	Document title	Authors	Source	Year	Citations
7	**Interactive effects of noise and odor on construction workers' environmental perception and cognitive performance amid outdoor thermal stress**	Li Z.,Zhang X.,Peng Z., ... Li A.,Hong B.	Building and Environment , 297, 114576	2026	0

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Construction workers chronically exposed to dynamic *outdoor* environments constitute a critical cohort for investigating multisensory environmental stress responses and cognitive ergonomics. This study systematically

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deconstructs the interactive effects of *thermal* stress gradients and noise–olfactory multimodal stimuli on workers' subjective *thermal* perception and neurocognitive performance in a cold region of China. Using a controlled field experiment with synchronized micrometeorological monitoring, we simultaneously acquired subjective environmental evaluations, multidimensional cognitive performance metrics, and electroencephalogram (EEG) data from 72 construction workers under nine factorial combinations of *sound* pressure levels (50/70/85 dBA) and odor types (Mentha piperita, Citrus sinensis, and Rosmarinus officinalis). A multidimensional benefit assessment model quantitatively ranked intervention efficacy. The results indicate that: (1) noise significantly modulated *thermal* sensation votes ($p < 0.05$), while *thermal* stress drove auditory sensation votes ($p < 0.01$). Neurophysiologically, escalating *thermal* stress enhanced low frequency (δ) activity and suppressed high-frequency (α, β) oscillations, with noise further diminishing frontal α-wave relative power ($p < 0.05$). (2) C. sinensis elevated significantly frontal-occipital θ-wave power ($p < 0.05$). (3) M. piperita broadly potentiated β-band activity under “neutral” conditions, yet R. officinalis yielded significantly lower *thermal* comfort votes than C. sinensis under “warm” conditions ($p = 0.02$). Our findings offer actionable climate adaptation frameworks for occupational health, safeguarding vulnerable workers against intensifying extremes in *outdoor* working environments.

Article • *Open access*

8 **Quantifying outdoor thermal comfort in different street vegetation arrangements: Insights from a study in tropical and subtropical Australian cities** — Fu J.,Tavares S.,King D.,Naylor S. — Urban Climate, 67, 102967 — 2026 — 0

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Street vegetation serves as a nature-based strategy for urban heat mitigation leading to improved *outdoor thermal* comfort (OTC). A *sound* understanding of spatial and temporal OTC variation in different street vegetation arrangements is essential to strategically develop urban heat mitigation planning and design. This study introduces a methodology framework that quantifies the *thermal* improvement in different street vegetation arrangements by using three indices - cooling intensity (CI), cooling area (CA) and cooling duration (CD). The framework is applied to four streets in tropical and subtropical Australian cities to quantify Physiological Equivalent Temperature (PET) over a 24-h period. Results indicate that the street with trees on the west-facing side experienced a longer duration of street *thermal* enhancement (up to 38% CD improvement). During daytime hours, the OTC improvement in the planting scenarios with additional trees in the street centre (median strip) incorporating multiple vertical layers of vegetation is more pronounced in the studied subtropical urban area (up to 2.02 °C CI in Ipswich) than that in the studied tropical urban area (up to 1.03 °C CI in Townsville). These findings can contribute to urban heat mitigation planning and design in tropical and subtropical cities worldwide. The methodology framework proposed in this study provides a foundation for quantifying spatial and temporal OTC variation in different street vegetation arrangements, and establishes an adaptive framework for landscape architects, urban designers and planners, and decision-makers worldwide to evaluate urban street *thermal* environments.

Article • *Open access*

9 **Theoretical and experimental study of a novel thermoacoustic engine for sound source application** — Zhang L.,Kang H. — Thermal Science and Engineering Progress, 74, 104699 — 2026 — 0

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This paper introduces a novel structures of standing-wave *thermoacoustic* engines designed for *outdoor sound* sources. The system is characterized by operational stability, enhanced SPL, and space-saving design. Compared to the conventional open-end system, the thin-plate *sound* source proposed is a closed-end one. The plate isolates the working gas from the external environment but remains acoustically transparent, enabling the use of alternative gases. Besides, it can effectively adjust the acoustic field distribution, enhance the acoustic power output. Experimental and theoretical research were conducted, the performance of systems using air or helium is analyzed and compared. The influence of thin plate properties on system behavior is investigated. Results show that with air as working gas, the system can achieve a higher SPL of 113.02 dB at 1 m away, marking a 5.4% increase in SPL compared to open-ended system. With helium adopted as the operational gas, the system can reach a highest SPL(1 m) of 123.49 dB with the condition of 0.3 mm SP and T_h = 675 K, indicating a 9.3% increase in SPL compared to system with air as

	Document title	Authors	Source	Year	Citations

working gas. The performance is significantly influenced by the thickness and Young's modulus. This system has potential for application as an *outdoor sound* source.

Article • *Open access*

10 **Auditory comfort and noise perception under thermal and visual conditions in urban parks** — Simsek O. — Scientific Review Engineering and Environmental Sciences, 35(2) — 2026 — 0

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This study investigates auditory comfort in *outdoor* urban environments by examining the combined influence of visual and *thermal* factors. Field research was conducted in two major urban parks in Adana, Turkey, representing the Mediterranean climate, where simultaneous environmental measurements and surveys were performed during the autumn season. The collected data included air temperature, relative humidity, wind speed, mean radiant temperature, *sound* pressure level, and both horizontal and vertical illuminance. Correlation and regression analyses revealed that *sound* pressure level was the most influential parameter, showing a strong negative association with auditory comfort, while higher vertical illuminance slightly decreased comfort levels. *Thermal* variables exhibited indirect effects on noise perception rather than on comfort itself. Participants expressed a strong preference for natural *sounds* such as birdsong and flowing water, while human and mechanical *sounds* were generally disliked. The findings highlight that auditory comfort is a multisensory phenomenon shaped by both acoustic and contextual environmental qualities.

Article

11 **Wearable Flexible Acoustic Alarm Device Based on MXene/Laser-Induced Graphene@Polyimide Thermoacoustic Film** — Zhang Y., Han D., Wang Y., ... Zhai W., Wang J. — ACS Sensors, 11(4) — 2026 — 1

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Flexible acoustic devices have advantages of high flexibility and large *sound* output area, capable of adhering to the surface of the skin or clothing, which shows great potential in the field of wearable electronics. However, the design of acoustic material structures and their functional applications are still difficult. Here, we fabricated a porous laser-induced graphene (LIG) microarray layer on a polyimide (PI) film by laser direct writing. Subsequently, we uniformly coated the few-layer MXene nanosheets onto the LIG surface to obtain an MXene/LIG@PI film featuring low specific heat capacity per unit area, large interlayer spacing, excellent electrical conductivity, and high *thermal* conductivity. Meanwhile, the MXene/LIG@PI-based acoustic device composed of flexible copper electrodes exhibits excellent acoustic performance with a high *sound* pressure level of 77.6 dB and maintained a stable acoustic spectrum as the frequency increased from 20.0 Hz to 20.0 kHz. Finally, the portable acoustic alarm device composed of the piezoelectric module and the MXene/LIG@PI-based acoustic module was designed, which can be worn on various parts of the human body clothing, to provide audible alarm reminders in case of accidental falls or drops during *outdoor* activities. This study provides a novel approach for designing new flexible acoustic materials and devices.

Article

12 **A highly resilient and large-strain wearable hydrogel sensor based on acrylamide/gelatin/Zr4+ and its application in human motion monitoring and information transmission** — Zhang Y., Liang M., Yan Z., Zhang S. — Polymer, 346, 129635 — 2026 — 5

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The application potential of flexible electronic technologies across diverse interdisciplinary domains has become increasingly evident. Owing to their unique combination of flexibility, biocompatibility, and skin-mimetic mechanical properties, hydrogels have emerged as a central material for wearable flexible sensors. Nevertheless, conventional hydrogel-based sensors often suffer from inadequate mechanical performance under high stress and large strain conditions, along with limited toughness. Gelatin, a natural protein, possesses a molecular structure abundant in

Document title	Authors	Source	Year	Citations

hydroxyl and amino functional groups, enabling the formation of a highly extensible, hydrogen-bonded flexible network and exhibiting excellent biocompatibility. In this study, a *thermally* induced free-radical polymerization approach was employed to construct a covalently cross-linked rigid framework using polyacrylamide (PAM), while incorporating gelatin to establish a hydrogen-bond-reinforced flexible network. Concurrently, Zr^{4+}were introduced to coordinate with anionic functional groups on both gelatin and PAM chains, resulting in the fabrication of a polyacrylamide/gelatin/zirconium ion composite hydrogel sensor (PMG_xZ_y). The PMG_xZ_y hydrogel sensor demonstrates exceptional toughness, rapid response dynamics, high sensitivity, and a broad sensing range. It achieves a strain of up to 924 % under a stress of 0.26 MPa, with a fracture energy of 1.12 MJ m^{-3}. The sensor not only enables real-time monitoring of large-amplitude human motions but also facilitates *sound* recognition, handwritten pattern detection, and information transmission via integration with Morse code encoding. This work effectively overcomes the mechanical limitations of traditional hydrogel sensors, offering a promising new material platform for wearable sensing applications in fields such as *outdoor* sports monitoring and rehabilitation training.

Article

13 **On the applicability of knowledge sources in climate-based design: Thermal-spatial micro-spaces in historical gardens versus contemporary outdoor design strategies** — Nouri-Horzvili N., Suzanchi K., Haghighat Bin M. — Urban Climate, 65, 102713 — 2026 — 0

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Quantitative research on *outdoor thermal* comfort often produces strategies that are technically *sound* but limited in systemic integration and practical design applicability. This study reveals the limitations of existing strategies in capturing the complexity of *thermal*-spatial mechanisms within historical gardens' micro-spaces. Using Alexander's pattern language as a shared framework, we translated pure science-based strategies and design-informed interpretation into a hierarchical *thermo*-spatial syntax to decode two comparable representations. We analyzed each representation using five systemic complexity indicators: four; Betweenness Centrality, Katz Centrality, Strongly Connected Components (SCCs), and feedback loops, capture horizontal integration and connectivity, while Hierarchy Depth captures vertical, multi-level dependencies. Accordingly, scientific decoding often reflects static, component-based logic, whereas designer-based decoding reveals dynamic, cyclic systems rich in integration. Additionally, bidirectional mapping shows how designer insights can enrich scientific structures, highlighting evolutionary micro-spaces as reservoirs of human-scale *thermal* strategies and revealing system flows that reduce the complexity of *outdoor*.

Article • *Open access*

14 **Prefabricated Co-Working Spaces' Window Design: Emotional Salience Scale-Based Optimisation** — Ciervo A., Masullo M., Maffei L., ... Morelli M.D., Scorpio M. — Buildings, 16(4), 875 — 2026 — 1

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Windows are key elements of the building's system; they connect workers with the *outdoor* environment, influence daylight penetration, *sound* insulation, and *thermal* exchanges of façades, but they also moderate the workers' well-being and productivity. This research investigates how the window-to-wall ratio, as well as the position and orientation of mullions, in movable offices affect the combination of workers' perceptual and emotional responses. A smart co-working prefabricated movable office was modelled in virtual reality to include dynamic visual elements and acoustic stimuli. Experiments were performed in a laboratory under controlled *thermal* conditions involving 32 volunteers. The Igroup Presence and Emotional Salience Questionnaires were used to collect subjective responses. ANOVA analysis and post hoc test with the Bonferroni correction were used for data elaboration. Results revealed that window design affects emotional salience. High window-to-wall ratio and no mullions achieved the highest scores. Increasing the number of mullions, particularly when they obstruct key visual elements, reduced the positive emotional salience rating. Horizontal mullions diminish the outdoors' spatial perception, interrupting visual continuity and restricting users' capacity to recognise variations in the views. Finally, the results suggest some valuable insights and suggestions that can help designers improve window design and people's well-being and satisfaction.

	Document title	Authors	Source	Year	Citations
15	Article **Acoustic enhancement and weathering durability of rice husk/PVC composites via microperforated design and nanocoating**	Ibrahim N.,Jaafar M., Che Hassan N.H., Garcia Sanz R.	Journal of Building Engineering , 119, 115253	2026	3

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Rice husk-reinforced polyvinyl chloride (rice husk/PVC) composites valorize agro-waste but suffer from poor durability and weak acoustic absorption, limiting their use in construction. This study adopts a dual approach, a microperforated panel (MPP) design was introduced to enhance acoustic performance, and a dual-layer nanocoating of multi-walled carbon nanotubes (MWCNTs) and silicon dioxide (SiO_2) was applied to improve weathering resistance. Rice husk/PVC composites were fabricated via extrusion and compression molding. For acoustic evaluation, samples with different perforation patterns were tested using an impedance tube, where MPP panels achieved a maximum *sound* absorption coefficient (SAC) of 1.0 at 1045 Hz, with resonance tunable by varying rear air gaps (10–30 mm). For durability evaluation, coated and uncoated composites were exposed to four weeks of accelerated UV-moisture weathering. Coated samples showed superior retention of tensile strength (only 18.5 % reduction compared to 56 % in uncoated composites) and maintained high hydrophobicity (contact angle 144° vs. 75° for uncoated). Fourier Transform Infrared Spectroscopy and Scanning Electron Microscope confirmed reduced oxidation and microcracking, while *thermal* conductivity values (0.312–0.327 W/m·K) remained comparable to other agro-waste-based insulation materials. Overall, the findings demonstrate that microperforation effectively addresses acoustic limitations, while nanocoating significantly enhances environmental durability. Together, these strategies highlight a sustainable pathway to transform rice husk into multifunctional composites for building engineering applications, particularly acoustic cladding, facade panels, and *outdoor* noise barriers.

16	Article • *Open access* **Prediction of outdoor ground effect**	Attenborough K., Taherzadeh S.	Applied Acoustics, 242, 111067	2026	3

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A classification of *outdoor* ground surfaces for calculating the attenuation due to destructive interference between direct and ground-reflected *sound* is proposed which uses a physically admissible ground impedance model with parameters of porosity, flow resistivity, and, for some ground surfaces, layer depth. This impedance model is shown to enable good fits to short range level difference measurements over a wide range of ground surfaces. Ranges of parameter values are suggested in each class to allow for variations in soil depth profiles, moisture content and surface roughness. Also, methods are proposed to account for mean ground roughness heights of less than 0.1 m and for waves on a water surface. The influence of atmospheric turbulence on ground effect is included through a coherence factor which assumes a Kolmogorov turbulence spectrum with parameters for which values can be calculated if heat flux and friction velocity are known. To account for changes in impedance along the propagation path, a Fresnel zone method is proposed which weights pressures squared since this method has been shown to compare better with 2D Boundary Element Method predictions than a method that weights excess attenuation.

17	Article • Article in Press **Outdoor comfort in historic spaces under thermoacoustic interactions**	Zhang R.,Cui L.,Liu J., ... Wang Z.,Zhen M.	Building Simulation	2026	0

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Outdoor comfort in historic spaces has received increasing attention under the pressures of urban heat and environmental noise, yet *thermoacoustic* research has focused mainly on modern urban settings. This study investigated Jinci Temple, a historic site in China's cold region. Field measurements and questionnaire surveys were conducted in four open-space types to examine the relationships among *thermal* conditions, acoustic conditions, and *outdoor* comfort. Representative optimization scenarios were simulated using ENVI-met and CadnaA and prioritized through an exploratory assessment informed by Heritage Impact Assessment (HIA). Respondents exhibited relatively high summer heat tolerance. The neutral physiological equivalent temperature (NPET) was 29.4 °C, the neutral PET range was 23.7–35.1 °C, and the upper acceptable PET threshold was 34.3 °C. Acoustic evaluation varied markedly with perceived *sound*-source type. At a neutral mean acoustic sensation vote, *sound* pressure levels were 72.2, 68.8, and

	Document title	Authors	Source	Year	Citations

60.9 dB for positive, neutral, and negative *sound* sources, respectively, indicating greater acoustic tolerance under positive *sound* conditions. *Thermoacoustic* associations were context-dependent: *thermally* comfortable conditions were associated with lower sensitivity to SPL variation, whereas moderate acoustic conditions were associated with weaker *thermal* discomfort. Scenario simulations predicted that reversible shading with a solar transmittance of 0.3 would reduce mean radiant temperature and PET by 12.5 and 5.3 °C, respectively, whereas the predicted *sound* pressure level reductions for the 3-m and 5-m multilayer green buffers were only 0.2 and 0.4 dB, respectively. HIA identified vegetation shading and greening as priority strategies, water–vegetation synergy as a localized conditional strategy, and visitor noise management as a supplementary measure.

Conference Paper

18 **Review on Soundproof Ventilation Techniques: Study on Ventilation and Acoustic Performance** — Baskara S.A., Trihamdani A.R. — Lecture Notes in Civil Engineering , 875 LNCE — 2026 — 0

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Natural ventilation has been known as a cost-effective and sustainable solution to provide good indoor air quality (IAQ) and *thermal* comfort in the hot-humid climate. Nevertheless, in the urban setting, opening windows also allows *outdoor* noises to penetrate the indoor spaces. This problem forces occupants to close the window, which will lead to poor IAQ and *thermal* comfort. This research aims to provide a solution to the aforementioned conflicting issues, which are ventilation performance and *outdoor* noise reduction. Firstly, several noise resonator techniques is compared. Then, a noise silencer based on the best resonator performance is developed. Finally, the detailed assessments of ventilation and noise reduction performances of selected (*Soundproof* Ventilation) SPV techniques, i.e. noise resonator, window plenum, and acoustic louver were conducted through simulation. The simulation results show that the *soundproof* ventilation is 24.1 dBA, 9.1 dBA, 2.7 dBA, 2.6 dBA SPL reduction and 0, 0.036, 0.209, and 0.362 discharge coefficients for noise silencer, plenum window, chevron acoustic louver, and single bank acoustic louver respectively. Discussion to improve the SPV technique was also conducted.

Article • Article in Press

19 **Threshold-based multi-scale analysis of thermal-acoustic comfort in urban public spaces: identifying critical thresholds and individual heterogeneity** — Xu L., Jing W., Ding W., ... Ge Z., Ai Y. — Environment Development and Sustainability — 2026 — 0

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Outdoor users encounter *thermal* and acoustic conditions simultaneously, and pooled comfort scores may obscure differences among respondents. We analysed 975 valid questionnaires paired with concurrent environmental measurements at a university plaza in Xi'an to examine response thresholds, conditional associations and heterogeneity. The analysis combined established approaches: piecewise regression, regression trees, K-means clustering, random forest interpretation using Shapley additive explanations (SHAP), and partial dependence analysis. The *thermal*-sensation curve changed slope near 30 ℃. A *thermal*-comfort breakpoint was estimated at 29.95 ℃, but the comparison with a linear model did not meet the 0.05 significance criterion. Acoustic sensation had an estimated breakpoint of 64.8 dB, while acoustic comfort declined without a significant breakpoint. Clustering yielded three response profiles. Under the combined high-temperature and high-*sound*-level condition, *thermal* comfort was 0.63 units below the additive prediction. Temperature accounted for the largest reported SHAP contribution to *thermal* comfort. The findings motivate local tests of coordinated shading and *sound* management, together with alternative *outdoor* settings for users with different preferences. Validation at other sites is needed before applying the thresholds or profile-based recommendations more widely.

Article • Article in Press

20 **Embodied eco-consciousness in paragliding: a multi-method phenomenological approach to a cosmic sport** — Schirrer M., Aubin N., Schnitzler C., Lefevre L. — Sport Ethics and Philosophy — 2026 — 0

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Anchored in the philosophy of sport, this article adopts a phenomenological approach centred on the embodied experience of free flight. It explores paragliding as a cosmic sport, in which the body enters into a continuous dialogue with the invisible forces of air, gravity, and the rhythms of the sky. Departing from performance or control-oriented conceptions of sport, this research makes visible the sensory and pre-reflective experience of pilots—both novice and expert—in their ways of inhabiting the air. Three qualitative field studies were conducted: the accompaniment of high-school students during initial slope training, a longitudinal exploration of novice high-school students' first major paragliding flights and explicitation interviews with experienced cross-country pilots. The methodology combines multisensory experience narratives, eco-consciousness drawings and interviews supported by GPS flight traces, providing access to the fine textures of lived experience, like tensions in the lines, wind pressure on the skin and the *sound* of rising *thermals*. Our findings reveal a progressive development of embodied eco-consciousness, from an anthropocentric perspective towards a more cosmic attunement, in which body, wing and elements become experientially intertwined. This research contributes to the philosophy of sport by offering an empirical phenomenological account of how nature-based practices cultivate embodied eco-consciousness, challenging performance-oriented models of sport and demonstrating how cosmic sports can foster ecological awareness through sensory attunement rather than environmental mastery. These experiences invite a rethinking of *outdoor* sports ethics: rather than seeking to dominate the environment, the aim becomes one of the attunement and transformation through it.

Article • *Open access*

21 **Multisensory Interactions in Greenway Plazas of Differing Openness and Effects on User Behaviors** — Peng Z., Liu W., Teng M., ... Keyhani A.B., Wang P. — Urban Science, 10(1), 60 — 2026 — 3

Hide abstract Find It @ UCL View at Publisher Related documents

Spatial openness affects the subjective evaluation of *soundscape*, landscape, and *thermal* perceptions, leading to various restoration effects and recreational behaviors. However, the literature lacks studies investigating the effects of multisensory interactions under different levels of spatial openness in plazas on users' behaviors in urban greenways. Thus, this study contributes to the enhancement of recreational experiences and the environmental design of urban greenways by examining the interaction between multisensory evaluations and recreational behaviors in greenway plazas with different levels of spatial openness. Three types of plazas (enclosed, semi-enclosed, open) were selected along an urban greenway to analyze interactions through in situ measurements, questionnaires, and behavior observation. The results showed that people rated the environment as the quietest and coolest in enclosed plazas, although the *sound* pressure level of these plazas was the highest. Furthermore, the visual evaluation (VE) was mostly correlated with acoustic evaluation (AE) in plazas with high openness, while the correlation effect between AE and *thermal* evaluation (TE) was only significant in enclosed plazas. In other words, AE was the key factor targeting the improvement in comfort in greenway plazas. Secondly, improving AE was more effective for stimulating the frequency of interactive activities in enclosed plazas, compared to improving TE. However, AE had a negative effect on the time that people were willing to spend on interactive activities in semi-enclosed plazas. Finally, these findings provide corresponding strategies for creating comfortable audio, visual, and *thermal* environments in greenway plazas with different levels of openness, as well as strategies for enhancing the recreational experiences of visitors.

Article • *Open access*

22 **The role of campus green spaces in improving microclimate, reducing noise, and achieving thermal comfort | ةرارحلا ةحارلا قيقحتو ءاضوضلا ليلقتو يرغصلأا خانملا نيسحت يف يعماجلا مرحلل ءارضخلا تاءاضفلا رود** — Alojany Z.O., Al-Zurfi M.T.H., Abdulrazzaq Z.M. — Basrah Journal of Agricultural Sciences, 38(2) — 2025 — 2

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The importance of campus green spaces lies in their planting with various plants (trees, shrubs, green spaces, etc.) which work to purify the air, reduce the intensity of solar radiation and temperatures, increase humidity in the atmosphere, and decrease wind speed, which results in achieving *thermal* comfort for campus occupants. This is what the results of the study proved, as the temperature, amount of solar radiation, relative humidity, and wind

	Document title	Authors	Source	Year	Citations

speed under the canopy of trees in the gardens planted with grass in the morning and afternoon were 36.7 and 37.1 C°, 1031 and 1551 candelas, 43.3%, 34.3%, 0.9, and 1.2 m s^{-1}respectively, compared to what was measured in the paved streets recorded41.6 and 45.3 °C, 5587 and 6933 candelas, 23%, 20.6%, 2.9, and 2.8 m s^{-1}respectively, At midday, the microclimatic contrast between tree-lined areas and paved streets manifested in mean differences of up to 8 C° in air temperature, 5382 candelas in solar radiation intensity, 13.7% in relative humidity, and 1.6 m s^{-1}in wind velocity. The results also showed that plants had a significant impact on reducing *sound* intensity (noise) under the tree canopy in turf-planted gardens in the morning and afternoon, with levels of 52.7 and 51.8 dB, compared to levels of 66.7 and 65.6 dB, respectively. Plants were more effective at absorbing *sounds* at higher frequencies. The results thus demonstrate the significant impact of plants in improving the microclimate and reducing noise in *outdoor* environments.

Article

23 **Comprehensive effects of thermal and acoustic environments on overall comfort in urban parks of cold regions in China** — Wu X.,Yuan C.,Xue H.,Yan Y. — Energy and Buildings, 347, 116329 — 2025 — 9

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Urban parks play a crucial role in enhancing residents' quality of life; however, research on the combined effects of *thermal* and acoustic environments on subjective evaluation remains limited. This study explores these interactions in cold-region urban parks, focusing on People's Park in Zhangjiakou, China. A combination of field measurements and questionnaire surveys was conducted across three representative spaces characterized by three common *sound* types: human-generated, artificial, and natural *sounds*. The results indicate that, in terms of *thermal* perception, natural *sounds* significantly reduce heat stress by lowering the *thermal* sensation vote and improving *thermal* comfort, whereas artificial *sounds* exacerbate it. Additionally, high *sound* pressure levels (L_{Aeq} ≥ 60 dB(A)) exacerbate *thermal* discomfort, making individuals feel hotter in summer and colder in winter, thereby reducing *thermal* comfort. Regarding acoustic perception, a *thermally* comfortable environment enhances the acoustic sensation vote and improves acoustic comfort. For overall comfort, during summer and transitional seasons, the overall comfort vote is negatively correlated with both the Universal *Thermal* Climate Index (UTCI) and *sound* pressure levels, while in winter, overall comfort vote is positively correlated with UTCI but negatively correlated with *sound* pressure levels. The analysis of equivalent overall comfort vote further reveals that in summer and transitional seasons, rising temperatures under extreme heat conditions (UTCI ≥ 35°C) have a more pronounced negative impact on overall comfort vote than decreasing temperatures in winter. Conversely, in winter, variations in *sound* pressure levels exert a stronger influence on overall comfort vote compared to those in summer and transitional seasons. This study addresses a research gap in understanding the interaction between *thermal* and acoustic environments in cold-region urban parks and provides valuable insights for the future planning and optimization of urban green spaces in such climates.

Article • *Open access*

24 **A Combined Weighting Method to Assess Indoor Environmental Sub-Factors for Human Comfort in Offices in China's Severe Cold Regions** — Li Z.,Song G.,Zhang Q.,Yu J.,Liu Y. — Buildings, 15(19), 3529 — 2025 — 3

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Indoor environmental quality in offices, comprising *thermal*, acoustic, lighting, and air quality domains, is known to influence human comfort, yet the relative importance of their sub-factors—particularly in severe cold regions—remains unclear. This study addresses this gap by integrating objective (Criteria Importance Through Intercriteria Correlation, CRITIC) and subjective (Analytic Hierarchy Process, AHP) weighting methods, supported by field measurements and questionnaire surveys in open-plan offices in three provinces in northeastern China. Cluster analysis categorized acoustic sub-factors into *outdoor* traffic, *outdoor* entertainment, people conversation, burst *sound*, and people movement. Results show that temperature is the dominant *thermal* comfort driver (39.7% CRITIC; 45.5% AHP), exceeding air velocity and humidity, which had nearly equal influence. Indoor *sound* exerted greater impact than *outdoor sound*, with people conversation ranked highest among indoor noise sources, and burst *sound* and movement showing similar but slightly lower weights. Natural light outweighed artificial light in importance (54.2% CRITIC; 61.0% AHP), while air freshness and pollution were nearly equally influential. Compared to CRITIC,

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AHP produced more dispersed weights, reflecting subjective bias toward pronounced differences. These findings provide a quantitative basis for prioritizing environmental design interventions—such as controlling indoor conversational noise, optimizing natural lighting, and stabilizing temperature—to enhance comfort in offices in severe cold regions.

Article • *Open access*

25 **Outdoor Comfort Optimization in Historic Urban Quarters: From Multisensory Approaches to Operational Strategies Under Resource Constraints**

Su H.,Ma H.,Liu K.

Buildings, 15(19), 3616 2025 1

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During the transition from urban expansion to renewal, optimizing environmental comfort under resource constraints presents critical challenges. While existing research confirms that multisensory interactions critically shape environmental comfort, these insights are rarely operationalized into protocols for resource-constrained contexts. Focusing on historic urban quarters that need to balance modification and preservation, this study quantifies multisensory (acoustic, visual, *thermal*) interactions and integrations to establish operational resource-optimization strategies. Through laboratory reproduction of 144 field-based experimental conditions (4 *sound* sources × 3 *sound* pressure levels × 4 green view indexes × 3 air temperatures), systematic subjective evaluations of acoustic, visual, *thermal*, and overall comfort were obtained. Key findings demonstrate: (1) Eliminating extreme comfort evaluations (e.g., “very uncomfortable”) within any single sensory domain stabilizes cross-sensory contributions to overall comfort, ensuring predictable cross-domain compensations and safeguarding resource efficacy; (2) Accumulating modest improvements across ≥2 sensory domains reduces per-domain performance threshold for satisfactory overall comfort, enabling constraint resolution (e.g., visual modification limits in historic districts); (3) Cross-domain optimization of environmental factors (e.g., *sound* source and air temperature) generates mutual enhancement effects, maximizing resource economy, whereas intra-domain optimization (e.g., *sound* source and *sound* pressure level) induces competitive inefficiencies. Collectively, these principles establish operational strategies for resource-constrained environmental improvements, advancing sustainable design and governance through evidence-based multisensory approaches.

Article

26 **Seeking comfort in the urban heat: Unraveling the effects of thermal-acoustic environments on psychological restoration in urban greenways**

Gao M.,Zhu X.

Building and Environment , 282, 113269 2025 19

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As environmental pressures from high-density urbanization increase, the comfort of *outdoor* spaces and their restorative benefits for urban residents have become increasingly important. Although *thermal* and acoustic environments are key factors influencing *outdoor* comfort, evidence linking *thermal* and acoustic comfort with restorative benefits is scarce. This study explores the impact of the *thermal*-acoustic environment on psychological restorative benefits in urban greenways through on-site meteorological measurements and questionnaire surveys. The results indicate that, during the hot season, compared to densely vegetated greenways, sparsely vegetated greenways are perceived as hotter and noisier. Furthermore, the relationship between the *thermal*-acoustic environment and restorative benefits is more strongly correlated in sparsely vegetated greenways than in densely vegetated ones. In other words, the environmental comfort in different types of greenways can be targeted for improvement. Secondly, the interaction between *thermal* and acoustic environments influences restorative benefits. At the same *sound* pressure levels, restorative benefits decrease as the Universal *Thermal* Climate Index (UTCI) increases. However, on sparsely vegetated greenways, under conditions of 70–75 dB(A), the increase in UTCI does not significantly affect restoration. Finally, to identify the factors hindering psychological restoration, the study compares the veto power of *thermal*-acoustic comfort voting (TCV, ACV). This veto power varies according to greenway type. In densely vegetated greenways, ACV has greater veto power than TCV, while in sparsely vegetated greenways, the reverse is true. The findings offer valuable insights for urban planners and policymakers, suggesting that creating

	Document title	Authors	Source	Year	Citations

comfortable *thermal*-acoustic environments in *outdoor* spaces can enhance restorative benefits and contribute to the sustainability of high-density urban areas.

Article • *Open access*

27 **Impact of personal and contextual factors on multi-domain sensation in classrooms: Results from a field study**

Morandi F.,Gasparella A., Tzempelikos A.,Pittana I., Cappelletti F.

Building and Environment , 280, 112892

2025

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In indoor spaces shared by many people, such as school classrooms, the contextual and individual characteristics can affect the perceived indoor environmental quality and explain the distribution of the occupants' subjective comfort evaluations. Understanding the impact of these factors on the subjective assessment of the indoor environmental quality (IEQ) and multi-domain comfort is crucial for reliable post-occupancy evaluations and retrofit solutions. This work presents the impact of personal (i.e., gender, clothing level, body mass index, age, use of glasses) and contextual (i.e., sitting position inside the classroom, facade orientation/*outdoor* view) factors on the students' *thermal*/air quality/visual/acoustic sensation votes collected in 50 classes during regular lectures, through the administration of 825 questionnaires. Students' sensation votes regarding the four perceptual domains were collected and correlated with environmental parameters (i.e., indoor and *outdoor* air temperature and relative humidity, indoor mean radiant temperature, indoor CO_2 concentration and TVOC, indoor horizontal illuminance, indoor *sound* pressure level). Multi-linear regression analyses were performed to identify environmental variables significantly correlated with sensation votes, exploring cross-domain interactions. Then, the sensation votes were grouped according to different levels of personal and contextual factors to study correlations between votes and related binned environmental parameters for each set, by means of covariance analysis and post-hoc tests. Among personal factors, age and gender were proven to impact on the sensation votes in all the four domains. Furthermore, Body Mass Index and clothing level influence *thermal* sensation while the use of glasses do not affect visual sensation. The sitting position in the classroom impacts *thermal*, IAQ, visual sensation votes in relation to the proximity to the windows and the visual and acoustic sensation in relation to the proximity to the teacher and blackboard. Finally, groups identified by classroom orientation and occupant density significantly differ in terms of sensation votes.

Conference Paper • *Open access*

28 **Contributions by Richard Raspet on two aspects of the acoustics of rigid framed porous media**

Attenborough K.

Proceedings of Meetings on Acoustics , 56(1), 045001

2025

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Two papers co-authored by Richard Raspet are about the surface impedance of grounds with exponential porosity profiles [R. Raspet and J. M. Sabatier, J. Acoust. Soc. Am., 99, 147–152 (1996)] and *sound* propagation in porous media containing main pores with porous walls [W. P. Arnott, J. M. Sabatier and R. Raspet, J. Acoust. Soc. Am., 90, 3299–3306 (1991)]. The variable porosity ground impedance model enables better fits to *outdoor* short-range measurements over many types of grassland than the widely used semi-empirical single parameter model. Moreover, it has been found to represent the surface impedance of artificial surfaces including felt and perforated rubber mat. A better approximation than that used hitherto for a variable porosity ground is suggested. The theory for pores with porous walls which was developed in the context of *thermoacoustic* stack designs is shown to be related to several designs involving slit like pores with regularly spaced dead end pores, periodically varying widths or microporous walls in the somewhat different context of creating useful *sound* absorbers by additive manufacture.

Article • *Open access*

29 **3SqAir Project: A Living Lab Towards Sustainable Smart Strategy for Indoor Climate Quality Assurance in Classrooms**

Ogundiran J., Nyembwe J.-P.K.B., Ogundiran J.O., ... Pereira L.D., Gameiro da Silva M.

Atmosphere, 16(5), 584

2025

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	Document title	Authors	Source	Year	Citations

The indoor climate quality in classrooms at the University of Coimbra, Portugal, was investigated as part of the 3SqAir project, supported by the Interreg SUDOE program. This research focused on two equipped classrooms with different ventilation systems: natural and mechanical ventilation. Both classrooms were continuously monitored for IEQ parameters: *thermal* comfort, indoor air quality, noise, and lighting during heating and cooling seasons. Air temperature, relative humidity, CO_2 concentration, particulate matter, nitrogen dioxide, volatile organic compounds, formaldehyde, *sound* pressure level, and illuminance were measured. *Outdoor* weather conditions were also recorded. The primary focus was on air temperature, CO_2 concentrations, and relative humidity, while air change rates (ACH) were estimated using the Tracer Gas Method. The results showed inadequate *thermal* conditions in both classrooms, particularly during the heating season. Most weekly mean CO_2 concentrations were within acceptable limits, while ACH were below standard recommendations in four CO_2 decay phases. Simulations of CO_2 decay revealed further air quality gaps in each room. Corrective measures within the 3SqAir project framework were suggested for approval and implementation while monitoring continues. This work represents the first phase in an evolving study towards developing sustainable strategies for improving indoor air quality in classrooms.

Article

30 **How does campus-scape influence university students' restorative experiences: Evidences from simultaneously collected physiological and psychological data**

Zhang J.,Liu S.,Liu K., Bian F.

Urban Forestry and Urban Greening , 107, 128779

2025

29

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“Time in nature” is widely acknowledged as beneficial for physical and psychological health. The landscape environment of campus – referred to as “campus-scape” – plays a crucial role in influencing students’ mental health. It provides *outdoor* spaces where university students sought for relief and recreation, yet it remains under-researched. In this study, we address a limitation in the static assessment of restorative effects of campus-scape, and extend the focus from visual to non-visual and event landscape, using simultaneously collected psychological and physiological data from an on-site walking experiment (n = 40). This study examined how different types of campus-scape affect university students' restorative experiences by analyzing stress and attention restoration using wearable sensors, alongside perceptional psychological data from questionnaires and interviews. Our results revealed significant differences in the restorative effects of various campus-scape, and the influences were further characterized by visual, non-visual, and event landscape. We have several key findings: (1) 93 % of students reported above-medium stress levels and 72 % experience attention fatigue, highlighting a strong need for restorative campus-scape; (2) Patchy green-blue landscape performs best in stress recovery, followed by linear green-blue landscape and teaching spaces, while living spaces especially the commuting paths were the least restorative, as indicated by physiological electrodermal activity (EDA) and psychological perceived restorative scale (PRS).Meanwhile, most participants experienced attention restoration post-walk across campus-scape types, interestingly subjective self-reported data (FS-14) underestimated the effects compared with objective tests (DSB and TMTB); (3) Within visual landscape, lower landscape segmentation, fewer building views, more welldesigned landscape facilities, and abundant waterbodies, green, and sky views are significantly associated with better stress recovery; (4) Non-visual landscape like *thermal* conditions, *soundscape*, and olfactory landscape, along with event landscape–occasional, staged, and routine events–also influence the restorative effects of campus-scape. The aesthetic qualities of campus-scape and capacity for recreational activities are key to students' restorative experiences. Therefore, campusscape designers should prioritize the planning of patchy green-blue spaces and consider enhancing visual, non-visual, and event landscape simultaneously in design guidelines to improve university students' mental well-being.

Article • *Open access*

31 **Exploring the relationship between elderly women responses, multi-domain factors, and street geometry indicators: A case study in a Mediterranean context**

Mahia L.,Berkouk D., Bouzir T.A.K.,Pigliautile I., Pisello A.L.

Building and Environment , 273, 112647

2025

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Urban public spaces must ensure safe conditions for all, including vulnerable groups such as the elderly people. Since environmental characteristics of urbanized areas largely depend on cities' design-related urban geometry, grasping

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how multi-domain stimuli resulting from this variation impact elderly *outdoor* activities and overall health is fundamental to promoting more sustainable and healthier urban communities. This study combines objective and subjective methods to examine the relationship between multi-domain factors, elderly women's responses, and street geometry during a late afternoon walk in a Mediterranean context, Algeria. Forty-one elderly women engaged in walking activities across different streets, providing subjective feedback on environmental sensation and psychophysical health while being monitored in terms of physiological parameters and environmental conditions. Findings reveal that a high sky view factor, associated with high illuminance mean values and increased sunlight sensation vote (SSV), increased the skin temperature of the forehead (up to 39 °C) and induced an unrelaxed state among elderly women ($P < 0.001$). Moreover, a high height-to-width ratio, associated with high mean values of the physiological equivalent temperature (PET) and increased *thermal* sensation vote (TSV), increased heart rate ($P < 0.05$) and induced a weak state ($P < 0.001$). Additionally, high mean values of the *sound* pressure level (SPL) and increased acoustic sensation vote (ASV) caused a weak ($P < 0.001$) and a relaxed state ($P < 0.05$). Furthermore, our study highlighted some health parameters that could be used as predictor indicators of *outdoor* sensations for elderly women who are unable to objectively assess their environment.

Article • *Open access*

☐ 32 **Acoustic Fabry–Perot Resonance Detector for Passive Acoustic Thermometry and Sound Source Localization** — Yue Y.,Dong Z.,Qi Z.-M. — Sensors, 25(8), 2445 — 2025 — 1

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Acoustic temperature measurement (ATM) and *sound* source localization (SSL) are two important applications of acoustic sensors. The development of novel acoustic sensors capable of both ATM and SSL is an innovative research topic with great interest. In this work, an acoustic Fabry-Perot resonance detector (AFPRD) and its cross-shaped array were designed and fabricated, and the passive ATM function of the AFPRD and the SSL capability of the AFPRD array were simulated and experimentally verified. The AFPRD consists of an acoustic waveguide and a microphone with its head inserted into the waveguide, which can significantly enhance the microphone's sensitivity via the FP resonance effect. As a result, the frequency response curve of AFPRD can be easily measured using weak ambient white noise. Based on the measured frequency response curve, the linear relationship between the resonant frequency and the resonant mode order of the AFPRD can be determined, the slope of which can be used to calculate the ambient *sound* velocity and air temperature. The AFPRD array was prepared by using four bent acoustic waveguides to expand the array aperture, which combined with the multiple signal classification (MUSIC) algorithm can be used for distant multi-target localization. The SSL accuracy can be improved by substituting the *sound* speed measured in real time into the MUSIC algorithm. The AFPRD's passive ATM function was verified in an anechoic room with white noise as low as 17 dB, and the ATM accuracy reached 0.4 °C. The SSL function of the AFPRD array was demonstrated in the *outdoor* environment, and the SSL error of the acoustic target with a *sound* pressure of 35 mPa was less than 1.2°. The findings open up a new avenue for the development of multifunctional acoustic detection devices and systems.

Review • *Open access*

☐ 33 **Environmental comfort in urban spaces: A systematic literature review and a system dynamics analysis** — Ataee S.,Lopes M.,Relvas H. — Urban Climate, 60, 102340 — 2025 — 20

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Urban environments are intricate systems where the well-being of inhabitants is shaped by various interrelated factors. This study employs a combination of systematic literature review (SLR) analysis and system dynamics (SD) approach to systematically explore the multifaceted concept of environmental comfort (EC) within urban settings. By combining a comprehensive literature review with innovative SD modeling, this research uniquely examines the complex relationships between *thermal* comfort, human perceptions, and spatial settings. The results demonstrate that perceiving EC in terms of promoting livability and sustainability requires an integrated understanding of *outdoor* comfort that accounts for its systemic nature. Importantly, the study identifies significant gaps in the understanding of sensory aspects of EC, such as the *outdoor soundscape* and air quality, highlighting these as critical areas for future research. By adopting a holistic planning paradigm, this study provides valuable insights for urban planners and policymakers to create environments that enhance overall comfort, livability, and resilience.

	Document title	Authors	Source	Year	Citations
34	Article **Design and validation of the multi-parameter control strategies for fan-pad cooling systems in rabbit housing during summer | 兔舍夏季湿帘-风机降温系统多参数调控策略设计与验证**	Wang Q.,Lyu W.,Li W., ... Liu W.,Wu Z.	Nongye Gongcheng Xuebao Transactions of the Chinese Society of Agricultural Engineering, 41(5)	2025	0

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Automatic control systems have been widely used for the fan-pad cooling of livestock houses in summer. However, current regulation can be confined to the single-parameter temperature. In high-humidity and high-temperature regions, the premature or delayed activation and deactivation of water pumps can increase the indoor humidity levels on the heat stress of animals. In low-humidity and high-temperature areas, the activating water pumps can cause significant fluctuations in indoor temperature, thus resulting in *thermal* discomfort for the animals. In this study, a multi-parameter control strategy was designed to tailor for the different climatic conditions. A dual-parameter control strategy was also developed for the high temperature and high humidity climates. The water pump was regulated using two parameters, including the indoor temperature and *outdoor* humidity. Fan operation was then controlled for the optimal indoor temperature. The multi-parameter control strategy was implemented in the low-humidity and high-temperature climates. The water pump was then regulated using three parameters, including the indoor temperature, *outdoor* temperature, and *outdoor* humidity. While the variable frequency of fans was regulated using indoor temperature. Additionally, a “prevent sudden drop measure” was introduced to reduce the fan speed during the initial startup of water pumps. The “temperature fluctuation mitigation measure” was applied to adjust the water circulation time. These strategies were validated in the rabbit farms in Qingdao City, Shandong Province, and Qingyang City, Gansu Province, China. In Qingdao, the meat rabbit house was equipped with a controller to monitor the environmental regulations under different weather conditions. The results showed that the proportion of time experiencing heat stress (temperature-humidity index, THI > 27.8) was reduced to 2.2% during daytime and 0 at night. On high humidity cloudy days with the insufficient cooling potential of the cooling pad, the dual-parameter control strategy effectively cooled the indoor environment, where the water pump was closed to increase ventilation. On sunny days, the cooling pad achieved a theoretical cooling value of (3.3 ± 1.0) °C and an actual cooling value of (0.8 ± 0.5) °C. Importantly, no rise of indoor temperature was observed, when the water pump was deactivated, indicating the effective pump operation and control. In Qingyang, another rabbit farm was used to further validate the control strategies. An automatic controller was installed in a treatment house. While another meat rabbit house with a manually controlled cooling pad and fan operations served as the control house. Environmental parameters, rabbit body temperature, and production performance indicators were collected to compare the performance between them. The findings revealed that the indoor temperature near the cooling pad area decreased by 6.6 °C within 7 min after the initial activation of the water pump. The “prevent sudden drop measure” was then implemented to extend the time. A similar cooling effect was then achieved in 23 min. The “temperature fluctuation mitigation measure” effectively reduced the fluctuations of indoor temperature from 3 °C to within 1 °C, indicating a stable *thermal* environment. Compared with the control, the treatment house reduced the fluctuations of diurnal temperature, the daily average temperature of the rabbit body ($P < 0.05$), a 4.1% decrease in the cumulative mortality rate, and a 4.3% increase in the total sales revenue. These results demonstrate the effectiveness and versatility of the multi-parameter control strategy to enhance the cooling performance of fan-pad cooling systems under varying weather conditions in both high-temperature-humidity and high-temperature-low humidity regions. Stable indoor environments were maintained to reduce the rabbit's body temperature for economic returns. This strategy can provide a *sound* reference to precisely control the environmental temperature in livestock houses in summer.

35	Article • *Open access* **The Influence of Multisensory Perception on Student Outdoor Comfort in University Campus Design**	Touhami H.,Berkouk D., Bouzir T.A.K.,Khelil S., Gomaa M.M.	Atmosphere, 16(2), 150	2025	14

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Document title	Authors	Source	Year	Citations

The user's experience is critical in spatial design, particularly in *outdoor* spaces like university campuses, where the physical environment significantly influences students' relaxation and stress relief. This study investigates the combined impact of *thermal*, luminous, and auditory environments on students' perceptions within recreational areas at Bordj Bou Arreridj University Campus. A mixed-method approach combined field surveys and on-site measurements across eleven locations within three distinct spatial configurations. The findings from this study indicate that the auditory environment had the most substantial influence on overall perceptions, surpassing luminous and *thermal* factors. The open courtyard (Area 1) was perceived as less comfortable due to excessive heat and noise exposure. The shaded zone (Area 2) was identified as the most vulnerable, experiencing significant *thermal* stress and noise disturbances. In contrast, the secluded patio (Area 3) achieved the highest comfort rating and was perceived as the most cheerful and suitable space. Correlation analysis revealed significant interrelationships between physical and perceptual dimensions, highlighting the critical role of factors such as wind velocity, sky view factor, and illuminance in shaping *thermal*, luminous, and acoustic perceptions. A fuzzy logic model was developed to predict user perceptions of comfort, suitability, and mood based on measured environmental parameters to address the complexity of multisensory interactions. This study highlights the importance of integrating multisensory evaluations into spatial design to optimize the quality of *outdoor* environments.

Article

36 **Effect of thermal-acoustic-air quality composite environments on overall comfort of urban pocket parks considering different landscape types** — Li W.,Tian L.,Jin X.,... Tao Y., Peng P. — Energy and Buildings , 328, 115167 — 2025 — 15

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To bridge the research gap of landscape types' impact on overall comfort, this study studied the relationship between the combined effect of *thermal*-acoustic-air quality and overall comfort in urban pocket parks during winter through surveys and monitoring. Environmental parameters monitored included *outdoor* air temperature, relative humidity, wind velocity, ambient noise etc. The (*thermal*, acoustic, and air quality) comfort survey data, universal *thermal* climate index (UTCI), weighted equivalent continuous *sound* level (LAeq), and particulate matter with a diameter of less than 2.5 μm (PM2.5) were utilized to objectively evaluate the overall comfort vote (OCV). Furthermore, the comfort index-based and the perception vote-based OCV prediction models were built. Results showed that the water landscape parks had the highest overall and *thermal* comfort, followed by mountain, square, and building landscapes. For water and square landscape parks, the *thermal* comfort vote (TCV) was the most important comfort perception, while the air quality sensation vote (AQSV) was key for building landscape parks. No comfort perception vote index significantly affected OCV for mountain landscape parks. For "acceptable" OCV (=0), minimum UTCI thresholds were −5°C, −1.5 °C, and 5 °C for water, mountain, and building landscape parks, respectively, when LAeq exceeded 56 dBA, OCV remained constantly low (< 0) regardless of UTCI. The comfort index-based model is accurate and reliable for OCV prediction. This study provides the threshold requirements for achieving an "acceptable" overall comfort level in different types of pocket park, and could guide the urban pocket park design.

Article

37 **Outdoor thermal comfort of urban plaza space under thermoacoustic interactions –Taking Datang Everbright City as an example** — Zhen M.,Yuan X.,Gao L., ... Nan K.,Zhang M. — Building and Environment , 267, 112327 — 2025 — 10

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The comfort of urban square environments in terms of *thermal* and acoustic conditions directly influences the quality of people's travel experiences and their overall well-being. This study, represented by Xi'an Datang Everbright City, investigates the impact of *thermal*-acoustic environments on human comfort in cold regions of China. Four typical spaces within the city square were selected for meteorological measurements and subjective questionnaire surveys to gauge the *thermal* perception of active crowds. Using the Universal *Thermal* Climate Index (UTCI) as a *thermal* benchmark, the study compares the interaction between *thermal* conditions and acoustic environments. Results indicate that the neutral UTCI in Xi'an urban squares is 14.9 °C, with a neutral UTCI range of 10.8∼19.0 °C and a preferred temperature of 17.1 °C. Positive types of *sounds* moderately reduce *thermal* sensation. Moderate environmental *sounds* alleviate *thermal* discomfort to some extent among respondents. In environments such as

	Document title	Authors	Source	Year	Citations

birdsong and flowing water, people prefer higher temperatures, thereby increasing *thermal* tolerance. Conversely, in noise from traffic or construction, individuals prefer lower temperatures, reducing their *thermal* tolerance. In environments with broadcasts or conversations, respondents exhibit slightly higher *thermal* tolerance compared to the average neutral temperature. Finally, based on meteorological characteristics at different survey points and respondent feedback, optimization strategies for design are proposed.

Article • *Open access*

38 **Atmospheric turbulence affects noise annoyance from aircraft flyovers** — Lincke D.,Kawai C., Pieren R. — Acta Acustica, 9, 69 — 2025 — 0

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Auralization of *outdoor sound* propagation has become an important tool for studying noise perception in the contexts of aircraft, wind farm, and drone noise. To achieve high realism, these auralizations consider amplitude fluctuations caused by atmospheric turbulence. In a recent publication, a semi-empirical model was introduced which is applicable to relatively long *outdoor sound* propagation such as aircraft and wind farm noise, as it considers the saturation effect for amplitude fluctuations in the partially-saturated regime. This article presents a study applying the semi-empirical model for investigating the impact of turbulence-induced amplitude fluctuations on annoyance in auralized aircraft flyovers. For this, two 2-alternative-forced-choice listening experiments were conducted in a controlled laboratory setting: the first tested whether participants could reliably detect audible differences between aircraft flyovers under different meteorological conditions; the second assessed which condition was perceived as more annoying. The results show that strong differences in meteorological conditions and the respective atmospheric turbulence can lead to salient audible differences. Relative annoyance ratings tend to increase with stronger atmospheric turbulence. Further, the data suggest that amplitude fluctuations can interact with other characteristics of aircraft noise such as fan tones and alter the perceptual impact of these characteristics. Therefore, the study highlights the importance of modeling turbulence-induced amplitude fluctuations in realistic aircraft auralizations, and presumably also wind farm and drone noise auralization, as the perceptual impression can be affected in several ways.

Article

39 **Enhancing autumn greenway walking experience: Exploring the combined effects of noise and thermal environment** — Li Y.,Li L.,Wang W. — International Journal of Biometeorology, 69(1) — 2025 — 2

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The evaluation of *outdoor* green spaces is influenced by diverse sensory perceptions. Traffic noise and *thermal* conditions significantly impact greenway-walking satisfaction; their optimization is vital for improving user experience and encouraging *outdoor* engagement. The study examines a typical Beijing greenway during autumn, focusing on strategies to enhance the walking experience under the combined effects of noise and *thermal* environments through mobile measurements and surveys. The results show that: 1) The interplay between noise and *thermal* factors varies depending on the walking state. Upon arrival, an increase in noise significantly worsens *thermal* comfort; higher *sound* levels intensify warm *thermal* sensations, though this effect is not consciously perceived. Upon departure, the effect of noise on *thermal* perception is not obvious. In both walking states, *thermal* sensation significantly affects subjective noise perception, yet the trends of influence differ. Subjective noise loudness increases as *thermal* comfort worsens, showing significant correlation only upon departure. 2) During autumn greenway walks, acoustic factors exert a greater impact on Overall Environmental Satisfaction (OES), with subjective noise loudness being more influential than noise level, followed by air temperature (T_a). Greater noise decreases OES, while OES increases initially with T_a and then decreases. The integrated effects of noise-*thermal* factors on OES show significant changes. 3) To enhance the autumn greenway-walking experience, the advised parameters are A-weighted *Sound* Level (ASL) ≤ 59.12 dBA and 15.17 °C ≤ T_a ≤ 18.75 °C. Finally, three design strategies are proposed: reducing subjective noise loudness, differentiating design based on walking status and balancing acoustic-*thermal* perceptual preferences.

Conference Paper

	Document title	Authors	Source	Year	Citations
40	**Microclimate Conditions in the SS. Salvatore Church of Bologna**	Saito H.,Manfren M., Fabbri K.,Tommasino M.C., Tronchin L.	Building Simulation Applications	2025	3

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This paper analyses the microclimate conditions in the Church of Santissimo Salvatore in Bologna and their influence on the acoustics of the Church and the *sound* of the pipe organ. It is commonly acknowledged that the variation of air temperature, median radiant temperature, relative and absolute humidity could provoke *thermal* expansions of metals which are used for the organ pipes. This paper analyses a monitoring campaign which lasted one year in which temperature and relative humidity were stored with a data logger. Moreover, the paper shows the effect in the variation of the microclimate conditions in the church by simulating different numbers of people in the church and different *thermal* conditions *outdoor*. Finally, the paper reports the variations of the acoustic parameters simulating the new values of temperature, relative humidity and air velocity.

Conference Paper • *Open access*

	Document title	Authors	Source	Year	Citations
41	**ALGORITHM FOR OPTIMAL CONTROL OF ENERGY-EFFICIENT SYSTEMS ENSURING ABIOTIC FACTORS FOR ENTOMOLOGICAL PRODUCTION**	Belchenko V., Pishchanska N., Bulgakov V.,... Rucins A., Chernysh O.	Engineering for Rural Development , 24	2025	0

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The article is dedicated to the development of an optimal control algorithm for energy-efficient systems that provide abiotic factors. The result of the algorithm operation is a final set of conditions, instructions, and actions executed in a specific order, serving as a tool for determining methods for forming energy-efficient systems that ensure abiotic factors in adaptive technologies for cultivating maternal entomocultures. The algorithm consists of six main blocks. The first block defines an automatic air parameter regulation system in microclimate systems based on the type of the air parameter being regulated, the magnitude of the parameter deviation, the method of maintaining the parameter, and the nature of the regulator's influence on the controlled object. The types of systems that meet all the constraints of entomoproduction facilities are considered. The second block determines the conditions for optimal system control using a *thermodynamic* model (TDM). The analysis of these conditions allows the formulation of fundamental principles by which TDM of air conditioning systems (AirCS) ensures optimal air treatment processes, a combination of specific *outdoor* air parameters and indoor air parameters that minimize cooling, heating, water, and air consumption. The third block involves processing the basic information and performing key calculations and operations. The fourth block determines the types of technological schemes for maintaining environmental parameters indoors, considering options such as preheating *outdoor* air, operating without preheating, or using preheating for recirculated air. The fifth block is responsible for presenting the algorithm's output, defining the conditions for controlling energy-efficient systems that provide abiotic factors. The final, sixth block repeatedly performs a specific operation until a predefined condition is met. The information in this block concerns methods of controlling humidification processes – regulating water consumption, bypassing the airflow, or managing intermittent water supply. Capital costs for AirCS in entomological production amount to 20% of the total construction cost, while operating expenses range from 30% to 50% of total operating costs. Therefore, energy saving has been one of the most crucial functions of automatic control systems (ACS) in air conditioning. During the design stage of ACS and the development of technological air treatment schemes, it is necessary to refine the possibilities of process control. As a result of the research, key schemes for maintaining environmental parameters in indoor spaces have been developed to minimize air heat and moisture treatment costs. The developed algorithm enables the generation of a set of conditions, instructions, and actions that must be performed in a specific order. Given detailed initial data and knowledge of ACS equipment management methods, following each step of the developed algorithm provides a technically *sound* ACS solution for air conditioning systems, reducing energy consumption by approximately 25%. The algorithm's outcome serves as a tool for determining methods for forming energy-efficient systems that ensure abiotic factors in adaptive technologies for cultivating maternal entomocultures.

Article

	Document title	Authors	Source	Year	Citations
42	**Indoor environmental quality in high school buildings during mild winter season: students' perception and field measurements**	Amoatey P.,Al-Saadi S.,Al-Harthy I.,Al-Jabri K.,Al-Azki S.	Advances in Building Energy Research, 19(2)	2025	5

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The state of indoor environmental quality (IEQ) on school students during the winter season in hot climates has received little attention. This study aims to evaluate the IEQ conditions, namely, *thermal*, visual, acoustic comfort, and air quality, for typical private (PSB) and government (GSB) school buildings during winter in a typical hot climate. Subjective and objective assessments were conducted on 304 students distributed in 14 classrooms. The results showed that winter *outdoor* climate might have favoured most of the measured IEQ factors, therefore, meeting the ASHRAE 55 and EN 16798-1 standards. *Thermal* comfort yielded neutral–cool classroom conditions in PSB according to predicted mean vote (PMV) and predicted percentage of dissatisfaction (PPD), whereas in GSB, a neutral *thermal* sensation environment of PMV (0.02–0.2) and a PPD of 5–6% was found across the classrooms. Despite the measured *sound* levels exceeding the critical level of 35 dBA, most of the students rated the classroom's *sound* levels as neutral (41.3–44.8%) and satisfying (27.9–31.3%), which might be due to the long-term acclimatization of the students to the background noise levels. The majority of the students (42.7–47.6%) described the indoor air quality as neutral, which raises potential health risks.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
43	**The effects of acoustic-light-thermal environment quality parameters on pedestrians' overall comforts in residential districts**	Zeng F.,Liang S.,Zhang J.,... Liu T.,Wang S.	Scientific Reports, 14(1), 19625	2024	13

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This study explored subjective responses towards various environmental quality parameters. It would be helpful to approach an overall comfort improvement relating to physics. In order to investigate the combined effects of *sound*, light and heat on the overall comfort of pedestrians, which could be evaluated by overall comfort vote, two residential areas in Yubei District, Chongqing (a Cfa city in China) were selected for field measurements and questionnaires. Three were three key findings were concluded based on subjective responses to environmental parameters via multiple linear regression. First, the average *outdoor* neutral temperature, mean neutral *sound* level and neutral illumination intensity were determined to be 26.6 °C (determined by physiologically equivalent temperature), 56.5 dBA (determined by A-weighted equivalent continuous *sound* level) and 21.4 klx (determined by LUX), respectively. Second, considering the effects of both physiology and psychology, *thermal* perceptions varied for acoustic and light environments. Residents partially had lower neutral temperatures at the noisy condition. Third, environmental quality factors had a significant effect on overall comfort. Positive correlation between physiologically equivalent temperature and overall comfort vote was found (sig < 0.000); while the effects of *sound* pressure and light intensity were not always significant. Hence, *thermal* stress played a significant role in people's overall comfort. This study has explored the effects of three environment quality parameters on human perceptions. It provided better understanding against dwellers' feelings under complex circumstances. Future urban design and planning works should consider the cooling factors regarding contextual acoustical and visual environments.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
44	**The Adjusting Effects of Trees on Cfa-Climate Campus Acoustic Environments and Thermal Comforts in the Summer**	Lu W.,Chen Y.,Zhou T.,... Yin H.,Liu T.	Acoustics, 6(4)	2024	3

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This study explores the effects of trees on the acoustic and *thermal* environment in addition to people's responses to trees in different contexts. Through field measurements conducted during the summer of 2023 at the campus of the Southwest University of Science and Technology in Mianyang, residents' neutral points were locally found to be 52.2 dBA (acoustic) and 23.8 °C (*thermal*). Further, at their maximum, the trees were able to reduce heat stress by 4 °C (indicated by the physiologically equivalent temperature—PET) and the noise level by 10 dBA (indicated by the A-weighted *sound* pressure—LAeq); this was achieved by trees with a crown diameter of 20 m. Subjective acoustic and *thermal* responses varied depending on the context. Acoustically, their neutral LAeq values toward the *sounds* of

	Document title	Authors	Source	Year	Citations

traffic, teaching, sports, and daily life were 46.9, 52.5, 51.0, and 52.7 dBA, respectively. *Thermally*, pedestrians' neutral PET values were 24.2, 26.1, 22.3, and 25.1 °C, respectively, under the same conditions. These phenomena might be a consequence of the effects of *sound* frequencies. Future urban forestry research should focus on planting for environmental quality improvement.

Article

45 **Exploring the natural ventilation potential of urban climate for high-rise buildings across different climatic zones** — Mao R.,Ma Z.,Ning H., Cao J. — Journal of Cleaner Production , 475, 143722 — 2024 — 14

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The natural cooling capacity of the environment can effectively reduce the building energy consumption by natural ventilation (NV). For architects, developing NV strategies requires a comprehensive understanding of the distribution characteristics of urban meteorology. This study investigates the vertical characteristics of urban local-scale meteorological parameters across climatic zones. It aims to assess the climatic potential for NV of *outdoor* air in high-rise buildings. Firstly, using Weather Research and Forecasting (WRF) model coupled with Local Climate Zone (LCZ) data and *sounding* data, the vertical distribution of meteorological parameters are explored in typical urban areas across different climatic zones. Then, based on an adaptive *thermal* comfort model, the climatic potential is assessed for NV in high-rise buildings via the indicator of annual natural ventilation hour (NVH). Finally, the characteristics of urban meteorology is further is analyzed in terms of its impact on the natural ventilation potential (NVP) in each climatic zone. Results indicate that air temperature decreases with height from 0 to 500 m, with more pronounced temperature gradients in compact building areas. Near the ground level, Kunming (Temperate Zone) has the most NVH, reaching 4,840 h annually. However, when building height exceeds 100 m, Guangzhou (Hot Summer and Warm Winter, HSWW) becomes more suitable for NV. In cold regions, low temperatures limit NVP during winter, while high humidity constrains it in Temperate and HSWW zones. Overall, the annual NVP in high-rise building areas exceeds that of low-rise building areas. These distribution characteristics of NVP are anticipated to promote the utilization of natural resources for sustainable urban development.

Article

46 **Estimation of the effect of thin-plated surface on a three-staged looped thermoacoustic engine for low-grade heat recovery** — Zhang L.,Kang H.,Ding X., ... Wen J.,Zhang P. — Applied Thermal Engineering , 248, 123366 — 2024 — 6

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This paper introduces two novel structures of looped *thermoacoustic* engines designed for *outdoor* low-frequency *sound* sources. The system consists of three *thermoacoustic* engines in a loop and a stub for *sound* radiating, which is characterized by reliability, high efficiency, and compactness. Two thin plates are used to prevent the Gedeon flow, facilitate *sound* propagation, and adjust the acoustic field. The system with a straight stub achieved a maximum *Sound* Pressure Level of 126.09 dB at 1 m away, surpassing highest reported records by 25.96 %. The system also achieves a high output efficiency of 3.86 %. The system with tapered stub and a 0.1 mm silicone plate, achieved an SPL of 120.76 dB at 1 m. To maintain the predetermined acoustic field, it is important to consider a thin plate with small Young's modulus and thickness. The effect on the acoustic field must be considered when the thickness exceeds 0.8 mm. The addition of the 0.28 mm thin plate in system with tapered stub resulted in a 7.8 W increase in acoustic power compared to that of system without the thin plate.

Article • *Open access*

47 **The Working Environment in Primary Healthcare Outpatient Facilities: Assessment of Physical Factors and Health Professionals' Perceptions of Working Environment Conditions** — Cezar-Vaz M.R., Bonow C.A.,Vaz J.C., ... de Souza J.L.,Vaz J. — International Journal of Environmental Research and Public Health , 21(7), 847 — 2024 — 1

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The objectives of this study were to assess the adequacy of physical parameters/factors (temperature, relative humidity, noise, and illuminance levels) of the work environment in PHC facilities, to evaluate the association

	Document title	Authors	Source	Year	Citations

between the adequacy of these measured physical parameters and the physical characteristics of the PHC facilities and their surroundings and to assess the association between health professionals' perceptions about exposure to physical risks in the PHC work environment and the adequacy of physical parameters measured in the same facilities. The study monitored 23 PHC facilities in southern Brazil and interviewed 210 health professionals. Data analysis involved Pearson's chi-square, Fisher's exact test, Spearman's correlation, and multivariate linear regression analysis was used to control for confounding factors. The significance level was set at 5% ($p \leq 0.05$). The combination of temperature and relative humidity presented *thermal* comfort levels outside the adopted criteria for adequacy in consultation (*outdoor* relative humidity, $p = 0.013$) and procedure rooms (front door open, $p = 0.034$). Inadequate *sound* comfort (noise) levels in the morning shift were found in the vaccination (front door open, $p = 0.021$) and consultation rooms (movement of people, $p = 0.016$). In PHC facilities where reception rooms had insufficient lighting, internal curtains were opened less frequently ($p = 0.047$). The analysis of health professionals' perceptions of physical factors demonstrated that physicians more frequently perceive the physical risk of temperature and humidity ($p = 0.044$). The higher the number of nurses ($p = 0.004$) and oral health technicians in the PHC facilities ($p = 0.031$), the greater the general percentage of adequacy of monitored physical parameters. It was also confirmed that the higher the perception of moderate or severe physical risk among health professionals, the lower the general percentage of the adequacy of the physical parameters of the work environment of the PHC facilities evaluated ($r_s = -0.450$, $p = 0.031$). This study's evidence contributes to a better understanding of physical conditions and future occupational interventions to ensure the comfort, safety, and well-being of PHC workers.

Article

48 **Sandwich composites from recycled plastic reinforced with krajood (Lepironia articulata) fiber for building applications**

Homkhiew C.,Srivabut C., Boonchouytan W., Rawangwong S.

Iranian Polymer Journal English Edition , 33(6)

2024

8

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Abstract: This research aimed to showcase an economical and environmentally friendly approach to producing sandwich-structured composites from recycled plastics and krajood waste. The effects of the contents and particle sizes on mechanical and physical properties, including *thermal* conductivity and *sound* absorption coefficient, were investigated. The experimental results were analyzed using Analysis of Variance (ANOVA). The results confirmed that the isocyanate glue at 25% (by weight) and krajood fiber at 75% (by weight) at 40 mesh enhanced the flexural, compressive, and screw withdrawal properties the most. The increase in krajood fiber contents ranged from 75 to 85% (by weight) and increased the water absorption percentages. Core layer samples with high conductivity at 0.085 W/mK are considered suitable for building insulation applications, whereas sandwich composites at 0.189 W/mK are used for *outdoor* materials. In addition, the *sound* absorption coefficient and noise reduction coefficient were used to evaluate the application efficiency of sandwich-structured composite materials. The highest *sound* absorption coefficient and noise reduction coefficient were found using a *sound* frequency of 1000 Hz. The highest noise reduction coefficient value of 0.78 was found in the core layer sample, meaning that the sample absorbs 78% of *sound* and reflects the remaining 22% of *sound* waves back into a room. Thus, the outcomes of this study offer recommendations for the manufacture of sandwich composite materials from recycled polypropylene reinforced with krajood fiber in applications for construction and building products. Graphical Abstract: (Figure presented.)

Article

49 **Creating comfortable outdoor environments: Understanding the intricate relationship between sound, humidity, and thermal comfort**

Qin Z.,Lu B.,Jing W., ... Qiu J.,Dong Y.

Urban Climate, 55, 101967

2024

17

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This study provides insight into the impact of interactive *sound* and humidity environments on *outdoor thermal* comfort in open spaces in cold regions. Executed in Xi'an, China, the research involved a synthesis of questionnaire-based subjective assessments and meteorological data collection. Four pivotal findings emerged from the study. Initially, environments characterized by concurrent acoustic and humid conditions demonstrated a proportional increase in *Thermal* Sensation Vote (TSV) corresponding to rising humidity levels. Secondly, in these environments, an increase in *sound* levels was linked to an escalated *thermal* sensitivity among individuals, with the most significant

	Document title	Authors	Source	Year	Citations

sensitivity, quantified at 0.1769, observed at *sound* intensities between 60 dB (A) and 80 dB (A). Thirdly, the influence of ambient *sound* on TSV and *Thermal* Comfort Vote (TCV) was observed to vary with ambient humidity. Variations in *sound* levels had a pronounced effect on both TSV and TCV within a humidity range of 25% RH to 70% RH, but this correlation dissipated in higher humidity conditions (70% RH to 85% RH). Finally, the study indicates that environments with lower *sound* levels contribute to the mitigation of extreme *thermal* stress, particularly when ambient humidity is maintained between 25% RH and 70% RH.

Article

50 **Effects of thermal and acoustic environments on human comfort in urban and suburban campuses in the cold region of China** — Chen P.,Dai Y.,Zhen M. — Environmental Science and Pollution Research, 31(21) — 2024 — 14

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In this study, we investigated the effects of *thermal*–acoustic environments on human comfort in a cold region by focusing on Xi'an in China as a representative city. Four typical open spaces were identified within two universities, with one located in an urban area and the other in a suburban area. Surveys were conducted using questionnaires and environmental parameters were measured to assess *thermal*–acoustic perception. The physiological equivalent temperature (PET) and noise *sound* pressure levels were employed to measure the *thermal*–acoustic environments in the four open spaces. The results showed that the neutral PET was 19.1 °C in Xi'an from March to May, and the neutral temperature range was 14.9–23.4 °C. The preferred temperature was 20.0 °C. Acoustic sensation votes were lower in suburban areas than urban areas. Respondents in urban areas exhibited significantly higher sensitivity to the *thermal* and acoustic environments compared with those in suburban areas. Sensitivity to the *thermal* environment decreased as the environmental noise levels decreased. The temperature perceived as most suitable increased as the sensitivity to the *thermal* environment decreased. Acoustic comfort evaluations were generally higher in colder and warmer environments. Warmer environments heightened the sensitivity to the acoustic environment and this effect gradually decreased as the *sound* pressure levels increased. Thus, based on empirical analyses, we compared the effects of *outdoor thermal*–acoustic parameters on comfort to provide experimental data support for further *outdoor thermal* comfort research.

Article • *Open access*

51 **Field Investigation on Adaptive Thermal Comfort in Rural Dwellings: A Case Study in Linyi (China) during Summer** — Tang H.,Gao Y.,Tan S.,Guo Y.,Gao W. — Buildings, 14(5), 1429 — 2024 — 16

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A large number of people in China still live in rural villages. The indoor environment of these rural dwellings directly affects the quality of life of the occupants. Nevertheless, constrained by the quality of dwelling construction, rural buildings have poorer indoor environments and, at the same time, have a higher operating energy consumption. However, inadequate attention has been given to the summer *thermal* environment in cold regions. This work has been carried out around the *thermal* environment of rural residences in cold regions during summer. Field measurements, questionnaires, and data analysis were used in this study. We recorded the indoor and *outdoor thermal* environment parameters on a typical summer day in the Linyi rural area. Moreover, the subjective sensations and *thermal* adaptive behaviors of the participants were recorded in detail with a questionnaire. Linear regression showed that the neutral temperature for residents in summer was 27.52 °C, with acceptable temperatures ranging from 25.14 °C to 29.9 °C. Age and gender differences were found to affect the occupants' sensation of *thermal* comfort and humidity, as well as their *thermal* adaptive behavior. In addition, a *thermal* adaptive model has been constructed in the study, which will further enrich the *thermal* adaptive investigation and provide a scientifically *sound* reference for the renovation and development of the local rural areas.

Article

	Document title	Authors	Source	Year	Citations
52	**Preparation and properties of high blending phosphogypsum-desulfurization ash-waste soil based functional prefabricated autoclaved aerated concrete slabs**	Wang C.-Q.,Cheng L.-X., Wang Z.-Y.,... Huang D.-M., Wei S.	Construction and Building Materials , 423, 135879	2024	25

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There is a huge stock of industrial solid waste piles such as phosphogypsum (PG), desulfurization ash (DA), and waste soil (WS) in China, and their utilization rate is insufficient. These have potential risk for environmental safety due to pollutants contained in them. In this study, after proportioning design, raw material pretreatment, bubble regulation and multi-temperature section maintenance process, successfully prepared a high blending solid waste based functional prefabricated autoclaved aerated concrete slabs (FPAS) with excellent functional performance, and the optimal ratio of each raw material is PG: WS: DA: cement: lime = 8:50:20:10:12, and the dry mass ratio is up to 75%, the compressive strength of the slabs is up to 5.2 MPa, the standard dry density is 606 kg/m^3, the dry shrinkage value is 0.36 mm/m, and the post-freezing strength is 4.0 MPa, which is in line with the basic requirements in the standard “Autoclaved aerated concrete slabs” (GB/T15762), and at the same time, its fire-resistant integrity is qualified for 240 min, and the coefficient of *thermal* conductivity is 0.15w(m.k), airborne *sound*-weighted *sound* insulation is 46 dB, also can show different functional characteristics. In order to investigate the environmental safety of heavy metals (HMs) in the slabs, the simulation experiment of HMs leaching from *outdoor* stacking of slabs was carried out. The results show that the HMs in the slabs do not pose obvious risks to the natural environment and human body, but three of the characteristic HMs, Cr, Cd and Hg, have significant leaching patterns, which can be expressed by a second-order kinetic model. Finally, the life cycle assessment and cost analysis of the high blending PG-DA-WS based FPAS and the conventional aerated slabs were carried out. The main environmental damage category of the high blending PG-DA-WS based FPAS is land ecological risk, and it has obvious environmental and economic benefits compared with the conventional aerated slabs.

Article

	Document title	Authors	Source	Year	Citations
53	**A study of acoustic-light-thermal effects on pedestrians’ overall comfort in a Cfa-climate campus during the summer**	Xiao A.,Cheng B.,Zhang J., ... Liu T.,Zhu F.	Journal of Thermal Biology , 121, 103839	2024	21

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The environmental quality, in terms of acoustic, visual, and *thermal* environments, significantly affects people's comfort levels. Along these lines, in this work, their comprehensive impact on people's overall comfort was systematically explored. Pedestrians' *outdoor* neutral points on various environmental parameters were found by performing linear regressions. Similarly, people's *thermal* perceptions (indicated by neutral temperatures, NT) were found to vary for both acoustic and light environments. They would be increasingly heat sensitive (R^2 increases) in a noisier environment while the NTs varied for either *sound* or light intensity levels. From our analysis, it was demonstrated that people's overall comforts were negatively correlated with these parameters in different degrees. This work provides valuable insights for future urban design and planning studies to create better *outdoor* environments.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
54	**Strategic Integration of a Vegetative Component on a Metal Roof Base: An Evaluation of Its Impacts on Thermal and Acoustic Performance in the Tropics**	Aw S.B.,Leng P.C., Ling G.H.T.,... Law N.H.K., Goh S.Y.Z.	Buildings, 14(4), 915	2024	2

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This paper attempts to ascertain the *thermal* and acoustic impacts of introducing a vegetative roof layer on insulated and uninsulated metal roofs for tropical climates, through field measurements in Skudai, Johor, Malaysia, that were conducted for both dry and wet days. Four small-scale roof modules were tested, namely an uninsulated metal roof (uiMDR), an insulated metal roof (iMDR), and two identical corresponding modules with an additional vegetative component (uiGR and iGR, respectively). *Outdoor* ambient temperature (T_{amb}) was the most influential correlated variable affecting the roof outer surface temperature (R_{TOS}) in 50% of the assessed scenarios. On the selected dry day, the inter-quartile ranges (IQR) of iGR, iMDR, uiGR, and uiMDR were 6.21 °C, 8.32 °C, 6.69 °C, and 1.66 °C, respectively;

Document title	Authors	Source	Year	Citations

the IQRs were 1.6 °C, 4.11 °C, 2.59 °C, and 1.78 °C, respectively, on the selected wet day. Based on design U-value calculations, iGR was better than iMDR and uiMDR for both dry and wet days. The U-value of uiGR was also better than iMDR under dry-day conditions. The Wilcoxon signed-rank test also indicated a statistically significant difference in the roof inner surface temperature (R_{TIS}) measurements (p-value = 0.00) during Malaysian daylight hours, between 8.00 a.m. and 6.00 p.m., regardless of the weather. In terms of *sound* level reduction under dry-day conditions, the Kruskal–Wallis and Wilcoxon signed-rank tests showed statistically significant differences in *sound* level reductions, with iGR and uiGR performing better than iMDR and uiMDR (p-values = 0.00). The *sound* level reductions for iGR, iMDR, and uiGR were 51%, 32%, and 31%, respectively, while uiMDR experienced *sound* level amplifications by 6%, possibly due to the acoustic resonance effect. This proof of concept may encourage a broader application of extensive GRs in Malaysia using metal roofs, beyond the conventional RC base construction method.

Article • *Open access*

55 **Optical System Design and Athermalization of Telephoto Lens | 长焦镜头光学系统设计及无热化研究** Zhang L.,Lu Q.,Duan F., Dai X.,Qiao D. Guangxue Xuebao Acta Optica Sinica , 44(8), 0822004 2024 7

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Objective The actual working environment of vehicle-mounted LiDAR is complex, including seasonal long-time limit temperature and rapid changes in indoor and *outdoor* high and low temperatures. These temperature variations possibly change the internal optics and structure of the lens, which results in image plane drift and reduces imaging quality. For telephoto lenses, the image plane drift can be more obvious with the ambient temperature changes. The current passive athermalization design method has the problems of complex structure and large volume caused by multi-layer lens barrels, or the introduction of diffractive elements and aspheric lenses to increase the production cost, and the narrow temperature range of *thermal*-free, which is difficult to adapt to the practical applications in the complex environment of vehiclemounted LiDAR. Therefore, it is necessary to reduce the image plane temperature drift and improve the environmental adaptability of telephoto lenses with an advanced athermal design method. Methods The telephoto lens which can accurately capture distant targets and magnify the details is integrated with a line array detector to improve the resolution. Meanwhile, based on the fact that the total length of the optical system of the telephoto structure is smaller than the focal length, the volume of the telephoto receiving optical system is compressed to a certain extent to realize the requirements of lightweight miniaturization and high resolution of the vehicle-mounted LiDAR. Aiming at the problem that the telephoto lens is susceptible to temperature, we improve the two-group compensation design method of the passive optical and mechanical athermalization to maximally offset the optical focal length change of the optical parts from the *thermal* difference brought by *thermal* expansion and contraction of the structural parts, and to reduce the image plane drift, thus realizing the athermalization of the telephoto lens. Finally, the image plane drift of the as-designed lens is less than the depth of focus over a wide temperature variation range from -40 to 100 ℃. This is conducive to ensuring the imaging quality of the lens, and the designed structure has a simple preparation process and is easy to engineer and produce. Results and Discussions Different combinations of optical materials and optical focal length distributions are determined, structural components of different *thermal* expansion coefficients (TCEs) are matched, and the *thermal* difference of the optical system compensates for each other, with the system athermalization design achieved. Without the *thermal* expansion and contraction of the barrel holder taken into account, the focal shift of the lens with temperature change is always minimized and the image plane drift is 0.075 mm when the temperature increases to 100 ℃ (Fig. 7). The *thermal* expansion and contraction of the barrel holder is considered as a material to compensate for *thermal* aberration to make the sensor detecting surface always in the image plane. In the wide temperature range from – 40 to 100 ℃, with the temperature change, the receiving optical system obtained from the selected optics and structural component materials has almost no significant focal shift, even when the temperature is as high as 100 ℃, and meanwhile the amount of focal shift is only 0.021 mm, smaller than its depth of focus at room temperature (0.074 mm), and the field curvature and distortion of this optical system have small changes (Fig. 8). The MTF at 30 lp/mm is all larger than 0.5 for each field of view (FOV), and the focal plane shifts are all small, which indicates that the designed lenses can maintain *sound* image quality over a wide range of temperatures from –40 to 100 ℃ (Figs. 9 and 10). The diffuse spot radius in the full FOV is smaller than 7 μm, which reveals that the focal shift of the lens is little affected by temperature (Fig. 11). The results of photographing vehicles traveling on the road show clear imaging of the vehicles and obvious feature areas such as the outer contours of the vehicles (Fig. 14). The above results prove that the imaging quality and temperature adaptability of the lens can be guaranteed by the above athermalization design

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to compensate the system *thermal* difference. Conclusions We employ the telephoto lens with a long focal length and small FOV to subdivide the scanning area and integrate a line array detector to achieve an image-level imaging effect. Based on the characteristic that the total length of the optical system of telephoto structure is smaller than the focal length, a receiving optical system with a telephoto ratio of 0.38 is designed, which has a smaller lens length and lower cost and meets the requirements of vehicle-mounted LiDAR in terms of high resolution, light weight, and small size. Given the large temperature difference in the working environment of vehicle-mounted LiDAR and the image plane drift of the telephoto lens, a passive optical and mechanical athermalization is implemented to confirm the reasonable combination of multi-plane spherical glass lens and structural components. Finally, a four-piece telephoto lens optical system with a simple structure and a focal shift of 0.021 mm less than the depth of focus of 0.074 mm over a wide temperature range from –40 to 100 ℃ is designed. The MTF of each FOV at 30 lp/mm is larger than 0.5 and the diffuse spot radius in the full FOV is smaller than 7 μm. The vehicle imaging is clear, and the outer contour of the vehicle and other characteristics of the area are obvious, which achieves athermalization and shows favorable environmental adaptability.

Article

56 **Design and application of all-day portable outdoor infrared detection aerosol lidar system | 全天时便携式户外型红外探测气溶胶激光雷达系统设计及应用**

Zhuang P.,Xie C.,Kang B., Liu J.,Xia X.

Hongwai Yu Jiguang Gongcheng Infrared and Laser Engineering , 53(3), 20230636

2024

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Objective Air pollution control has higher requirements for environmental monitoring equipment. As an active remote sensing instrument, lidar is currently a powerful tool for monitoring the three-dimensional distribution characteristics of atmospheric aerosols in the troposphere. Lidar can compensate for the insufficient spatial distribution rate of existing ground monitoring data, mainly using various monitoring modes such as horizontal, vertical, navigation, and networking. Among them, horizontal scanning can effectively monitor horizontal visibility, dynamically extract the location and transmission path of pollution sources. Moreover, vertical monitoring can analyze the spatiotemporal changes in aerosol vertical diffusion, sedimentation transport, optical characteristics, boundary layer changes and their cloud information, as well as sand and dust monitoring. This article introduces a self-developed infrared lidar that can detect aerosol distribution, visibility, and boundary layer height in real-time under environmental pollution and multiple sudden weather conditions. It has unique advantages in analyzing the diffusion and sedimentation trends of polluted air masses. In the context of increasing observation needs, it can provide more scientific and effective data support for environmental management decisions and meteorological services. Methods The detection principle is based on the Mie scattering lidar equation. The schematic diagram and appearance of the aerosol lidar product structure are shown (Fig.2). The system mainly consists of three parts of laser emission unit, optical receiving unit, and data acquisition and processing unit. The main technical specifications are shown (Tab.1). The fundamental frequency 1 064 nm linearly polarized laser of Nd: YAG laser is used as the detection light source, and the telescope uses an aspherical lens as the main mirror. The aspherical lens has the advantages of small aberration and short focal length, which can reduce the volume of the receiving module. The subsequent optical and detection units are composed of fiber coupling devices and small core diameter multimode fibers, which can effectively control the telescope's receiving field of view. The optical channel measures the scattering signal generated by the interaction between 1 064 nm outgoing laser and atmospheric particles. By using the measurement data of the channel and combining with the above inversion method, the distribution characteristics of optical parameters of tropospheric aerosols and clouds can be obtained. Results and Discussions This radar can be effectively applied in fields such as obtaining the spatiotemporal distribution characteristics of atmospheric aerosol pollution, monitoring horizontal visibility, and monitoring vertical boundary layers. The specific application conclusions are as follows. 1) Infrared detection lidar horizontal scanning monitoring can timely detect pollution sources in the area. During the monitoring period, a total of 10 main pollution source areas were discovered (Fig.3), distributed in the southeast and northeast sides of the radar points. Based on the comprehensive analysis of particulate matter concentration data from national urban environmental air quality monitoring stations within the scanning area, the pollution source spreads southwestward under the influence of northeast winds, resulting in a significant increase in particulate matter and carbon monoxide concentrations at downwind stations simultaneously. 2) Two infrared detection lidars were observed at the same location as the visibility meter, and the results showed that the correlation coefficient of

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visibility between the two lidars was 0.98, with a relative error of 7.76% (Fig.10-11). The trend of change was consistent, and the instrument performance was relatively stable. The relative error in comparison and analysis with the visibility meter data is less than 20%, which is consistent with the standard data. The daily variation of visibility shows a unimodal distribution, with higher daytime visibility compared to nighttime. From the 26th to the 27th, visibility decreases. Combined with environmental air quality analysis, an increase in particulate matter concentration is the main factor affecting visibility. 3) The infrared lidar and *sounding* balloon simultaneously inverted the boundary layer height, and the results showed that the absolute deviation between the lidar boundary layer height and the *sounding* balloon was 200 m, which can accurately invert the boundary layer height. During the monitoring period, the height of the boundary layer inverted by the lidar showed a diurnal trend, with an average height of 1.2 km (Fig.12). However, during the nighttime period, the height of the boundary layer was significantly higher than that of the *sounding* inversion, which was affected by residual pollution at night and was not a stable boundary layer height at night. Conclusions Infrared detection aerosol lidar has the characteristics of strong atmospheric penetration, little influence by sky background light, and sensitivity to large particles, and has achieved good results in the spatial and temporal distribution characteristics of atmospheric aerosol pollution, horizontal visibility monitoring, and vertical boundary layer monitoring. Horizontal scanning can obtain a map of particulate matter distribution in a large area, find pollution sources in time, and evaluate the impact of pollution sources by combining data such as wind speed, wind direction, particulate matter and carbon monoxide concentration near the ground. The system can accurately capture the distribution and transmission of atmospheric aerosols in real time, and accurately invert information such as horizontal visibility and boundary layer height, and has a wide range of application scenarios in the field of atmospheric parameter monitoring.

Article

☐ 57 **Exploring the natural ventilation potential for supertall buildings considering vertical meteorology: A case study in Harbin, China** — Cao J.,Mao R.,Ning H.,Shui T. — Applied Thermal Engineering, 239, 122163 — 2024 — 16

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The natural ventilation strategy is gaining increasing attention from architects due to its advantages in reducing energy consumption and maintaining a healthy indoor environment. For supertall buildings, the vertical variation in *outdoor* meteorology has inspired us to explore the energy-saving potential by adopting natural ventilation strategy. This study investigates the impact of vertical variation in urban meteorology on the natural ventilation potential (NVP) of supertall buildings based on numerical simulation and field observation. Firstly, the Weather Research and Forecasting (WRF) model is used to simulate the urban meteorology of the study region by integrating the local climate zone (LCZ) map. Next, meteorological data from *sounding* experiments by unmanned aerial vehicle (UAV) and monitoring stations are used to validate the simulated result, and thereby to explore the vertical meteorology in urban area. Finally, the vertical NVP in supertall buildings is quantitatively assessed considering the impact of LCZ in urban areas based on *thermal* comfort models. The findings reveal that atmospheric temperature decreases approximately linearly with increasing altitude, with a maximum temperature gradient of up to −0.82 °C/100 m. Static models underestimated the annual natural ventilation hours (NVHs) by at least 348 h compared to the adaptive *thermal* comfort model. Compared to low-density building regions, the higher NVP is indicated in high-density building regions. This study aims to illustrate the importance of considering the vertical variations of urban meteorology, and is expected to assist architects and policy makers in quantifying the energy saving potential of natural ventilation in supertalls.

Conference Paper

☐ 58 **Thermal Music: Exploring Sensation of Temperature as a Performance Parameter** — Snyder J.,Meggers F.,Polito D.,Hu G. — Proceedings of the International Conference on New Interfaces for Musical Expression — 2024 — 1

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Thermal Music is a project to explore the possibilities of coordinating control of temperature sensation with control of *sound* and light. Unlike convective heating technologies that change the temperature of the air around us, radiative

	Document title	Authors	Source	Year	Citations

heating works by directly heating the surface of our skin via infrared. The technology is widely used in portable heaters in *outdoor* dining areas of restaurants. One interesting advantage of radiative heating is that if the source of infrared radiation is shaded by a *thermally* reflective surface, the change in perceived temperature by a person near the heater is very rapid. We worked to create a system that could quickly control robotic shades in front of a radiative heater to synchronize changes in perceived temperature with music and light. We also explored the inverse, using *thermal* camera input as a control method for audio, and presented both of these techniques at a "*Thermal* Music" concert at Princeton University in October of 2023.

Conference Paper • *Open access*

59 **Review of urban water thermal environment cooling** — Qiu Y.F.F.,He B.J. — Iop Conference Series Earth and Environmental Science , 1363(1), 012073 — 2024 — 3

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The urban underlying surface undergoes constant changes due to the acceleration of urbanization. The increasingly prominent heat island phenomenon has led to higher energy consumption and worsened *thermal* conditions, significantly impacting people's daily *outdoor* activities and health. Water bodies, as a unique form of underlying surface, play a crucial role in enhancing the urban microclimate through their large heat capacity, high latent heat of evaporation, and low reflectivity. Previous research on the *thermal* comfort effect of water bodies primarily focuses on meteorology, environmental science, and landscape architecture. While quantitative studies have been conducted at both macro-scale city regions and micro-scale field domains to investigate the mechanisms behind these effects, there is a lack of translation from data into practical application strategies. In this study, we aim to adopt an interdisciplinary approach to review previous research achievements comprehensively. Firstly, we will examine the mechanisms and influencing factors involved in water body *thermal* regulation. Secondly, we will explore major methods and techniques for cooling urban water bodies while establishing an interactive model that considers landscape elements alongside physical factors affecting microclimate conditions. By developing a climate-adaptive design framework that addresses human *thermal* comfort requirements effectively, our goal is to create more livable and sustainable urban environments through scientifically *sound* strategies for cooling urban water bodies.

Conference Paper

60 **Semi-empirical model for amplitude fluctuations by atmospheric turbulence in aircraft flyover sound** — Lincke D.,Pieren R. — 53rd International Congress and Exposition on Noise Control Engineering Internoise 2024, 6 — 2024 — 1

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Long-range *sound* propagation in the turbulent atmosphere leads to audible fluctuations in the *sound* amplitude. So far, the best currently available theoretical model for predicting amplitude fluctuations is limited to situations with low turbulence and short propagation distances. A semi-empirical model was recently introduced, which extends the range of applicability of the theoretical model based on a large dataset of aircraft flyover measurements. The dataset encompasses field measurements in diverse meteorological conditions and for different *sound* propagation distances. The semi-empirical model accounts for partial saturation of amplitude fluctuations. The semi-empirical model is used to generate auralizations of aircraft flyovers, considering the impact of atmospheric turbulence. The meteo- and distance-dependent characteristics of amplitude fluctuations in aircraft flyover *sound* are successfully reproduced by the model. The realism of the auralization is significantly improved compared to auralizations without consideration of amplitude fluctuations. The semi-empirical model enhances modeling techniques for *outdoor sound* propagation of elevated noise sources such as aircraft, drones, and wind turbines.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
61	**Auralization of atmospheric turbulence-induced amplitude fluctuations in aircraft flyover sound based on a semi-empirical model**	Lincke D.,Pieren R.	Acta Acustica, 8, 47	2024	5

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Atmospheric turbulence causes well-audible fluctuations in *sound* amplitude as it propagates through the atmospheric boundary layer. However, the best currently available theoretical model describing amplitude fluctuations is limited to relatively short propagation distances and weak turbulence. This article presents a semi-empirical model for predicting the standard deviation of amplitude fluctuations, combining the existing theoretical model of Ostashev and Wilson with empirical data. Leveraging a dataset of over 5000 aircraft flyovers across diverse meteorological conditions, the model incorporates factors such as frequency- and distance-dependent saturation of amplitude fluctuations, an effective propagation length based on the boundary layer height, turbulence decay time during transitional states of the atmosphere, and turbulence production by nocturnal low-level-jets. The mean absolute error of the semi-empirical model compared to the measurements is 0.4 dB. The application of the semi-empirical model to aircraft auralization is demonstrated for a flyover in two meteorological conditions. The findings will enhance the modeling of *outdoor sound* propagation for elevated *sound* sources like aircraft and wind turbines. Enhanced modeling of these fluctuations can considerably improve the realism of the listening experience in aircraft flyover auralization. Further, it helps estimating the measurement uncertainty in measured aircraft noise, particularly for maximum levels.

	Document title	Authors	Source	Year	Citations
	Article • *Open access*				
62	**Causal Artificial Intelligence-Driven Approach for HVAC Preventive Maintenance Explanation**	Kliangkhlao M., Haruehansapong K., Yeranee K.,Sahoh B.	IEEE Access, 12	2024	8

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Preventive maintenance of heating, ventilation, and air conditioning (HVAC) systems aims to monitor and detect early-stage failures, thereby enhancing *thermal* comfort satisfaction and reducing energy consumption. However, it is a complex and uncertain factor, and technicians must understand the root cause of the problem to interpret prefailure events accurately. This study addresses this concern using a causal machine learning model, which encodes HVAC system behavior, indoor ambiance, and the *outdoor* environment based on random variables, with a particular focus on air conditioning units. It models their causal representation concerning prefailures using the structural causal model. It employs d-separation and d-connection to justify cause-and-effect relationships with the model and applies the expectation maximization algorithm to fit model parameters given observational data. The causal model is verified using a do-operator to rationalize the *sound* explanations (d-separated and d-connected) and employing odds ratio and confidential interval to prove their statistical strength. The model achieves high causal significance, with odds ratios of the submodel lying between 0.3-0.9 and 1.4-6.0 (exclude 1.0). The results indicate that the model and its explanations can encode human-like interpretation and achieve high causal significance aligned with real-world prefailure events. This refers to a reasoning process that is similar to that of humans, where information is included or excluded based on explanations of cause and effect. This confirms that the contribution of this study is suitable for causal-based decision-making systems for HVAC preventive maintenance.

	Document title	Authors	Source	Year	Citations
	Article • *Open access*				
63	**A prenatal acoustic signal of heat reduces a biomarker of chronic stress at adulthood across seasons**	Udino E., Oscos-Snowball M.A., Buchanan K.L., Mariette M.M.	Frontiers in Physiology , 15, 1348993	2024	1

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During development, phenotype can be adaptively modulated by environmental conditions, sometimes in the long-term. However, with weather variability increasing under climate change, the potential for maladaptive long-term responses to environmental variations may increase. In the arid-adapted zebra finch, parents emit “heat-calls” when experiencing heat during incubation, which adaptively affects offspring growth in the heat, and adult heat tolerance. This suggests that heat-call exposure may adjust individual phenotype to hot conditions, potentially compromising individual sensitivity to cool weather conditions. To test this hypothesis, we manipulated individual prenatal acoustic

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and postnatal *thermal* experiences during development, and sought to assess subsequent chronic responses to *thermal* fluctuations at adulthood. We thus measured heterophil to lymphocyte (H/L) ratios in adults, when held in *outdoor* aviaries during two summers and two winters. We found that birds exposed to heat-calls as embryos, had consistently lower H/L ratios than controls at adulthood, indicative of lower chronic stress, irrespective of the season. Nonetheless, in all birds, the H/L ratio did vary with short-term weather fluctuations (2, 5 or 7 days), increasing at more extreme (low and high) air temperatures. In addition, the H/L ratio was higher in males than females. Overall, while H/L ratio may reflect how individuals were being impacted by temperature, heat-call exposed individuals did not show a stronger chronic response in winter, and instead appeared more resilient to *thermal* variability than control individuals. Our findings therefore suggest that heat-call exposure did not compromise individual sensitivity to low temperatures at adulthood. Our study also reveals that prenatal *sound* can lead to long-term differences in individual physiology or quality/condition, as reflected by H/L ratios, which are consistent with previously-demonstrated reproductive fitness differences.

Article • *Open access*

☐ 64 **Developing a Feasible Integrated Framework for Occupational Heat Stress Protection: A Step Towards Safer Working Environments** — Gourzoulidis G.A.,Gofa F., Ioannou L.G., Konstantakopoulos I., Flouris A.D. — Medicina Del Lavoro, 114(5), e2023043 — 2023 — 7

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Background: Specialized occupational health and safety (OHS) issues are covered at the EU level through detailed legislation and guidelines. Unfortunately, this does not extend to occupational heat stress, not only in Greece but also (with few exceptions) internationally. One possible explanation could be the difficulty in accurately identifying the dangerous conditions, as many environmental and individualized elements are involved, and hundreds of "*thermal* stress indicators" are available. Another explanation could be the difficulty in adequately measuring hazardous conditions for workers affected more (i.e., *outdoor* and high intensity) since the biological protection framework is based on the human body's internal temperature. Methods: The Wet Bulb Globe Temperature (WBGT) has been proposed as the most efficacious *thermal* stress indicator. Since 2021, the Hellenic National Meteorological Service has provided 48-h WBGT forecast predictions to serve as a first level of alert. Real-time measurements and 48-h forecasts of WBGT are also available through a smartphone application. Additionally, as revealed when developing the occupational heat stress legislation in Cyprus and Qatar, crucial first steps are identifying the specific characteristics of worker exposure and the tripartite collaboration between employers, workers, and the State. Results: Evaluating the simplified WBGT forecasted values and the smartphone application estimates proved well-established. The *sound* scientific basis can be effectively combined with administrative measures based on the EU OHS legislative experience to produce practical solutions. Conclusions: As the climate crisis exacerbates, worker productivity and well-being will decline, underscoring the urgent need for an integrated protection framework. Such a framework is proposed here.

Article

☐ 65 **Impact of Indoor Environmental Quality on Students' Comfort in High School Buildings during the Summer Season in an Extreme Climate** — Amoatey P.,Al-Jabri K., Al-Saadi S.,Al-Harthy I., Al-Khuzairi M. — Journal of Architectural Engineering , 29(3), 04023014 — 2023 — 14

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Maintaining a satisfactory indoor environmental quality (IEQ) is an integral element of improving occupant health and well-being. In particular, IEQ is an essential factor in some facilities, such as school buildings, to improve student academic performance. This study employed an objective and subjective measurement approach to assess IEQ in a government high school (GHS) and a private high school (PHS) located in Oman, which is characterized by its extreme hot climate. A total of 212 high school students were surveyed during the summer season across 10 different classrooms to ascertain their perception of IEQ situations. The results suggested that most of the observed IEQ factors, including air temperature, relative humidity, and lighting levels across the classrooms in GHS and PHS, satisfied the ASHRAE standard, except the indoor air temperature, which was slightly above the maximum threshold of the European Standard (EN). The average indoor *sound* levels (>60 dB) exceeded critical limits of 35 dB for school

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buildings, which may pose the risk of high annoyance (HA) to 30% of students. Satisfactory predicted mean vote (PMV) and predicted percentage of dissatisfied (PPD) scores were obtained from both schools when they were evaluated against the ASHRAE standard compared with the EN standard. The maximum proportion of the students reported their classroom's air temperature (33%), relative humidity (25%), and air movement (47%) levels as not acceptable, and *thermal* comfort (33%), visual comfort (22%), and air quality (41%) as bad ($P < 0.001$). Multiple regression analysis revealed a strong linear relationship between indoor air temperature ($R2 = 0.89$) and other IEQ parameters, while PPD ($R2 = 0.45$) and PMV ($R2 = 0.17$) showed a weak relationship with *outdoor* environmental factors. *Outdoor* air temperature and humidity levels accounted for greater changes in indoor air temperature levels in the PHS than in the GHS. Providing increased urban vegetation, green façade, and retrofitting and the utilization of passive cooling could help improve IEQ in Oman schools.

Article • *Open access*

66 **Physics-based scintillations for outdoor sound auralization** — Bresciani A.P.C., Maillard J., de Santana L.D. — Journal of the Acoustical Society of America, 154(2) — 2023 — 12

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The *sound* propagating in a turbulent atmosphere fluctuates in amplitude and phase. This phenomenon, known as acoustic scintillation, is caused by random fluctuations in the acoustic refractive index of the air induced by atmospheric turbulence. Auralization techniques should consider this phenomenon to increase the realism of the synthetic *sound*. This paper proposes a physics-based formulation to model sequences of log-amplitude and phase fluctuations of a *sound* propagating in a turbulent atmosphere. This method applies to slanted and vertical propagation of the *sound*, which is useful for simulating elevated noise sources such as aircraft, drones, and wind turbines. The theoretical framework is based on the spatial correlation functions for the log-amplitude and phase fluctuations for spherical waves, the von Kármán spectrum, and similarity theories to model atmospheric turbulence. Two applications with audio files are presented to demonstrate the applicability of this method to tonal and broadband noise.

Article • *Open access*

67 **Influence of ground blocking on the acoustic phase variance in a turbulent atmosphere** — Ostashev V.E., Wilson D.K., Hart C.R. — Journal of the Acoustical Society of America, 154(1) — 2023 — 5

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Sound propagation through atmospheric turbulence is important in many applications such as localization of low flying aircraft, sonic boom disturbances, and auralization of aircraft during takeoff and landing. This article extends an isotropic turbulence model in the atmospheric boundary layer to account for ground blocking of buoyancy-produced velocity fluctuations. The extended, anisotropic turbulence model is needed to correctly predict the effect of the largest velocity eddies on the statistical characteristics of *sound* signals. This model and geometrical acoustics are then employed to derive a closed-form expression for the variance of the phase fluctuations of a spherical *sound* wave for vertical and slanted propagation, without the use of the Markov approximation. A numerical analysis of this expression indicates significant anisotropy of the phase variance due to the buoyancy-produced velocity fluctuations with ground blocking such that it decreases in the vertical direction and increases in the near-horizontal directions. The newly formulated phase variance is compared with data from an *outdoor* experiment on vertical and slanted *sound* propagation. By accounting for ground blocking, much better agreement is obtained between the theoretical predictions and experimental data.

Review

68 **Acoustic tomographic reconstruction of temperature and flow fields with focus on atmosphere and enclosed spaces: A review** — Othmani C., Dokhanchi N.S., Merchel S., ... Voelker C., Takali F. — Applied Thermal Engineering, 223, 119953 — 2023 — 35

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	Document title	Authors	Source	Year	Citations

Acoustic travel-time TOMography (ATOM) technique is being widely used as a remote sensing method to determine the air temperature and air flow velocity distribution for both *outdoor* and most recently indoor applications. The first process of ATOM measuring system corresponds to compiling the travel-times of a *sound* signal between a set of loudspeakers (transmitters) and microphones (receivers) distributed over the tomographic area, whose positions are known. These measured travel-times would then be used as the input data for inverse algorithms to reconstruct the airflow velocity and air temperature fields. Accordingly, the accuracy of the measuring system depends heavily on both the accuracy of the estimated travel times and the tomographic algorithms. Recent investigations on arrays of acoustic tomography are pointed towards optimization and reliability purposes. Many experimental layouts have been identified that could allow reducing the number of transceivers while keeping the accuracy of measurements. This paper gives a detailed review and discussion of recent advances in ATOM development applied to *outdoor* and indoor applications covering 90 articles published since 1990. It discusses the relevant challenges, opportunities, pros, and cons of existing approaches in relation to both travel time estimations and reconstruction algorithms, providing strategies to improve the sector of acoustic tomography. Accordingly, a comparative study of various reconstruction techniques is performed based on its systematic errors and theoretical strategies. This review inspires more successful explorations to improve the estimation of the air temperature and the airflow velocity using the ATOM technique.

Article

69 **Multiple effects of visual-acoustic-thermal perceptions on the overall comfort of elderly adults in residential outdoor environments** — Du M.,Hong B.,Gu C.,Li Y.,Wang Y. — Energy and Buildings, 283, 112813 — 2023 — 95

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We selected typical residential *outdoor* spaces and five common *sounds* (birdsong, dance music, traditional opera, conversation and traffic *sound*) with three A-weighted equivalent continuous *sound* pressure level (LAeq) (60, 70 and 80 dBA) as visual-acoustic-*thermal* conditions. 458 elderly adults completed a subjective questionnaire while environmental parameters (meteorological parameters, illumination intensity (LUX), visible green index (VGI) and LAeq) were simultaneously measured to determine effects of visual-acoustic-*thermal* perceptions on overall comfort. We found that: 1) When physiological equivalent temperature (PET) was above 43.80 °C, the elderly felt *thermally* uncomfortable. Older adults perceived traffic *sound* as acoustic uncomfortable when LAeq was higher than 66.1 dBA. A higher VGI decreased the sensitivity of respondents to LUX. 2) *Thermal* sensation vote (TSV) and *thermal* comfort vote (TCV) were susceptible to the acoustic and visual environments. The influence of the visual environment and PET on acoustic sensation vote (ASV) and acoustic comfort vote (ACV) were not significant. We found a significant correlation between PET and sunlight sensation vote (SSV). 3) Acoustic and *thermal* comfort had a one-vote veto tendency relative to overall comfort, but no absolute veto power. *Thermal* comfort was the most important factor affecting overall comfort in summer while acoustic comfort was the most important in spring. 4) A binary logistic regression to predict overall comfort of elderly adults had 84.7 % accuracy, indicating a good performance.

Article

70 **Climate Change and Its Influence on the Energy Engineering Complex (Taking St. Petersburg as an Example)** — Mil'kov D.A.,Yuferev Y.V.,Tyutyunnikov A.I.,Gorshkov A.S. — Thermal Engineering, 70(3) — 2023 — 1

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Abstract: Observations of the Earth climate for the last 100–150 years have revealed that the atmospheric air temperature in the ground level shows quite a rapid growth. Megalopolises and other settlements behave as large sources producing *thermal* pollution of the Earth air basin. Transport facilities, heated buildings, and heat and electric networks release a significant amount of energy, which accumulates in the air medium of cities and then dissipates into the surrounding space. The most rapid change in the air temperature has been noted for the last 50 years. The greatest increase of the *outdoor* air temperature, e.g., in St. Petersburg, is observed in the cold period of the year. It has been determined that the average air temperature in the cold period of the year (from October to April) has increased by 1.51°C, whereas the average air temperature increased by 0.72°C for the same interval of time but in the warm period of the year (from May to September), i.e., half as much as in the cold period. It has been found from studies of regulatory and reference documents that the standard climatic data specified in them lag significantly

Document title	Authors	Source	Year	Citations

from the currently occurring climate changes. The article presents an analysis of the *outdoor* air temperatures in St. Petersburg for the period from 1992 to 2021 and an assessment of air temperatures in the coldest day and the coldest 5-day spell with reliabilities of 0.92 and 0.98 and also the absolute minimal air temperature for the considered 30-year interval of time. It is pointed out that the actual climatic indicators derived from statistical processing of daily data arrays are by 3–4°C higher than those specified in the relevant standards. As a result, when buildings are designed for the standard conditions specified by the existing regulations, the contractual loads turn out to be overestimated essentially in comparison with the actual ones, which is revealed in designing the configurations of settlement heat-supply systems. The consideration of actual climatic indicators opens up the possibility to decrease the contractual heat loads of consumers on a *sound* basis and, as a consequence, to achieve *thermal* energy saving without degrading the heat-supply quality.

Article

71 **Active Acoustic Sensing for 'Hearing' Temperature under Acoustic Interference** — Cai C.,Pu H.,Ye L.,Jiang H.,Luo J. — IEEE Transactions on Mobile Computing, 22(2) — 2023 — 20

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Though measuring ambient temperature is often deemed as an easy job, collecting large-scale temperature readings in real-Time is still a formidable task. The recent boom of network-ready (mobile) devices and the subsequent mobile crowdsourcing applications do offer an opportunity to accomplish this task, yet equipping commodity devices with ambient temperature sensing capability is highly non-Trivial and hence has never been achieved. In this paper, we propose Acoustic *Thermometer* (AcuTe+) as an interference-resilient ambient temperature sensor empowered by a single commodity smartphone. AcuTe+ utilizes on-board dual microphones to estimate air-borne *sound* propagation speed, thereby deriving ambient temperature. To accurately estimate *sound* propagation speed, we leverage the phase of chirp signals to circumvent the low sample rate on commodity hardware. In addition, we propose to use both structure-borne and air-borne propagations to address the multipath problem. Most importantly, we equip AcuTe+ with a mask-based desnoising algorithm to handle intensive acoustic interference. As a mobile, economical, highly accurate sensor, AcuTe+ may potentially enable many relevant applications, in particular large-scale indoor/*outdoor* temperature monitoring in real-Time. We have conducted extensive experiments on AcuTe+; the results demonstrate a median error of 0.6°°C even under severe acoustic interference (overall median 0.3°°C), and they also showcase the practical ability of AcuTe+ in real-Time distributed temperature sensing.

Article

72 **The influence of outdoor thermal comfort on acoustic comfort of urban parks based on plant communities** — Mohammadzadeh N.,Karimi A.,Brown R.D. — Building and Environment, 228, 109884 — 2023 — 85

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As human beings live in an environment where various factors interact and integrate, it seems vital to examine the effects of these factors on environmental perceptions. *Outdoor thermal* and acoustic environments are the main factors affecting the overall comfort of urban spaces. This research uses measurements and questionnaire surveys to explore the effects of *outdoor thermal* comfort dimensions on acoustic comfort, emphasizing the plant community in a city park (the El Goli Park of Tabriz, Iran). In this regard, the general comfort of the psychological equivalent temperature (PET) under different conditions of the *thermal*-acoustic environment has been evaluated and predicted. The results showed that a higher *thermal* sensation vote (TSV) and PET result in a significant decrease in *soundscape* quality (SQ), overall impression (OI), and the percentage of reported natural dominant *sound* (%RNDS), while an inverse trend has been reported in *thermal* comfort vote (TCV). On the other hand, by increasing TSV and PET, reported loudness (RL) has recorded an upward trend. However, this process was completely reversed in TCV. Furthermore, the analyzes showed that PET has a lower effectiveness compared to TSV and TCV and, it should be stated that the three indicators of *thermal* comfort did not show any significant relationship with landscape quality (LQ) and *sound* pressure level (SPL). These findings are important for managing and improving *outdoor* acoustic comfort and *soundscapes*, and it is expected that future studies will consider the acquired results in various green urban environments and provide a practical framework for landscape architects and urban planners.

	Document title	Authors	Source	Year	Citations
73	Conference Paper **Multiple effects of visual-acoustic-thermal perceptions on the overall comfort of elderly adults in a humid subtropical city**	Du M.,Hong B.	Healthy Buildings 2023 Asia and Pacific Rim	2023	0

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We selected typical residential *outdoor* spaces and five common *sounds* with three A-weighted equivalent continuous *sound* pressure level (LAeq) as visual-acoustic-*thermal* conditions. 458 elderly adults completed a subjective questionnaire while environmental parameters were simultaneously measured to determine effects of visual-acoustic-*thermal* perceptions on overall comfort. We found that: 1) *Thermal* sensation vote (TSV) and *thermal* comfort vote (TCV) were susceptible to the acoustic and visual environments. The influence of the visual environment and physiological equivalent temperature (PET) on acoustic sensation vote (ASV) and acoustic comfort vote (ACV) were not significant. 2) Acoustic and *thermal* comfort had a one-vote veto tendency relative to overall comfort, but no absolute veto power. *Thermal* comfort was the most important factor affecting overall comfort in summer while acoustic comfort was the most important in spring. 3) A binary logistic regression to predict overall comfort of elderly adults had 84.7% accuracy, indicating a good performance.

74	Conference Paper • *Open access* **Integrating Simulation and Measurement Techniques to Model Outdoor Noise and Heat in Airport Neighbourhoods with Varying Urban Geometries**	Wuite G.,Peng Z.,Kim K., Lugten M.,Tenpierik M.	Building Simulation Conference Proceedings , 18	2023	0

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This study aims to evaluate the impact of different urban building geometries (six courtyards, two canyons, two slabs) on heat mitigation and aircraft noise attenuation, in order to support an evidence-based retrofit plan for future airport neighborhoods. Using'Pachyderm + ENVI-met simulations + field measurements', we found that the slanted-roof, low-rise courtyard exhibited optimal acoustic-*thermal* performance (SPL_{min} = 71.1 dB(A), σ_{UTCI} < 5 ˚C), while the mid-rise canyon demonstrated limited performance (SPL_{min} = 93.4 dB(A), σ_{UTCI} > 10 ˚C). These findings were observed under averaged boundary conditions of a 140 dB(A) aircraft *sound* source and a diurnal MRT range of 60 ˚C on a heatwave day in July 2022.

75	Conference Paper **Reproducibility of an Emulator-Type Assessment Method for the Evaluation of the Dynamic Performance of Air Conditioners**	Mori R.,Matsui S., Giannetti N.,... Togashi E., Saito K.	ASHRAE Transactions , 129	2023	1

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The operation of air conditioners in practice responds to *thermal* loads with variable or intermittent modulations of the compressor speed and the expansion valve opening according to the native control system. Consequently, standard performance ratings obtained at fixed compressor speed benefit from ease of applicability and strong reproducibility but are not representative of the performance of actual installations and may be misleading for guiding the development of commercial units. This study aimed to establish a new performance evaluation method that accounted for the native control of the system in response to given *thermal* loads. Accordingly, the "emulator-type testing facility"was developed by combining an emulator that could simulate arbitrary indoor and *outdoor* conditions and a performance testing facility using the air enthalpy method for dynamic measurement of the cooling capacity supplied by the air conditioner. The reliability of the equipment was estimated, and the *soundness* of the evaluation technique was confirmed. Subsequently, a round-robin test was conducted on the same air conditioning unit in a separate testing facility from a third-party organization to prove the reproducibility of the emulator-type testing methodology while considering the actual controllability of the air conditioner. As a result, consistent Coefficient of Performance (COP) results and differences of approximately 5% were achieved when the system was tested in different testing facilities under the same load and *outdoor* conditions. Accordingly, the reproducibility of the emulator-type dynamic performance evaluation results was verified.

	Document title	Authors	Source	Year	Citations
76	Book Chapter **Aerogel-Inspired Materials Derived from Industrial Waste**	Duong H.M.	Springer Handbooks, Part F1485	2023	1

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In this chapter, we discuss fabrication methods, properties, and applications of eco-friendly macroporous aerogel-inspired materials made from recycled fibers of industrial waste feedstocks such as paper waste, fabric waste, plastic waste, and rubber tire waste, herein referred to as eco-aerogels. Properties afforded by such eco-aerogels are surveyed, including ultralow density, high porosity, large *sound* transmission loss, low *thermal* conductivity, high elasticity, and the ability to be made hydrophilic or superhydrophobic. The effects of fiber and crosslinker content on material microstructure are summarized. Applications including heat and *sound* insulation, oil spill and chemical remediation, *outdoor* apparel, *thermal* protection jackets, medical devices, personal care products, and packaging materials are reviewed.

	Document title	Authors	Source	Year	Citations
77	Conference Paper • *Open access* **Cozie Apple: An iOS mobile and smartwatch application for environmental quality satisfaction and physiological data collection**	Tartarini F.,Frei M., Schiavon S.,Chua Y.X., Miller C.	Journal of Physics Conference Series , 2600(14), 142003	2023	34

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Collecting feedback from people in indoor and *outdoor* environments is traditionally challenging and complex to achieve in a reliable, longitudinal, and non-intrusive way. This paper introduces Cozie Apple, an open-source mobile and smartwatch application for iOS devices. This platform allows people to complete a watch-based micro-survey and provide real-time feedback about environmental conditions via their Apple Watch. It leverages the inbuilt sensors of the smartwatch to collect physiological (e.g., heart rate, activity) and environmental (*sound* level) data. This paper outlines data collected from 48 research participants who used the platform to report perceptions of urban-scale environmental comfort (noise and *thermal*) and contextual factors such as who they were with and what activity they were doing. The results of 2,400 micro-surveys across various urban settings are illustrated in this paper, showing the variability of noise-related distractions, *thermal* comfort, and associated context. The results show that participants experienced at least a little noise distraction 58% of the time, with people talking being the most common reason (46%). This effort is novel due to its focus on spatial and temporal scalability and the collection of noise, distraction, and associated contextual information. These data set the stage for larger deployments, deeper analysis, and more helpful prediction models toward better understanding the occupants' needs and perceptions. These innovations could result in real-time control signals to building systems or nudges for people to change their behavior.

	Document title	Authors	Source	Year	Citations
78	Conference Paper **Study on laser transmission characteristics of atmospheric turbulence disturbed by acoustic waves**	Wang M.,Li Y.	Proceedings of SPIE the International Society for Optical Engineering , 12554, 1255414	2023	0

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When laser propagates in the atmosphere, due to the influence of atmospheric turbulence, it will produce spot drift, light intensity flicker, beam spread and other phenomena, which restrict the development of laser engineering. Acoustic wave is a mechanical wave, when it propagates in the atmosphere, it will change the atmospheric density, affect the internal structure of atmospheric turbulence, and then affect the transmission of laser in atmospheric turbulence. In this paper, a coherent acoustic wave generator is designed, and the laser transmission experiment in the *outdoor* atmospheric turbulence environment disturbed by coherent acoustic waves is carried out. The variations of the beam drift fluctuation variance, scintillation index and beam diameter with coherent acoustic wave frequency and acoustic pressure level are studied. The results show that coherent acoustic waves will change the original structure of atmospheric turbulence and affect the variance of beam drift fluctuation, scintillation index and beam diameter when laser transmitted in atmospheric turbulence; The variance of beam drift fluctuation, scintillation index and beam diameter change with the change of *sound* source frequency and acoustic pressure level. This paper

Document title	Authors	Source	Year	Citations

preliminarily explores the influence of acoustic wave disturbing atmospheric turbulence on laser transmission characteristics, and the research results provide a new idea for improving or interfering laser transmission characteristics in atmospheric turbulence.

Book Chapter

79 **Climate Parameters, Heat Islands, and the Role of Vegetation in the City** Xhexhi K. Urban Book Series 2023 11

Hide abstract Find It @ UCL View at Publisher Related documents

Climate has a strong influence on urban planning and also plays a fundamental role in soil composition affecting the character of plants and animals. The climate is a combination of different meteorological factors that characterized a specific region over a specific time. The movement of the Sun and Earth inclination toward it is the most important factors which determine the characteristics of the climate. The global movement of the air from equator toward poles and vice versa influences also drastically the climate change. In this section are also included breezes. Also air relative humidity plays an important role shaping the climate. It plays a very important role in cloud formation. The built environment can influence the urban climate of the city through height, form, and applied building materials. Building form and shape, building materials, openings in the façade, and their orientation often are determined by climate parameters. Urban density, directions, and street network and their shape are related to the climate change, apart from topography. Density of urban structures produces important effects on the microclimatic conditions in external spaces. The bioclimatic behavior of each type of *outdoor* spaces depends on the relationship with its surrounding elements. The city of Tirana is included in Mediterranean climate characterized by hot dry summers and mild cold wet winters with high daily *thermal* excursions. All of the countries bordering the Mediterranean Sea benefit from the Mediterranean climate. Specifically spotted, Tirana is part of Mediterranean coastal zone (<22 °C in temperature) with an annual temperature of 15.3 °C. In the inner city can be found areas with high pollution, which normally contain a high percentage of business and industry, settlements, and traffic, so-called “sealed” areas. They constitute “hyperthermic areas” or “heat islands”. The term “heat island” describes built-up areas which are hotter than nearby rural areas. Heat islands can affect communities by increasing summertime peak energy demand, air-conditioning costs, air pollution, greenhouse gas emissions, heat-related illness and mortality, and water quality. The aim of the bioclimatic design is also the reduction of the heat-island phenomenon. The use of materials with different energy absorption capabilities and integration of green spaces into urban life is important. The vegetation effects on microclimate and in the built environment are present. Trees are necessary and indispensable to human life. They provide shelter for us, provide food, and their fibers cloth us. Implementation of the green belts in the city is one of the strategies in order to reduce heat-island effect. Vegetation also can provide a deep feeling of attachment to certain places and trees which are part of it. The use of trees avoids monotony and ensures diversification. They play a crucial role creating microclimates which influence the *thermal* comfort of people reducing heating or cooling loads. Vegetation also creates cooler microclimates around the building through evapotranspiration, reduces wind speed, regulates humidity, and reduces temperature. It is observed that tree species have a greater influence on humidity than on temperature. Trees can also absorb and reduce the reverberation time of the *sound*, behaving as *sound* buffers. The benefits of tree plantation in the inner city are enormous and they can’t be neglected.

Article • *Open access*

80 **Heat Stress During Arba’een Foot-Pilgrimage (World’s Largest Gathering) Projected to Reach “Dangerous” Levels Due To Climate Change** Choi Y.-W., Eltahir E.A.B. Geophysical Research Letters, 49(19), e2022GL099755 2022 9

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Arba'een, a Muslim pilgrimage, is one of the largest annual mass gatherings in the world, with a date fixed according to the lunar calendar. Most pilgrims start their long walking journey from Basra/Najaf toward Karbala (about 70–500 km) and are significantly affected by *outdoor* weather conditions during this period. Here, based on simulations performed using carefully-selected climate models, we project that heat stress during the pilgrimage is likely to reach a “dangerous” level, defined according to the US National Weather Service criteria, by the end of this century. Moreover, a significant increase in consecutive occurrence of hot days and hot nights is expected within the coming decades, which may cause a high incidence of heat-related disorders as the human body may not recover from the

	Document title	Authors	Source	Year	Citations

daytime heat loads. Our study suggests that *sound* adaptation measures and stringent mitigation actions must be established to ensure a safe pilgrimage in the future.

Article • *Open access*

81 **Effects of Acoustic Perception on Outdoor Thermal Comfort in Campus Open Spaces in China's Cold Region**

Chen H.,Hong B.,Qu H.,Geng Y.,Su M.

Buildings, 12(10), 1518

2022

36

Hide abstract Find It @ UCL View at Publisher Related documents

To explore the effects of acoustic perception on *outdoor thermal* comfort, acoustic perception in five typical open spaces in a campus in China's cold region with common *soundscapes* was evaluated by using meteorological measures, *sound* level devices, and a questionnaire survey. Eight adjectives were used to evaluate an individual's acoustic perception, and the physiological equivalent temperature (PET) was selected as a *thermal* index. The correlations between acoustic perception and *thermal* comfort in these typical open spaces were further analyzed. We demonstrated that: (1) Acoustic sensation and comfort varied significantly among *sound* types (STPs). Respondents reported pleasant perceptions of broadcasting music, running water and birdsong, and wind (a gentle breeze) and insects. (2) Although there was no significant difference in the *thermal* sensation vote (TSV), we found large differences in the *thermal* comfort vote (TCV) among STPs. (3) The respondents' neutral PET (NPET) varied among STPs. In autumn, the ranked order of NPET was machines > running water and birdsong > crowds > broadcasting music > wind (a gentle breeze) and insects. In winter, the order of the NPET was machines > wind (a gentle breeze) and insects > crowds > broadcasting music > running water and birdsong. (4) When people perceived "acoustic comfort", their TSV improved, and vice versa.

Article

82 **Pre-dimensioning of ventilation openings in double façades | Vordimensionierung von Lüftungsöffnungen in Doppelfassaden**

Friedrich M.,Schweers K.,Wellershoff F.

Bauphysik, 44(4)

2022

0

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Pre-dimensioning of ventilation openings in double façades. Double façades improve *sound* insulation while natural ventilation and offer the shading in the space between the façades, protection from higher wind loads. With regard to natural ventilation, however, it should be noted that the air flows must pass through two façade layers and the air exchange rate cannot be planned with the existing assumptions for single-skin façades. The potentially higher temperatures in the intermediate façades space also require special consideration when planning summer heat protection. Climate data were recorded, and air exchange measurements were taken on 10 buildings in Hamburg with corridor double façades and box window double façades. Based on these data, a calculation approach of the *outdoor* air exchange was created and applied in transient *thermal* building simulations. A method was created with which the required size of the ventilation openings can be determined quickly and without software application in early design phases.

Article • *Open access*

83 **Post-Occupancy Evaluation of Green Technologies for a High-Rise Building Based on User Experience**

Ye C.,Yao L.,Meng Y.,Zhang Y.,He G.

Sustainability Switzerland , 14(15), 9538

2022

14

Hide abstract Find It @ UCL View at Publisher Related documents

Post-occupancy evaluations of buildings help us understand the effectiveness of building designs. Most of such evaluations focus on the overall building performance and on the indoor air environment, and few focus on individual technologies. In this paper, we report a user experience-based evaluation on multiple green technologies applied in a high-rise building constructed 17 years ago. These technologies include a circular building design, innovative floor plans, a corridor-type, double-skin facade with cavity shading, and three-dimensional greening design. Data were obtained through a questionnaire for the building occupants and on-site measurements of environmental variables. The results show that the occupants were most satisfied with the three-dimensional greening, natural lighting and ventilation in the public spaces, and the *sound* and *thermal* environment of the rooms with the double facade. They were not satisfied with room ventilation, solar shading, or lighting adjustment provided by the double facade. In

	Document title	Authors	Source	Year	Citations

addition, the aesthetic appearance of the circular building was not recognized by the occupants, who appeared to pay more attention to its space efficiency than the external appearance. This was true especially for older occupants. The users expressed a strong demand for *outdoor* leisure spaces and green vegetation. However, the practicality and functionality of three-dimensional greening, along with the interference of the glass curtain wall on vegetation growth, calls for more attention to the design. The female occupants were found to be more sensitive to the *thermal* and acoustic environment. Based on these results, we provide recommendations on the maintenance of the building and the future design of these measures.

Article • *Open access*

☐ 84 **Design of Multimodal Sensor Module for Outdoor Robot Surveillance System** — Uhm T.,Park J.,Lee J., ... Ki G.,Choi Y. — Electronics Switzerland, 11(14), 2214 — 2022 — 10

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Recent studies on surveillance systems have employed various sensors to recognize and understand *outdoor* environments. In a complex *outdoor* environment, useful sensor data obtained under all weather conditions, during the night and day, can be utilized for application to robots in a real environment. Autonomous surveillance systems require a sensor system that can acquire various types of sensor data and can be easily mounted on fixed and mobile agents. In this study, we propose a method for modularizing multiple vision and *sound* sensors into one system, extracting data synchronized with 3D LiDAR sensors, and matching them to obtain data from various *outdoor* environments. The proposed multimodal sensor module can acquire six types of images: RGB, *thermal*, night vision, depth, fast RGB, and IR. Using the proposed module with a 3D LiDAR sensor, multimodal sensor data were obtained from fixed and mobile agents and tested for more than four years. To further prove its usefulness, this module was used as a monitoring system for six months to monitor anomalies occurring at a given site. In the future, we expect that the data obtained from multimodal sensor systems can be used for various applications in *outdoor* environments.

Article • *Open access*

☐ 85 **Compact Decentral Façade-Integrated Air-to-Air Heat Pumps for Serial Renovation of Multi-Apartment Buildings** — Ochs F.,Monteleone W., Dermentzis G.,Siegele D., Speer C. — Energies, 15(13), 4679 — 2022 — 7

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To address the huge market of renovation of multi-apartment buildings, minimal-invasive decentral serial-renovation solutions are required. One major challenge in the design of decentral heat pumps is to find the optimal balance between, on one hand, compactness and pleasant design, and on the other hand, efficiency and minimal *sound* emissions. A comprehensive holistic design and optimization process for the development of decentral heat pumps, from the component level, to the system level, and up to the building level, is developed. A novel façade-integrated speedcontrolled exhaust air to supply air heat pump combined with a mechanical ventilation system with heat recovery and recirculation air was developed and simulated in a reference flat. Compared to a traditional supply air heat pump without recirculation, it shows only slight performance improvement, but allows significantly better *thermal* comfort and control, independently from the hygienic air flow rate and from the heating and cooling loads. Detailed measurement and simulation results are presented for several functional models with heating power of around 1 kW up to 2.5 kW. The design was optimized by means of CFD simulations to allow for low pressure drop, homogeneous flow, and low *sound* emissions. Moreover, mock-ups of innovative façade-integrated heat pump *outdoor* units are presented.

Article

☐ 86 **Holistic Operational Signatures for an energy-efficient district heating substation in buildings** — Hong Y.,Yoon S. — Energy, 250, 123798 — 2022 — 23

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District heating (DH) is recognized as a sustainable energy infrastructure in cities and buildings. DH substation operation is essential to bridging the gap between plants and *thermal* clients in achieving the intended DH values and visions. Therefore, in this study, an operation analysis method was developed for building-side DH substations.

Document title	Authors	Source	Year	Citations

Specifically, Holistic Operational Signature (HOS) was suggested with multiple signature elements (x-OS) and representations to consider comprehensive operational correlations. A HOS-based analytics is proposed to provide more insights and *sound* reasoning about the DH substation design, operation, control, and heating efficiency potentials, compared with the existing Energy Signature. In a case study for winter, the proposed HOS method was applied to a target DH substation serving actual residential buildings using the real operational datasets. The six HOSs (HOSs 1–6) were derived with their six elements to identify the representative heating consumption levels and patterns, partial heating load ratios, and operation and control efficiency. Excessive water flow rates were identified based on the partial heating load elements (less than 50% in most cases) of all signatures. Inefficient signatures (HOSs 1–4) that do not follow the conventional control principle according to *outdoor* air temperatures were captured, especially for high heating demand patterns, which accounted for 72.5% of the total in winter. According to the signatures, the demand-based control can decrease the heating supply temperature setpoint by about 1.4 °C and 0.6 °C, respectively for HOS2 and the winter. It is recommended that the valve operation should be improved to lower the hunting levels (within ±1.5 °C) of heating water temperature control, especially in the afternoon with low heating demands.

Article • *Open access*

87 **Influence of atmospheric conditions on measured infrasound from wind turbines** — D'Amico S., Van Renterghem T., Botteldooren D. — Journal of Wind Engineering and Industrial Aerodynamics, 225, 105021 — 2022 — 5

Hide abstract Find It @ UCL View at Publisher Related documents

Scientific interest towards low frequencies and infrasound from wind turbines has renewed over the last decade as the size of wind turbines and the number of wind farms have continuously increased. However, few comprehensive studies are found assessing such noise emitted by wind turbines in response to meteorological parameters, including turbulent characteristics of the atmospheric boundary layer. The current work presents a detailed long-term measurement campaign combining infrasound recordings close to wind turbines and high-resolution meteorological parameters at three heights. An artificial neural network (ANN) model, developed based on the measurement data, was able to adequately predict both blade passing frequency levels and broadband infrasonic noise. The model allowed discriminating between the broadband inflow noise emitted by the wind turbines and the remaining microphone-induced wind noise under the wind-shielding dome for a wide range of atmospheric conditions. At the same time, the ANN allowed deriving physically plausible relationships between meteorological and turbulence parameters and wind turbine emitted infrasound components. Dominant predictors for the infrasound production were mean wind speed, turbulence intensity and turbulent vertical heat flux.

Article • *Open access*

88 **Intercomparison of Vaisala RS92 and RS41 Radiosonde Temperature Sensors under Controlled Laboratory Conditions** — Rosoldi M., Coppa G., Merlone A., Musacchio C., Madonna F. — Atmosphere, 13(5), 773 — 2022 — 10

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Radiosoundings are essential for weather and climate applications, as well as for calibration and validation of remote sensing observations. Vaisala RS92 radiosondes have been widely used on a global scale until 2016; although in the fall of 2013, Vaisala introduced the RS41 model to progressively replace the RS92. To ensure the highest quality and homogeneity of measurements following the transition from RS92 to RS41, intercomparisons of the two radiosonde models are needed. A methodology was introduced to simultaneously test and compare the two radiosonde models inside climatic chambers, in terms of noise, calibration accuracy, and bias in temperature measurements. A pair of RS41 and RS92 radiosondes has been tested at ambient pressure under very different temperature and humidity conditions, reproducing the atmospheric conditions that a radiosonde can meet at the ground before launch. The radiosondes have also been tested before and after fast (within ≈ 10 s) temperature changes of about ±20 °C, simulating a scenario similar to steep *thermal* changes that radiosondes can meet when passing from indoor to *outdoor* environment during the pre-launch phase. The results show that the temperature sensor of RS41 is less affected by noise and more accurate than that of RS92, with noise values less than 0.06 °C for RS41 and less than 0.1

	Document title	Authors	Source	Year	Citations

°C for RS92. The deviation from the reference value, referred to as calibration error, is within ±0.1 °C for RS41 and the related uncertainty (hereafter with coverage factor k = 1) is less than 0.06 °C, while RS92 is affected by a cold bias in the calibration, which ranges from 0.1 °C up to a few tenths of a degree, with a calibration uncertainty less than 0.1 °C. The temperature bias between RS41 and RS92 is within ±0.1 °C, while its uncertainty is less than 0.1 °C. The fast and steep *thermal* changes that radiosondes can meet during the pre-launch phase might lead to a noise increase in temperature sensors during radiosoundings, up to 0.1 °C for RS41 and up to 0.3 °C for RS92, with a similar increase in their calibration uncertainty, as well as an increase in the uncertainty of their bias up to 0.3 °C.

Article • *Open access*

89 **Exploring the Multisensory Interaction between Luminous, Thermal and Auditory Environments through the Spatial Promenade Experience: A Case Study of a University Campus in an Oasis Settlement**

Berkouk D.,Bouzir T.A.K., Boucherit S., ... Matallah M.E.,Mazouz S.

Sustainability Switzerland , 14(7), 4013

2022

29

Hide abstract Find It @ UCL View at Publisher Related documents

This paper aimed to develop a multisensory approach in a university campus, based on quantitative and qualitative approaches, investigating sense walk experiences (*thermo*-visual *sound* walk) under interactions of luminous, *thermal*, and auditory environments. The study was conducted in October 2021, in Chetma university campus in Biskra city, southern Algeria, which remains a famous oasis settlement of arid regions over the country. A comparative and correlation analysis was performed between the physical dimensions collected through a walking experience in three campus routes (*outdoor*, semi-*outdoor* and indoor). In addition, a multisensory survey of the walking experience on perceptual dimensions was evaluated in parallel to the empirical contribution. The paper shows that walkers' *thermal* levels were balanced between neural and slightly hot in different spatial aspects. The glare was almost unperceived regarding the luminous conditions in the study site. The auditory experience reveals that the conducted points were generally quiet and well placed for educational requirements. Findings also show a strong relationship between the physical dimensions of the luminous and auditory environment. Furthermore, the findings suggest that the *thermal* and luminous environments are more perceptible than the auditory environment for the walkers of the *outdoor* and indoor routes. In contrast, the semi-*outdoor* route is often perceptible by the perceptual dimensions of the luminous and auditory environments. The findings on sensorial thresholds and spatial adaption are essential for the educational practices' architectural and urban strategies for the Saharan cities and oasis settlements.

Article • *Open access*

90 **A Sustainable Approach to Build Insulated External Timber Frame Walls for Passive Houses Using Natural and Waste Materials**

Georgescu S.-V.,Sova D., Campean M.,Cosereanu C.

Forests, 13(4), 522

2022

11

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This paper presents structures of timber-framed walls designed for passive houses, using natural and waste resources as insulation materials, such as wool, wood fibers, ground paper, reeds (Phragmites communis), and Acrylonitrile Butadiene Styrene (ABS) wastes. The insulation systems of stud walls composed of wool–ABS composite boards and five types of fillers (wool, ABS, wood fibers, ground paper, and reeds) were investigated to reach U-value requirements for passive houses. The wall structures were designed at a thickness of 175 mm, including gypsum board for internal wall lining and oriented strand board (OSB) for the exterior one. The testing protocol of *thermal* insulation properties of wall structures simulated conditions for indoor and *outdoor* temperatures during the winter and summer seasons using HFM-Lambda laboratory equipment. In situ measurements of U-values were determined for the experimental wall structures during winter time, when the temperature differences between outside and inside exceeded 10 °C. The results recorded for the U-values between 0.20 W/m^{2}K and 0.35 W/m^{2}K indicate that the proposed structures are energy-efficient walls for passive houses placed in the temperate-continental areas. The vapour flow rate calculation does not indicate the presence of condensation in the 175 mm thick wall structures, which proves that the selected *thermal* insulation materials are not prone to degradation due to condensation. The research is aligned to the international trend in civil engineering, oriented to the design and construction of low-energy buildings on the one hand and the use of environmentally friendly or recycled materials on the other.

	Document title	Authors	Source	Year	Citations
	Article • *Open access*				
91	**Associations between indoor soundscapes, building services and window opening behaviour during the COVID-19 lockdown**	Torresin S.,Albatici R., Aletta F.,... Oberman T., Kang J.	Building Services Engineering Research and Technology , 43(2)	2022	28

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Results of an online survey conducted during the COVID-19 lockdown among 848 home workers living in London (United Kingdom) and in Italy are reported with a focus on (1) the impacts of building services on the perception of the acoustic environment while working and relaxing at home and (2) the factors associated with window opening behaviour. The analyses showed no significant difference in *soundscape* appropriateness for relaxation depending on the heating, ventilation and cooling system typologies, and in *soundscape* appropriateness for working from home (WFH) based on the ventilation strategy. Higher *soundscape* appropriateness for WFH was associated with houses equipped only with radiant floors for heating in Italy and with air-cooling systems in London. In London, air systems resulted in higher perceived dominance of noise from building services compared to other systems. Overall, rooms with less dominant *sounds* from building services were evaluated as more appropriate for working and relaxing. The dominance of sky or buildings from the window view, *outdoor* noisiness, noise sensitivity, age and gender were not significantly associated with participants' window opening behaviour while WFH. Differently, participants viewing more vegetation from windows in Italy were more likely (odds ratio: 1.279) to keep the window open while WFH.

	Article				
92	**Combined effects of visual-acoustic-thermal comfort in campus open spaces: A pilot study in China's cold region**	Geng Y.,Hong B.,Du M., Yuan T.,Wang Y.	Building and Environment , 209, 108658	2022	108

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To investigate the combined effects of *thermal*, acoustic and visual comfort in open spaces, this study selected five typical spaces on a university campus with five common *sounds* (broadcasting music, running water & birdsong, wind (a gentle breeze) & insects, crowds, and machines) in a cold region of China. 418 volunteers were asked to complete a subjective sensory questionnaire in a randomly combined audio-visual environment while meteorological measures were taken on-site. Four primary outcomes emerged. First, broadcasting music improved individual subjective *thermal* comfort votes (TCV). Under the moderate heat stress ($0.5 \leq$ *thermal* sensation vote [TSV] < 2.5), a comfortable acoustic environment helped reduce individuals' subjective TSV and increase TCV. Second, the subjective acoustic comfort vote (ACV) of broadcasting music, wind & insect *sound* and machine noise decreased with the increase of Universal *Thermal* Climate Index (UTCI); respondents perceived crowds, machines and wind & insects louder when TCV = 2 and –2. Third, under no *thermal* stress ($-0.5 \leq$ TSV < 0.5)/moderate heat stress ($0.5 \leq$ TSV < 2.5), TSV increased with increasing illumination intensity (LUX). Under the strong heat stress ($2.5 \leq$ TSV ≤ 3), there was no significant difference in TCV among subjective visual comfort vote (VCV) levels. Finally, when the LUX class was neutral/slightly bright, respondents felt darker in *thermo*-neutrality. Additionally, VCV and TCV were positively correlated under each LUX class.

	Article				
93	**Outdoor environmental investigation of old communities during summer in hot summer and cold winter regions**	Li D.,Zhao K.,Ge J.,Ren M.	Indoor and Built Environment , 31(1)	2022	8

Hide abstract Find It @ UCL View at Publisher Related documents

The *outdoor* physical environments of many old communities are unable to meet modern living requirements and require improvement urgently. This is especially true for communities in regions with extremely hot summers and cold winters, which may cause additional difficulties. This study used measurements and questionnaire data to investigate *outdoor* environment (the *thermal*, *sound*, light environments and air quality) and resident sensations in four such communities. Results revealed some key factors affecting *outdoor* environmental comfort, including summer shading, winter sunlight, air temperature, air quality, the *sound* environment and odour. The results also revealed that the

	Document title	Authors	Source	Year	Citations

main problems were noise, insufficient illumination and high air temperatures in summer, all of which had seriously affected comfort during *outdoor* activities. Specifically, noise values during the most unfavourable periods were between 57.4 and 80.6 dB(A), while average air temperatures were between 32.3°C and 35.8°C, and average illuminances were below 2.9 lx. As such, this paper proposes improvement measures such as *sound* barriers and greening. Further, we propose a method for prioritizing these factors for such improvements based on a combination of subjective and objective factors. This study provides data and technical references for the reconstruction of old communities in regions with hot summers and cold winters.

Article

94 **Indoor noise A-level assessment related to the environmental noise spectrum on the building facade** — Zaporozhets O., Fiks B., Jagniatinskis A., ... Karpenko S., Mickaitis M. — Applied Acoustics, 185, 108380 — 2022 — 14

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There is a growing body of research reporting that unwanted indoor noise can negatively affect the well-being of the occupants. According to the World Health Organisation (WHO), indoor guideline values in dwellings are 30 dBA ($L_{Aeq,night,indoor}$), which in turn should also include appropriate airborne and impact *sound* insulation of façades. Such low noise values in dwellings are difficult to measure, which is why indirect methods of indoor noise assessment (including the accurate calculation and measurement procedures for *sound* insulation of dwelling façades) are an important issue worth investigating. Therefore, in this paper, a new methodology of how indoor noise A-level may be assessed by applying the *outdoor* noise monitoring and standardized *sound* insulation measurement of the façade is proposed. It also provides an overview of the existing difficulties in performing accurate calculations of the *sound* insulation on the façades. The proposed calculation procedure for indoor equivalent noise A-level estimation is also evaluated. The highest accuracy was achieved by adding to the calculation a specific spectrum adaptation term $C_{spec,100\text{-}3150}$ for the noise emitted façade in-situ instead of a standard spectrum adaptation term $C_{tr,100\text{-}3150}$, defined by the unified spectra corresponding to road traffic, and suggested by the ISO 717–1 for the calculations. The specific spectrum adaptation term was calculated based on the averaged spectrum of the specific noise events (different from road traffic noise) that contribute to the *outdoor* environment at the locations where measurements were carried out. When assessed in such a way, the equivalent *sound* levels of indoor noise, which are produced by different types of *outdoor* noise sources, show a good agreement with the directly measured levels indoors.

Article • *Open access*

95 **S-shaped dependence of the sound pressure level in outdoor propagation on the effective sound speed gradient** — Hohenwarter D., Mursch-Radlgruber E., Kirisits C. — Acta Acustica, 6, 13 — 2022 — 3

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The effective *sound* speed gradient is used to describe the meteorological conditions during *sound* measurements at roads and railways. Meteorological parameters were assessed up to a height of 10 m. The *sound* level differences between a reference point close to the passing vehicles and at distances of 100 500 m from motorways and railway tracks were determined. These differences were found to correlate well with the effective *sound* speed gradient determined from the measured temperature and wind speed gradients which follow the day/night cycle as a result of the reversing air temperature gradient, incoming solar radiation and wind conditions. The correlation with *sound* level differences can be approximated by an S-shaped function which is constant for large positive or negative gradients of the effective *sound* speed. These threshold values are a consequence of the local meteorological and attenuation conditions. The analysis shows that large effective *sound* speed gradients are mainly a result of the wind conditions whereas gradients without a substantial wind speed contribution are low and a result of the temperature gradient. In the distance range between 150 m and 250 m, the meteorological influences cause a level increase of 3 4 dB(A) for downward refracted *sound* rays (favourable *sound* propagation) compared to a situation without meteorological influence (effective *sound* speed gradient of zero). In the same distance range, meteorological conditions cause a maximum *sound* level attenuation of 5 10 dB(A) for upward curved *sound* rays (unfavourable *sound* propagation).

Conference Paper

	Document title	Authors	Source	Year	Citations
96	**Multi-Objective Optimization Of Thermo-Acoustic Comfort Of School Buildings**	Colarossi D.,Di Loreto S., Tagliolini E.,Principi P., Serpilli F.	Building Simulation Applications , 2022-June	2022	3

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The reduction of the environmental impact of the building sector is one of the top priorities in the “climate change challenge”. As the primary energy consumption of the building decreases, a high level of indoor comfort must be maintained. Both *thermal* parameters, lighting, acoustic level, and indoor air quality affect indoor comfort. These aspects are fundamental, especially in school buildings, where a good level of indoor comfort can help student to stay focused. This paper proposes a methodology for a combined optimization of the energetic and the acoustic performance of a school building. A case study, located in the center of Italy, was analyzed. Firstly, the *thermal* and acoustic performance was determined. Then a list of interventions was hypothesized and simulated, involving both the building envelope, the lighting and *thermal* plants. Normalized acoustic insulation of partitions between adjacent rooms, acoustic insulation of the façade and reverberation time were evaluated. The *outdoor* and ambient noise levels were based on the main characteristics of the façade (type and stratifications of opaque and transparent components, ventilation system, etc.). Results show that the optimal combination of interventions reduces the CO2 emissions of 88.55 % and the global energy performance index of 85.2 %. The indoor *sound* pressure level due to traffic noise is reduced by 19 dB after acoustic insulation of the façade, while further treatments to indoor surfaces should be implemented to reduce internal reverberation time and to improve speech intelligibility. The combined optimization shows that the highest reduction of the global impact (89.2 %) is obtained by weighting 80 %/20 % the acoustic/*thermal* performance.

Article

	Document title	Authors	Source	Year	Citations
97	**Performance of externally bonded fiber-reinforced polymer retrofits in the 2018 Cook Inlet Earthquake in Anchorage, Alaska**	Tatar J.,Sattar S., Goodwin D.,... Dukes J., Segura C.	Earthquake Spectra, 37(4)	2021	5

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As part of the effort to improve the seismic performance of buildings in Alaska (AK), many of the deficient structures in Anchorage, AK, were retrofitted—some with externally bonded fiber-reinforced polymer (EBFRP) composite systems. The 2018 magnitude 7.1 Cook Inlet earthquake that impacted the same region offered an opportunity to evaluate the performance of EBFRP retrofits in a relatively high-intensity earthquake. This study summarizes the following findings of this field investigation: (1) the performance of EBFRP-retrofitted structures in the Cook Inlet earthquake and (2) the observations concerning the condition of FRP retrofits from over a decade of exposure in a subarctic environment. A deployment team from the National Institute of Standards and Technology (NIST) in collaboration with the University of Delaware (UD) Center for Composite Materials conducted post-earthquake inspections of EBFRP retrofits in multiple buildings to assess their performance during the earthquake and condition with respect to weathering. EBFRP debonding was documented with infrared *thermography* and acoustic *sounding* and the bond quality between EBFRP and concrete was assessed using pull-off tests. Visual inspections showed no major signs of earthquake damage in the EBFRP-retrofitted components. However, evaluation of debonding and pull-off test results suggested that *outdoor* conditions may have led to bond deterioration between EBFRP and concrete from installation defects that grew over time, freeze–thaw expansion from moisture present at the FRP/concrete interface, differences in *thermal* expansion of the materials, or a combination thereof. The carbon fiber–reinforced polymer (CFRP) bond to concrete was found to be more vulnerable to *outdoor* exposure than the glass fiber–reinforced polymer (GFRP) bond. Earthquake effects on FRP/concrete bond could not be assessed due to the lack of baseline data.

Article

	Document title	Authors	Source	Year	Citations
98	**ROSI: A Robotic System for Harsh Outdoor Industrial Inspection - System Design and Applications**	Rocha F.,Garcia G., Pereira R.F.S.,... Costa R.R., Lizarralde F.	Journal of Intelligent and Robotic Systems Theory and Applications , 103(2), 30	2021	40

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Document title	Authors	Source	Year	Citations

Belt Conveyors are essential for transporting dry bulk material in different industries. Such structures require permanent inspections, traditionally executed by human operators based on cognition. To improve working conditions and process standardization, we propose a novel procedure to inspect conveyor structures with a ground robot composed by a mobile platform, a robotic arm, and a sensor-set. Based on field experience, we introduce ROSI, a new robotic device designed for long-term operations in harsh *outdoor* environments. The mobile robot has a hybrid locomotion system, using wheels to reduce energy consumption while covering long distances, and also flippers with tracks to improve mobility during obstacle negotiation. A mechanical passive switch allows decoupling tracks' traction, reducing components wear and energy consumption without raising mechanical complexity. Aiming the robot-assisted operation, control strategies help to (i) command both the mobile platform and a robotic manipulator considering the system whole-body model, (ii) adjust the contact force for touching the conveyor structure during vibration inspection, and (iii) climb stairs while automatically adjusting the flippers. Machine Learning algorithms detect conveyors' dirt build-ups, roller failures, and bearing faults by processing visual, *thermal* and *sound* data as inspection functionalities. The algorithms training and validation use a dataset collected from running conveyors at Vale, presenting detection accuracy superior to 90%. Field test results in a mining site demonstrate the robot capabilities to stand for the harsh operating conditions while executing all the required inspection tasks, stating ROSI as a disruptive solution for Belt Conveyor inspections and other general industrial operations.

Article

99 **Heat recovery ventilation system combined with boiler room preheating to improve the indoor environment of rabbit house in severe cold areas | 热回收通风系统结合锅炉房预热改善严寒地区兔舍室内环境**

Liu P.,Guo Y.,Ni J.,... Tian J., Wu Z.

Nongye Gongcheng Xuebao Transactions of the Chinese Society of Agricultural Engineering , 37(18)

2021

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Hide abstract Find It @ UCL View at Publisher Related documents

Heat recovery ventilation (HRV) is type of energy saving ventilation. It has been widely used in animal house to alleviate contradiction between *thermal* comfort and ventilation in recent years. However, air heat exchanger freeze of HRV caused by severe cold weather limited HRV application in northeast China. It is necessary to find an economic way to ensure HRV being operated without freezing. Establish a more reasonable integration of heating and ventilation system in severely cold area. An experiment was conducted in rabbit house during winter in Arun Banner, northeast China. The rabbit house was heated by a coal boiler in the next room. Hot water produced by boiler was supplied to heating radiator inside the rabbit house. The boiler room was warm because of some extra heat release by boiler. To prevent freezing, the boiler room was used to preheat cold fresh air before it entered the heat exchanger. The HRV was installed on the ground of the rabbit house. It was an integration of HRV and boiler heating system. Temperature, relative humidity and concentration of noxious was measured with automatic recording sensor instrument. Data was collected to evaluate the decrease in air pollutants and air temperature of the rabbit house. Preheating effect of the boiler room and the HRV were tested when the system was in operation. The performance of the HRV and prevention of freezing in heat exchanger was evaluated in typical cold weather in northeast China. The economic benefit of the system was analyzed. The results showed that when the HRV and boiler system began to operate, concentrations of ammonia (NH_3) and carbon dioxide (CO_2) in rabbit house dropped by 46% and 64%, respectively. Moreover, indoor air temperature only dropped by an average of 1.8 ℃ when the *outdoor* air temperature varied from -15.8 to -11.8 ℃. It means that the system can significantly improve indoor air quality with barely a slight decline of indoor air temperature. Respiratory and skin disease decreased by 16% and 4%, respectively. It is great potential for better health and production of rabbits. The boiler room heated fresh air by 8.1 ℃, and the HRV heated fresh air by 6.4 ℃ on average. No frozen of the heat exchanger was observed. Heated load of the boiler room and the heat recovery load of the HRV were 6.8 and 5.6 kW, respectively. Sensible heat recovery efficiency of the HRV was 75.6%. Coefficient of performance of the HRV was 6.2. The system was operated efficiently. The method of freezing prevention ensure HRV in good condition among the whole cold season in northeast China. It is an economic option because the system can operated without extra energy consumption. The return of investment is 3 years in this work condition. The integration of HRV and boiler allowed HRV and boiler heating to operate with high efficiency and solve the contradiction between heating and ventilation in severely cold weather in the rabbit house. This finding can

	Document title	Authors	Source	Year	Citations

provide a *sound* reference for the design of animal house heating and ventilation system in severely cold climate northeast China.

Article

100 **Energy-efficient smart window based on a thermochromic microgel with ultrahigh visible transparency and infrared transmittance modulation**

Zhang R.,Xiang B.,Shen Y., ... Guan Q.,Tang S.

Journal of Materials Chemistry A , 9(32)

2021

102

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Both high visible transparency and strong solar modulating ability are highly required for energy-saving smart windows, but conventional responsive materials usually have low transparency and a narrow solar transmittance range. Herein, we report a significant advance toward the design and fabrication of responsive smart windows by trapping a novel V0.8W0.2O2@SiO2-doped poly(N-isopropyl acrylamide) (PNIPAm) *thermochromic* microgel within two glass panels. The smart window is highly transparent to allow solar transmittance at low temperatures, while it turns opaque automatically to cut off the solar energy gain when exposed to sunlight. With a remarkably low content (1.0 wt‰) of dopant, the V0.8W0.2O2@SiO2/PNIPAm (VSP) microgels exhibited an ultrahigh luminous transmittance Tlum of 92.48% and solar modulation ΔTsol of 77.20%. This superior performance was mainly attributed to the V0.8W0.2O2@SiO2 doping inducing a reduction in the PNIPAm particle size and a change in the internal structure. Also, W-doping decreased the phase-transition temperature (Tc) of VO2 from 68 °C to ∼30 °C (close to the Tc of PNIPAm), contributing to infrared transmittance modulation. In particular, the smart window showed excellent energy saving during daytime *outdoor* demonstrations, where the practically achievable cooling temperature reached up to 15.1 °C. This new type of *thermochromic* microgel offering unique advantages of shape-independence and scalability together with *soundproof* functionality has potential applications in energy-saving buildings and greenhouses.

Review • *Open access*

101 **A review of cultural background and thermal perceptions in urban environments**

Naheed S.,Shooshtarian S.

Sustainability Switzerland , 13(16), 9080

2021

18

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Thermal comfort is among the chief indicators of the sustainability of *outdoor* spaces. However, the complex nature of comfort represents the interaction of several determinants that leads to a perception of the *thermal* environment. Recently, researchers have paid particular attention to non-physical factors to understand the mechanisms involved in *thermal* perceptions in urban environments. The extant literature has contended that culture and cultural background are determinants to individuals' *thermal* perceptions. Therefore, this study aimed to review how the link between *outdoor thermal* comfort (OTC) and cultural background is investigated. This is, to the best of our knowledge, the first review study on the subject. The study used a systematic literature review approach based on secondary data available in relevant and contemporary literature. The findings first showed the scarcity of research on cultural background and OTC; however, all studies identified corroborated the significant impact of cultural background on *thermal* perceptions. Notably, the cultural background was found to be the source of variation in *thermal* perceptions, tolerance to, and preference for certain *thermal* conditions, *thermal* comfort requirements and expectations, choice of clothing, and environmental attitudes. The findings provide a *sound* basis for future researchers to address the research gaps identified. The study also raises policy makers' and designers' awareness of urban environment users' genuine needs and requirements.

Article

102 **Extracting boundary conditions for sound propagation calculations using augmented reality | Ermittlung von Randbedingungen für Schallausbreitungsberechnungen mittels Augmented Reality**

Sporr A., Blank-Landeshammer B., Kasess C.H.,... Waubke H., Reichl C.

Elektrotechnik Und Informationstechnik , 138(3)

2021

2

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	Document title	Authors	Source	Year	Citations

The number of air-to-water heat pumps has increased over the last years, resulting in a rising number of installed *outdoor* units. These objects constantly emit noise into the environment during the heat pumps' runtime. To avoid conflicts with neighbors or even a relocation of the already placed heat pump, an app is developed to estimate the emitted *sound* of the virtual heat pump on site. This publication presents adapted *sound* calculation methods suitable for *outdoor* use and on mobile devices. Based on this, boundary conditions are specified that must be met by the *sound* data of the respective *sound* source and the objects recorded on site using augmented reality in real time. Both the determination of the *sound* data and the identification of objects using AR are explained in detail. Furthermore, different virtual objects that alter the *sound* propagation can be placed in the real environment and configured with *sound*-relevant properties. This allows the estimation of the usefulness of *sound*-reducing measures already before realization. An example of the identification of individual objects, the placement of virtual walls and heat pumps, and the final visualization of the *sound* propagation in different ways on the mobile device is presented.

Conference Review

☐ 103 **20th Sustainable, Environment and Architecture** [No Authors Found] Iop Conference Series Earth and Environmental Science , 738(1) 2021 0

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The proceedings contain 80 papers. The topics discussed include: *thermal* environment control of buildings using installation of plants and metal panels on brick walls; *thermal* comfort of the *outdoor* transition space in the dean's office building; *thermal* comfort analysis of residential home in coastal city based on physiological equivalent temperature (PET) index and operative temperature zone; *thermal* and visual comfort analysis using CBE tools and Velux simulation of classroom in R.21 in Gedung Kuliah Bersama V University of Bengkulu; design of sustainability of solar panel integration in a green building complex of Wonogiri regent office; the influence of the *sound* absorber application to the acoustics conditions and the *thermal* transfer value of the building envelope in open plan office; optimization of *thermal* comfort in Rifain digital printing: vertical garden with water wall system concept; energy efficiency in church building based on Sefaira energy use intensity standard; and estimation of energy consumption efficiency in office rooms cooling systems to create *thermal* comfort for the user.

Article

☐ 104 **Cooling operation analysis of multienergy systems in a nearly zero energy building** Li H.,Zhang S.,Yu Z.,Wu J.,Li B. Energy and Buildings , 234, 110683 2021 9

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A 4025 m^2 office building was built in Beijing, China, in 2014 as a living laboratory and prototype to explore the technical energy systems of nearly zero energy buildings (NZEBs) in China. Not only the building itself but also the heating, ventilation and air conditioning (HVAC) systems were designed with flexibility to explore their performance. Solar-assisted air conditioning (SAAC) and ground source heat pump (GSHP) systems with different terminal units were designed as HVAC systems for either experimental, comparison, or measurement purposes. In this paper, the cooling operation of multienergy systems is discussed on the basis of actual measurement data. HVAC systems have three possible types of operation configuration types, namely, SAAC priority, GSHP only and multimode. Operation scenarios for these configurations are presented first, and the small-scale simulation was carried out to analyze the performance of SAAC priority and GSHP#1 only mode. It indicates that SAAC could save about 47% of energy than that of GSHP#1. Then, the actual operation performance in 2018 was analyzed by reference to the operation data. Through this study, it is found that 1) multienergy systems of SAAC and GSHP operated well in this NZEBs during the summer season; 2) a 10 RT absorption chiller can work well for an NZEB building of approximately 1000 m^2 when *outdoor* enthalpy is low and the solar radiation is stronger than 400 W/m^2 as a result of high building *thermal* performance and air tightness; 3) SAAC can save about 40 to 50% of energy compared to that of GSHP system when providing services for the same terminal system under the same *outdoor* weather conditions and 4) The GSHP system is a good complement to the SAAC system which works well in June, September and early July when *outdoor* enthalpy is low. This study indicates that a combination of GSHP and SAAC systems is highly promising for NZEBs in cold climate zones such as Beijing, where the solar radiation is *sound*.

	Document title	Authors	Source	Year	Citations
	Article				
105	**Thermal performance experiment of heat storage wall with flat micro heat pipe array in solar greenhouse | 日光温室平板微热管阵列蓄热墙体热性能试验**	Guan Y.,Wang T.,Wei M., ... Hu W.,Duan S.	Nongye Gongcheng Xuebao Transactions of the Chinese Society of Agricultural Engineering , 37(3)	2021	6

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Composite hetero-structure wall is widely used for the north wall of solar greenhouse in China. A *thermal* stable thick layer in this type of wall can greatly weaken the heat storage capacity of the greenhouse wall. This study aims to improve the temperature of the *thermal* stable layer, and thereby promoting the heat storage capacity of north walls in a solar greenhouse. A new type of wall was proposed for the solar greenhouse with a flat Micro Heat Pipe Array (MHPA) wall or heat storage wall. The wall was made by bending the flat micro heat pipe array into a zigzag-shape while mixing the cement mortar to form an MHPA block. Then, the MHPA block and ordinary blocks were built into the MHPA wall. The evaporation section of the flat micro heat pipe array was placed on the inner surface of the MHPA wall, and the condensation section was placed inside the MHPA wall to achieve the purpose of enhancement heat transfer. Two small-scale greenhouse comparison test-benches were set up on site with the MHPA wall (named as MHPA greenhouse) and the ordinary block wall (named as ordinary greenhouse). The evaluation parameters included the temperature of the greenhouse wall, the daily heat storage/release of the wall, and the greenhouse environmental temperature. A systematic investigation was made to compare the characteristics of heat storage/release of the MHPA wall, and the effect of improving the greenhouse *thermal* environment on representative days. The results showed that the average internal surface temperature of the MHPA wall on typical sunny days (from 17: 00 to 9: 00 the next day) was 1.6-2.3℃ higher than that of an ordinary wall. From indoor to *outdoor*, the average temperature inside the MHPA wall for the 0-370 mm area along the wall thickness direction increased by 2.7-4.0℃. During the heat storage period (from 9: 00 to 17: 00), the average environmental temperature of the MHPA greenhouse was 0.2-0.5℃ lower than that of an ordinary greenhouse, and the average ground surface temperature was 1.8-1.9℃ lower. During the heat release period, the average environmental temperature of the MHPA greenhouse was 1.2-1.5℃ higher than that of an ordinary greenhouse, and the average ground surface temperature was 0.6-1.0℃ higher. The MHPA wall can play the role of "peak cutting and valley filling" in the *thermal* environment of a greenhouse. The total daily heat storage of the MHPA wall was 8.92%-14.35% higher than that of an ordinary wall, and the total daily heat release was 2.24%-8.07% higher. In the heat storage and release rates of a wall on December 27, the maximum and average heat storage rates of the MHPA wall were 9.63% and 10.99% higher than those of the ordinary wall, and the maximum and average heat release rates were 9.23% and 5.60% higher than those of the ordinary wall, respectively. Moreover, the heat release rate of the MHPA wall was higher than that of an ordinary wall between 23: 00 and 7: 00 the next day, with an average increase of 11.53%. Therefore, the introduction of the MHPA wall into the wall of a solar greenhouse can effectively improve the heat storage and release capacity of the wall materials for a better *thermal* environment. These findings can provide a *sound* reference for the application of a flat micro-heat pipe array to the heat storage wall in a solar greenhouse.

	Book Chapter				
106	**Influence of internal thermal insulation on the sound insulation of walls**	Zeitler B.	Energy Efficient Retrofit of Buildings by Interior Insulation Materials Methods and Tools	2021	0

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Adding any sort of lining to a building element, e.g., to a wall or floor, influences the *sound* transmission through that building element. In this chapter, the change of *sound* transmission or *sound* insulation caused by adding an internal *thermal* insulation lining to exterior wall is discussed. First, a short review of the physical phenomena related to the *sound* insulation of a wall with and without internal *thermal* insulation is given. This should help understand the wall's ability to insulate indoor spaces from *outdoor* noise. Following, the measured *sound* insulation, described by the

	Document title	Authors	Source	Year	Citations

standardized weighted *sound* reduction index R_w, is presented for three types of linings on a masonry wall. Of the three types, the detached stud linings perform best and give the highest improvement when added to a wall (up to 20 dB), linings attached to the wall with furring strips show a lesser improvement (up to 4 dB), and the lowest acoustically performing linings are the panelized composite linings that even result in a worsening of *sound* insulation when they are attached over the entire surface. The role, the spectra of the *outdoor* noise sources, e.g., traffic noise, can play on the indoor levels is also touched upon. The *sound* insulation of the exterior wall with lining is put into perspective with the other elements of the exterior wall, such as windows and ventilators. The internal *thermal* linings do not only influence the *sound* transmission from outdoors to indoors but also the *sound* transmission from indoors to indoors. An example suggests how and why this should also be taken into consideration when selecting appropriate interior *thermal* insulation.

Conference Paper

☐ 107 **Propagation loss of low frequency horn loudspeakers: Is "throw" a real phenomenon?** Hipperson J. 150th Audio Engineering Society Convention AES 2021 2021 0

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Horn loading is frequently used in *sound* reinforcement to increase efficiency and directivity of high and mid frequency transducers. Low frequency horn loudspeakers are less common due to their large size. Increases in available amplifier power and *thermal* dissipation in transducers have led to widespread use of high power, low efficiency dual 18" bass reflex loudspeakers. However, some manufacturers and enthusiasts continue to develop and use low frequency horn loudspeakers for their high efficiency and subjective audio quality. In the fields of live event production and noise control, there is sometimes a perception, or "urban myth" that horn low frequency loudspeakers project or "throw" *sound* a further distance than direct radiating low frequency loudspeakers. This is either considered to be beneficial, or problematic depending on the context. Considering the relevant acoustic theory, it is not immediately apparent why this should be the case, providing the loudspeakers are level matched and of similar physical dimensions. Unfortunately, there is very little investigation of low frequency horns in previous literature to aid in providing a definitive answer. Measurements in this paper demonstrate that horn and direct radiating low frequency loudspeakers and arrays closely follow the theory, and the difference in propagation loss is within measurement uncertainty. The implication for noise control of *outdoor* events, is that bass loudspeaker size and type are not especially relevant factors, and focus should instead be on system/array design and site layout.

Article • *Open access*

☐ 108 **Liquid Thermo-Responsive Smart Window Derived from Hydrogel** Zhou Y., Wang S., Peng J., ... Boey F.Y.C., Long Y. Joule, 4(11) 2020 435

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

Buildings account for 40% of global energy consumption, while windows are the least energy-efficient part of buildings. *Thermochromic* window responds to temperature automatically to regulate the solar transmission solely. In this work, a smart window that combines good *thermochromic* performance and large *thermal* energy storage capability was introduced. At lower temperature, the window is transparent to let in the solar transmission; when heated, the window blocks sunlight automatically to cut off solar gain. The added function of heat storage further reduces the energy consumption and shift the electricity load peak to lower price period. This first *thermo*-responsive liquid encapsulated window panel offers a think-out-of-box strategy, giving unique advantage of easy fabrication, good uniformity, scalability, and *soundproofing*. Buildings account for 40% of global energy consumption, while windows are the least energy-efficient part of buildings. Conventional smart windows only regulate solar transmission. For the first time, a smart *thermochromic* window with high *thermal* energy storage was developed to cut off building energy consumption as demonstrated by experiments and simulations. The first liquid encapsulated glass panel gives unique advantages of high uniformity, easy fabrication, shape independence, scalability, and *soundproofing*. Buildings account for 40% of global energy consumption, while windows are the least energy-efficient part of buildings. Conventional smart windows only regulate solar transmission. For the first time, we developed high *thermal* energy storage *thermo*-responsive smart window (HTEST smart window) by trapping the hydrogel-derived liquid within glasses. The excellent *thermo*-responsive optical property (90% of luminous transmittance and 68.1% solar modulation) together with outstanding specific heat capacity of liquid gives the HTEST smart window excellent

	Document title	Authors	Source	Year	Citations

energy conservation performance. Simulations suggested that HTEST window can cut off 44.6% heating, ventilation, and air-conditioning (HVAC) energy consumption compared with the normal glass in Singapore. In *outdoor* demonstrations, the HTEST smart window showed promising energy-saving performance in summer daytime. Compared with conventional energy-saving glasses, which need expensive equipment, the *thermo*-responsive liquid-trapped structure offers a disruptive strategy of easy fabrication, good uniformity, and scalability, together with *soundproof* functionality that opens an avenue for energy-saving buildings and greenhouses.

Conference Paper

☐ 109 **AcuTe: Acoustic thermometer empowered by a single smartphone** — Cai C.,Chen Z.,Pu H., ... Hu M.,Luo J. — Sensys 2020 Proceedings of the 2020 18th ACM Conference on Embedded Networked Sensor Systems — 2020 — 42

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Though measuring ambient temperature is often deemed as an easy job, collecting large-scale temperature readings in real-time is still a formidable task. The recent boom of network-ready (mobile) devices and the subsequent mobile crowdsourcing applications do offer an opportunity to accomplish this task, yet equipping commodity devices with ambient temperature sensing capability is highly non-trivial and hence has never been achieved. In this paper, we propose Acoustic *Thermometer* (AcuTe) as the first ambient temperature sensor empowered by a single commodity smartphone. AcuTe utilizes on-board dual microphones to estimate air-borne *sound* propagation speed, thereby deriving ambient temperature. To accurately estimate *sound* propagation speed, we leverage the phase of chirp signals to circumvent the low sample rate on commodity hardware. In addition, we propose to use both structure-borne and air-borne propagations to address the multipath problem. Furthermore, to prevent disruptive audible transmissions, we convert chirp signals into white noises and propose a pipeline of signal processing algorithms to denoise received samples. As a mobile, economical, highly accurate sensor, AcuTe may potentially enable many relevant applications, in particular large-scale indoor/*outdoor* temperature monitoring in real-time. We conduct extensive experiments on AcuTe; the results demonstrate a robust performance, a median accuracy of 0.3° C even at a varying humidity level, and the ability to conduct distributed temperature sensing in real-time.

Article

☐ 110 **Indoor environmental quality in existing public buildings in China: Measurement results and retrofitting priorities** — Miao Y.,Ding Y. — Building and Environment , 185, 107216 — 2020 — 32

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Through a large-scale questionnaire survey across China and indoor physical measurements of typical cities in the four climatic zones, this study investigates the current status and retrofitting priorities of the IEQ of existing public buildings in China. The public buildings in China mainly present uneven indoor hot and cold distributions and draft sensations. The utilization ratio of natural light is low, and glare often exists. HCHO and TVOC are not major concerns of indoor air quality, but affected by the region and season, the $PM_{2.5}$ and PM_{10} concentrations exceed the limits of the Chinese standard. All the buildings present excessive noise. Respondents generally demand improved indoor air quality and *thermal* environment, and different control strategies should be adopted across regions with different local climatic conditions and behavioural habits. For the *thermal* environment, the indoor air distribution should be optimized and the indoor humidity ratio increased in winter in northern regions of China. For the luminous environment, the indoor illuminance value should be decreased and the lighting quality improved. For the acoustic environment, the *sound* insulation and *sound* absorption performance of the building envelope should be improved. For indoor air quality, the purification capacity of the air conditioning system and the airtightness of the building envelope should be strengthened. According to the *outdoor* air quality, different channels can be opened to introduce fresh air. For example, when the *outdoor* air quality is good, fresh air can be directly introduced to improve the service life of the filtering system and reduce energy consumption.

	Document title	Authors	Source	Year	Citations

Article • *Open access*

111 **Post-occupancy evaluation of indoor environmental quality in ten nonresidential buildings in Chongqing, China**

Tang H.,Ding Y.,Singer B.C. | Journal of Building Engineering, 32, 101649 | 2020 | 50

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Indoor environmental quality (IEQ) is an essential element of sustainable building performance. To advance knowledge of indoor environmental quality in Chinese nonresidential buildings, this study conducted post-occupancy evaluation (POE) in varied use areas within schools, hospitals, offices, hotels and shopping malls through environmental measurements and occupant surveys in 10 buildings over four seasons in Chongqing, China. Measured *thermal* conditions were more frequently within comfort standards in winter than in summer according to the predicted mean vote – predicted percentage dissatisfied model; but the model did not reliably predict *thermal* satisfaction. There was insufficient illuminance in 28% of studied areas and high illuminance in shopping malls led to complaints about glare. *Sound* pressure exceeded the Chinese guideline in all buildings with shopping malls having significantly higher levels than other buildings. CO_2 was generally below the Chinese national guideline of 1000 ppm; the two percent of measurements that exceeded the guideline occurred in classrooms, hospital wards and underground supermarkets. Formaldehyde did not exceed the limit level of 0.1 mg/m^3 in any space. Most of the buildings did not provide effective protection from *outdoor* fine particulate matter ($PM_{2.5}$), and infiltration from outdoors was the main factor for exceeding health-based guidelines indoors. Shopping malls had lower IEQ satisfaction than offices, schools and hospitals. *Sound* was the factor with the lowest satisfaction level among the four principal environmental factors. Overall, this study demonstrates value of POE on identifying deficiencies in indoor environment and provides a reference for assessment of IEQ in nonresidential buildings.

Article • *Open access*

112 **Effects of sound types and sound levels on subjective environmental evaluations in different seasons**

Jin Y.,Jin H.,Kang J. | Building and Environment, 183, 107215 | 2020 | 48

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Human beings live in an environment where various factors act together, therefore, it is essential to study the comprehensive effects of multiple factors, particularly the effects of their interaction on environmental evaluations. In this study, the effects of the interaction between *sound* types and *sound* levels on acoustic, *thermal*, and overall evaluations were explored by simulating typical *outdoor* temperatures in different seasons in a controllable environmental chamber. The results indicated that the acoustic evaluations were significantly higher for birdsong and slow-dance music than for dog barking, conversation, and traffic *sound*; additionally, the acoustic evaluations at the low *sound* level were always higher than those at the high *sound* level. In terms of the *thermal* evaluations, there was a significant variation in different seasons. In summer, birdsong and slow-dance music effectively improved subjects' *thermal* evaluations, while a high *sound* level of dog barking, conversation, and traffic *sound* resulted in a decrease; in the transition season, all types of *sounds* resulted in a decline in the *thermal* evaluations; meanwhile, in winter and summer, dog barking, conversation, traffic *sound* and slow-dance music at the low *sound* level produced higher *thermal* comfort and *thermal* acceptability. In terms of the overall evaluations, birdsong and slow-dance music at the low *sound* level improved overall comfort, while dog barking, conversation, and traffic *sound* resulted in a significant decrease. For dog barking, conversation, traffic *sound* and fast-dance music, the overall evaluations at the low *sound* level were higher than those at the high *sound* level.

Review • *Open access*

113 **On the colloidal and chemical stability of solar nanofluids: From nanoscale interactions to recent advances**

Sharaf O.Z.,Taylor R.A.,Abu-Nada E. | Physics Reports, 867 | 2020 | 125

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The *outdoor* deployment of solar nanofluids in practical solar photothermal conversion devices for prolonged periods of time has been an ongoing challenge. This is due to their exposure to intense solar radiation, elevated temperatures, *thermal* and solar cycling, and other rough operation conditions that compromise their dispersion and chemical

Document title	Authors	Source	Year	Citations

stability. This work is an attempt to address this challenge in light of state-of-the-art advances in understanding, characterizing, and mitigating issues pertaining to the stability of solar nanofluids. We first defined the concept of nanofluids, along with their types and applications, merits and limitations, and preparation techniques. This was followed by presenting the basic physical principles that govern the interparticle interactions, clustering and deposition kinetics, and theories of colloidal stability for nanoparticles in aqueous environments. Building on a *sound* understanding of these necessary fundamentals, the destabilization factors and instability effects on solar nanofluids were systematically identified. A critical evaluation of the most recent stability characterization techniques and stabilization strategies for solar nanofluids was then provided. A coverage of selected state-of-the-art works in the literature that target the enhanced stability of solar nanofluids in different photothermal conversion processes based on the direct volumetric absorption of solar radiation was presented and comparatively evaluated. Finally, recommendations were drawn regarding current knowledge gaps and future research directions in order to overcome the stability limitations hindering the deployment of solar nanofluids in practical solar photothermal conversion applications.

Article • *Open access*

☐ 114 **Green Composite of Instant Coffee and Poly(vinyl alcohol): An Excellent Transparent UV-Shielding Material with Superior Thermal-Oxidative Stability** — Lyu Y., Gu X., Mao Y. — Industrial and Engineering Chemistry Research, 59(18) — 2020 — 35

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Producing ultraviolet-light-shielding (UV-shielding) materials via green chemistry has been attracting researchers' attention. Herein, a polymer composite produced by hybridizing instant coffee with poly(vinyl alcohol) (PVA) is reported. It has been shown that adding coffee extract can significantly improve the UV-shielding property of the PVA material in the wavelength range between 200 and 400 nm, and the polymer composite remains largely transparent in the visible region. The UV-shielding performance of the composite can be maintained under the accelerated aging condition with a UV light dose of 40 MJ/m^2 for 68 h (approximately equivalent to a 45-day *outdoor* exposure in Miami, Florida). In addition, enhanced *thermal*-oxidative stability has been observed for the composite, as indicated in the *thermal* gravimetric analysis (TGA). The composite material can be used in the form of polymer films processed by solution casting or be served as a protection layer by dip coating. The mechanisms of UV shielding and *thermal*-oxidative stability are discussed based on effective components such as melanoidin, caffeine, and chlorogenic acid. The work demonstrates yet another route toward fabricating biocompatible and environment-friendly UV-shielding materials with *sound* overall performance.

Review

☐ 115 **Review of parameters used to assess the quality of the indoor environment in Green Building certification schemes for offices and hotels** — Wei W., Wargocki P., Zirngibl J., Bendzalova J., Mandin C. — Energy and Buildings, 209, 109683 — 2020 — 119

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Fourteen Green Building (GB) certification schemes were reviewed to examine the parameters they used to assess indoor environmental quality (IEQ). Ninety different parameters were identified. They were classified into four major IEQ components defining the *thermal*, acoustic and visual environments, and indoor air quality (IAQ). For the *thermal* environment, the most commonly used parameters were PMV, PPD, room operative temperature, room air temperature, room air relative humidity, and air speed. For the acoustic environment, the most commonly used parameters were ambient noise and reverberation time. For the visual environment, the most commonly used parameters were illuminance level, daylight factor, and spatial daylight autonomy. For IAQ, the most commonly used parameters were ventilation rate (*outdoor* air supply rate), TVOC, formaldehyde, CO_2, CO, PM_{10}, $PM_{2.5}$, ozone, benzene, and radon. Credits are used to rate the importance of different parameters for the overall level of IEQ in the reviewed schemes. Using these credits and the figures published in peer-reviewed papers, it was found out that the average contribution of the *thermal*, acoustic, luminous environment and air quality parameters to the overall IEQ rating of a building was respectively 27%, 17%, 22%, and 34%. The present work can be regarded as a reference for selecting parameters that are commonly used to characterize IEQ.

	Document title	Authors	Source	Year	Citations
	Conference Paper				
116	**Indoor environmental quality evaluation of an institutional building**	Zhong L.	ASHRAE Transactions , 126	2020	1

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Indoor environment in an institutional building, including ventilation, indoor air quality (IAQ), acoustic and *thermal* conditions, as well as lighting directly influence the students' study and health performance. In this paper, all five aspects were studied to evaluate the indoor environmental quality (IRQ) in an institutional building. This study examines IEQ parameters, including pressure, illuminance level, acoustics, carbon dioxide (CO_2) levels, temperature, and humidity, with appropriate monitors allocated during a lecture duration (50min or 80 min) in four lecture classrooms (59187 ± 1554 ft^3, 1676 ±44 m^3) repeatedly from October 2018 to March 2019 with the objectives of providing a comprehensive analysis and updating the literature. *Outdoor* temperature was-13°F to 54°F (-25°C to 12°C) during two-season measurements, while the classroom environment was maintained at 73 ± 2°F (23 ± 1°C) and 33% ± 3% RH. Indoor CO_2 had mean concentrations of 550-1055 ppm and a mean *sound* level of 58 ± 3 dBA. Using the one-spne steady-state box model, the air change rates were configured at 1.3-6.5 per hour based on continuous CO_2 measurements and occupant loads in the lectures. Results were compared to ventilation guideline for lecture classrooms by ASHRAE. Buildingfeatures and mechanical operation details were discussed to analyse the relationships between building design/operation and IEQ performance.

	Conference Paper				
117	**The environmental performance of temporary urban interventions: Technical assessment of regeneration initiatives in the city centre of São Paulo, with focus on thermal and acoustic performance**	Michalski R.L.X.N., Rodrigues L., Goncalves J.C.S., ... Bilesky D.F., Guimaraes M.M.	PLEA 2020 35th PLEA Conference on Passive and Low Energy Architecture Planning Post Carbon Cities Proceedings, 1	2020	0

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Temporary urbanism is an approach to reactivate urban spaces through short-term interventions in a range of urban contexts. In central São Paulo, the Luz and Santa Ifigênia neighbourhoods, characterized by deprivation of their physical environments and social structures, were the focus of this investigation. The Mungunzá Container Theatre and the General Osório Square, located within these neighbourhoods, were selected as case-studies. Whilst the *thermal* performance of the container theatre itself was the main interest, in the case of the Square the fundamental issue was the environmental noise. The objective was to identify adequate strategies to improve environmental conditions in these locations in order to enhance positive social impact, and, then, contribute to the regeneration of these neighbourhoods. This research was based on fieldwork and analytical procedures of *thermal* and acoustic performances. In the container theatre building, the adoption of external shading and wider openings for ventilation reduced its indoor peak temperatures and delivered *thermal* comfort during the warmest period of the year. In the Square, *sound* absorber road surface material and an acoustic shell were proposed to reduce noise and promote better acoustic quality for *outdoor* performances.

	Conference Paper				
118	**Analysing Energy Savings and Overheating Risks of Retrofitting Chinese Suburban Dwellings to the Passivhaus EnerPHit Standard**	Liu C.,Sharples S., Mohammadpourkarbasi H.	PLEA 2020 35th PLEA Conference on Passive and Low Energy Architecture Planning Post Carbon Cities Proceedings, 3	2020	0

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Chinese suburban housing largely consists of uninsulated reinforced-concrete apartment blocks with poor insulation performance. Much energy is required to reach indoor *thermal* comfort in Chinese regions which have hot

	Document title	Authors	Source	Year	Citations

summer/cold winter climates. Retrofitting existing dwellings to a better energy performance standard might be preferable to new build as existing buildings are mostly structurally *sound* and local demand for new homes is low. This paper examines the viability and benefits of applying the German Passivhaus EnerPHit standard to a suburban dwelling in Hunan, which is in the hot summer/cold winter climate zone of China. The indoor and *outdoor* hygrothermal data of the dwelling was monitored for 12 months, and the dwelling was then modelled and validated against the measured data. Incremental EnerPHit retrofitting measures were applied to the modelled building, and the *thermal* performance of the retrofitted building was evaluated using multiple approaches, including a PMV model, the Passivhaus standard and the Chinese passive house standard. The simulation results suggest that the EnerPHit standard is achievable, producing a high energy saving efficiency. Indoor *thermal* comfort was hugely improved after retrofitting, but two of the multiple comfort evaluation approaches highlighted that overcooling was a bigger problem than overheating for the retrofitted dwelling.

Conference Paper

☐ 119 **Phase change materials ranking by using the analytic hierarchy process** | Nadeem A.,Rakhman K., Hossain A. | Proceedings of International Structural Engineering and Construction, 7(1), CPM-09 | 2020 | 5

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Many phase change materials (PCMs) are available with building envelope use potential. PCM in a building envelope stores heat under warm climate as it changes from solid to liquid phase at the melting temperature. Mainly, the latent heat of fusion is stored, which reduces the daily heat flux from the outdoors into the building. Thus, the indoor cooling requirement is reduced. When the *outdoor* temperature falls, PCM releases the stored heat into the atmosphere over the phase change from liquid to solid that may offset the internal heating needs depending on the design of the PCM application. The PCMs exhibit a great variety in their properties. While many researchers are optimizing the use of PCM for a building application under specific climates, most of them focus on the energy-saving potential of PCMs. Other criteria, such as economic feasibility, environmental hazards, stability, fire resistance, and nonthermal physical properties, should also be considered. Therefore, with many criteria to consider, a *sound* selection method is necessary for choosing a suitable alternative PCM in an application. Some phase change materials were examined and ranked by taking into consideration their *thermal* and non-*thermal* properties. These properties were assessed using a multi-criteria decision-making tool called the Analytic Hierarchy Process (AHP). The results of this paper illustrate a pathway for selecting suitable PCMs and provide support to PCM application in building envelopes.

Conference Paper

☐ 120 **Phase change materials ranking by using the analytic hierarchy process** | Nadeem A.,Rakhman K., Hossain M.A. | 3rd European and Mediterranean Structural Engineering and Construction Conference 2020 Euro Med Sec 2020, CPM-09 | 2020 | 3

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Many phase change materials (PCMs) are available with building envelope use potential. PCM in a building envelope stores heat under warm climate as it changes from solid to liquid phase at the melting temperature. Mainly, the latent heat of fusion is stored, which reduces the daily heat flux from the outdoors into the building. Thus, the indoor cooling requirement is reduced. When the *outdoor* temperature falls, PCM releases the stored heat into the atmosphere over the phase change from liquid to solid that may offset the internal heating needs depending on the design of the PCM application. The PCMs exhibit a great variety in their properties. While many researchers are optimizing the use of PCM for a building application under specific climates, most of them focus on the energy-saving

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Article • *Open access*

121 **Indoor environmental quality evaluation of lecture classrooms in an institutional building in a cold climate** — Zhong L.,Yuan J.,Fleck B. — Sustainability Switzerland, 11(23), 6591 — 2019 — 43

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In this paper, ventilation, indoor air quality (IAQ), *thermal* and acoustic conditions, and lighting were studied to evaluate the indoor environmental quality (IEQ) in an institutional building at the University of Alberta in Edmonton, Canada. This study examined IEQ parameters, including pressure, illuminance, acoustics, carbon dioxide (CO_2) concentration, temperature, and humidity, with appropriate monitors allocated during a lecture (duration 50 min or 80 min) in four lecture classrooms repeatedly (N = 99) from October 2018 to March 2019 with the objectives of providing a comprehensive analysis of interactions between IEQ parameters. The classroom environments were maintained at 23 ± 1 °C and 33% ± 3% RH during two-season measurements. Indoor mean CO_2 concentrations were 550-1055 ppm, and a mean *sound* level of 58 ± 3 dBA was observed. The air change rates were configured at 1.3-6.5 per hour based on continuous CO_2 measurements and occupant loads in the lectures. A variance analysis indicated that the within-lecture classroom variations in most IEQ parameters exceeded between-lecture classrooms. A multilayer artificial neural network (ANN) model was developed on the basis of feedforward networks with a backpropagation algorithm. ANN results demonstrated the importance of the sequence of covariates on indoor conditions (temperature, RH, and CO_2 level): Air change rate (ACR) > room operations (occupant number and light system) > *outdoor* conditions.

Article • *Open access*

122 **Real time monitoring of indoor environment quality and energy consumption in a residential building** — Toderasc M.,Iordache V., Petcu C.,Petran H. — Environmental Engineering and Management Journal, 18(7) — 2019 — 6

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Several theoretical studies were oriented towards the relationship between the main two building design estimators (building energy consumption and indoor environmental quality) but very little experimental research has been carried out in order to underline the correlation between this two parameters. This paper is focused towards the experimental study of the correlation between these two design estimators. Several indoor comfort parameters (air temperature, radiant mean temperature, *sound* pressure level, *outdoor* ventilation rate and lighting level) as well as energy consumption were monitored in different rooms inside a laboratory experimental house at real scale located in Bucharest, Romania. Time variations of these parameters and space distribution maps inside the house were analysed in order to understand the indoor comfort variations. Four comfort indexes (*thermal* comfort index, acoustic comfort index, indoor air quality index and visual comfort index) were determined and their time variations were used to emphasize the correlation between the indoor environment quality and the energy consumption. The overall indoor environment quality was found to be inversely correlated with the energy consumption and that this general trend can be easily influenced by the weight of the four comfort types.

Article

	Document title	Authors	Source	Year	Citations
123	**Assessing noise with provisions for ventilation and overheating in dwellings**	Harvie-Clark J.,Chilton A.,Conlan N.,Trew D.	Building Services Engineering Research and Technology, 40(3)	2019	24

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In the design of residential developments, it has been common practice for façade *sound* insulation (to protect against *outdoor* noise) to be considered separately from the provisions for ventilation and for mitigating overheating. This fragmented approach has led to different designers making different, incompatible assumptions about the internal environmental quality conditions: the acoustic designer assumes that windows are closed to control external noise ingress, while the mechanical designer assumes that windows are open for ventilation or mitigating overheating. This leaves occupants with a choice between reasonable noise levels or *thermal* comfort, but not both. This problem is exacerbated by increased overheating risk in modern buildings and future climate scenarios. In response to this issue, the Association of Noise Consultants has produced the draft Acoustics Ventilation and Overheating Residential Design Guide – the 'AVO Guide'. The AVO guide recommends an approach to acoustic assessment that takes regard of the interdependence of provisions for external noise ingress, ventilation and overheating. This paper describes the context, background and content of the AVO Guide. There are references to passive attenuated options for ventilative cooling to help designers avoid simply specifying mechanical cooling. Practical application : This paper is entirely focused on the practical application of the guidance in the Association of Noise Consultants' Acoustics, Ventilation and Overheating Residential Design Guide. It describes the current practical problem that designers face, and the mistakes that are the common practice across the industry, of failing to integrate the design for noise with the ventilation strategy and provisions for overheating. It outlines how this can be achieved to enable better internal environmental quality conditions for occupants, supported by the current English Planning regime and professional good practice guidance available.

Article

124	**Noise-reducing vents for windows in warm, humid, tropical countries**	Satwiko P.,Prasetya A.,Suhedi F.	Architectural Engineering and Design Management, 15(2)	2019	2

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Sustainable building design requires the application of natural ventilation, which has three prerequisites–a comfortable *outdoor* air temperature (20–25°C), unpolluted *outdoor* air and a low-noise environment (under 55 dBA). There are many locations where the first two prerequisites are met, but the environmental noise makes applying natural ventilation challenging. This research developed an auxiliary noise-reducing vent for windows (NRVW) that allows daylight and *outdoor* air to enter the room while reducing the penetration of *outdoor* noise. As an auxiliary, the NRVW is intended to be integrated into any (double) glazed windows with a *sound* transmission class (STC) above 25 dBA. The research used physical model experiments to measure the NRVW's STC and *outdoor*–indoor transmission class (OITC). Digital model simulation was used to measure the NRVW's ventilation performance. It was found that NRVW has a rating of STC 18 and OITC 16, and it generates indoor air velocity of 0.015–0.086 m/s in the occupants' zone for an *outdoor* wind speed of 1–5 m/s. Modification of the room interior increases the indoor air velocity to 0.147 m/s. Computer simulation also demonstrated that, with an occupant inside the room, an *outdoor* air temperature of 25°C and *outdoor* wind speed of 1–5 m/s maintained the indoor air temperature at 25.309–25.258°C, indicating that the resulting airflow could keep the indoor air temperature low. Although the NRVW was originally developed for warm–humid climate conditions, it is also applicable for other climates if the required *outdoor* conditions are met.

Review

125	**Smart Animal Agriculture: Application of Real-Time Sensors to Improve Animal Well-Being and Production**	Halachmi I.,Guarino M.,Bewley J.,Pastell M.	Annual Review of Animal Biosciences, 7	2019	329

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	Document title	Authors	Source	Year	Citations

Consumption of animal products such as meat, milk, and eggs in first-world countries has leveled off, but it is rising precipitously in developing countries. Agriculture will have to increase its output to meet demand, opening the door to increased automation and technological innovation; intensified, sustainable farming; and precision livestock farming (PLF) applications. Early indicators of medical problems, which use sensors to alert cattle farmers early concerning individual animals that need special care, are proliferating. Wearable technologies dominate the market. In less-value-per-animal systems like sheep, goat, pig, poultry, and fish, one sensor, like a camera or robot per herd/flock/school, rather than one sensor per animal, will become common. PLF sensors generate huge amounts of data, and many actors benefit from PLF data. No standards currently exist for sharing sensor-generated data, limiting the use of commercial sensors. Technologies providing accurate data can enhance a well-managed farm. Development of methods to turn the data into actionable solutions is critical.

Article

☐ 126 **A modified approach to estimating thermodynamic impact on buildings: A case study of poor urban setting in Lagos state** — Emetere M.E. — Journal of Engineering Design and Technology, 17(1) — 2019 — 4

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Purpose: Managing the urban housing plan of a very fast-growing city may be difficult if the scientific input, i.e. *thermodynamic* architecture and the climate change challenges, is not factored into its initial framework. Recent building plan in some parts of a growing city located in a developing country was adopted for the purpose of this research. The purpose of this study is to investigate the impact of poor urban planning on humans. Design/methodology/approach: The reverberation time analysis was carried using the Ecotect software. In total, 15-year surface temperature data were obtained (1999-2013) from the Global Land Data Assimilation System. *Thermal* distributions were calculated using beta probability and Gaussian distribution. Also, the parametric study of the solar constant was accomplished using possible mathematical outcomes. Findings: It was discovered that irrespective of the fabrics of building, air properties and materials within a building, the total heat and *sound* absorptions are high for the life form. Necessary recommendations were made for further study. Research limitations/implications: Only the *outdoor* impact was calculated. Practical implications: There should be more proactive measures by the urban planning authorities. Social implications: There would be wide spread of diseases and very low *thermal* comfort. Originality/value: This paper illustrates on the most ignored parameter in environmental architecture.

Conference Paper

☐ 127 **Simulation of noise propagation of outdoor HVAC/R unit in surrounding space** — Poysat P.,Robinet B., Lenaerts S. — Inter Noise 2019 Madrid 48th International Congress and Exhibition on Noise Control Engineering — 2019 — 3

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The proposed paper will describe how acoustic simulation technique will improve the quality of installation of *outdoor* HVAC units and its acceptance. The HVAC/R *outdoor* units are *thermodynamic* machines having 2 major sources of noise, a refrigerant compressor and a fan to move the air and make the *thermal* exchange. The installation of such equipment in urban area becomes more challenging due to restricting regulation and neighborhood acceptance. Furthermore, a modification of the installation after commissioning is not easy and certainly costly. A system manufacturer can now use acoustic solver to predict the *sound* propagation of a unit, anticipate various configuration of field installation and provide recommendations to contractor. The objective is to minimize the claims for noise and improve the quality of installation and acceptance of HVAC *outdoor* units.

Conference Paper

☐ 128 **Atmospheric propagation of aircraft acoustic signature from high altitude** — Ghinet S.,Price A., Daigle G.A.,... Grewal A., Wickramasinghe V. — Inter Noise 2019 Madrid 48th International — 2019 — 3

	Document title	Authors	Source	Year	Citations
			Congress and Exhibition on Noise Control Engineering		

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A large number of computational approaches for acoustic atmospheric propagation have been developed and extensively validated during the past decades. Generally, the model of choice to predict *outdoor sound* propagation is the Parabolic Equation (PE) because it can simultaneously account for all the factors that can affect the propagation. These factors include geometrical spreading, atmospheric absorption, ground effects, refraction and turbulence. However, in the case of a high-altitude aircraft flying overhead, a microphone near the ground would be located outside the solid angle of validity of the PE. In this paper an impedance plane formulation that does not account for refraction and a Fast Field Program (FFP) that does not account for turbulence were implemented to predict the acoustic spectra of an aircraft propagating through the atmosphere at various altitudes, distances and angles. The paper aims at presenting the two theoretical approaches, the numerical and testing validation results and discussing the influence of various factors on the aircraft *sound* propagation from high altitude. In particular it will be shown that refraction is negligible, but that meteorological measurements made only at ground level are insufficient to estimate the atmospheric absorption.

Conference Paper

☐ 129	**Measurement of the sound transmission loss of rain screen wall assemblies**	Connelly M.,Stehling E., Tamanna O.,Mahn J.	Proceedings of the 26th International Congress on Sound and Vibration Icsv 2019	2019	0

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As part of an initiative to adapt building facades for the future weather expected in British Columbia due to climate change, the *sound* insulation of rainscreen wall assemblies with superior moisture and *thermal* performance targets was investigated. A hybrid field-test method was used to screen for the critical material and design variables which affect the *sound* transmission loss of rainscreen wall assemblies. The hybrid field-test method used an *outdoor* source as per ASTM E966, the indoor *sound* levels were measured in a semi-anechoic environment as per ISO 15186. The prerequisite to evaluating the hybrid field test method was a specific investigation on the inclusion of the rainscreen drainage and ventilation cladding gaps in the test specimens, a deviation from the standard practice of testing a sealed wall assembly. A total of 57 wall assemblies were screened at the hybrid field lab and a further eleven test were completed using the ASTM E-90 test method. While the results of the hybrid field-test method did not align with ASTM E90 test results, the hybrid field-test method was shown to be viable as a quick and lower cost means of obtaining results for the initial screening of designs. The results of the testing demonstrated that the use of split-insulation in the rainscreen cavity with increased cladding mass increased transmission loss

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Conference Paper • *Open access*

☐ 130	**Self-Interference Channel Measurements for In-Band Full-Duplex Street-Level Backhaul Relays at 70 GHz**	Haneda K.,Jarvelainen J., Karttunen A.,Putkonen J.	IEEE International Symposium on Personal Indoor and Mobile Radio Communications PIMRC , 2018-September, 8580687	2018	16

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This paper reports measurements of self-interference (SI) channels in an in-band full-duplex (IBFD) relay scenario for millimeter-wave street-level backhauling. The street-level backhauling suffers from clutters surrounding the relay devices and hence SI originating from the environment may be dominant. An ultrawideband channel *sounder* operating at 70 GHz carrier frequency is used to measure the SI channels in a chamber, indoor window-side, indoor and *outdoor* corner scenarios. The transmit and receive antennas are waveguide-fed lens antennas. The

	Document title	Authors	Source	Year	Citations

measurements showed that cross-polarized feeding of waveguides to the lenses provided higher isolation than co-polarized feeding. In the corner scenario, observed minimum isolation level is 85 dB in the frequency domain, and 100 dB in the delay domain. The results confirmed that isolation is not limited by the direct coupling because of the high directivity of the lens antenna, but by the environmental coupling. Further link budget analysis for *outdoor*-to-indoor backhauling showed that the IBFD relay with practical transmit power is capable of enhancing the signal-to-noise ratio and hence extending the coverage, as SI at the relay receiver can be almost the same level as *thermal* noise.

Article

131 **Identification of urban park activity intensity at different thermal environments and visible sky by using sound levels** — Tsai K.-T.,Lin Y.-H. — International Journal of Biometeorology , 62(11) — 2018 — 20

Hide abstract Find It @ UCL View at Publisher Related documents

An efficient urban environmental design considers appropriate *thermal* comfort condition, shaded space, and activity intensity. Physiologically equivalent temperature (PET) and visible sky, i.e., the sky view factor (SVF), are usually used as indicators to determine the *outdoor thermal* comfort and amount of shaded space, respectively. The activity intensity in urban parks, which is dependent on culture and micrometeorological conditions, was represented inappropriately by attendance density in Taiwan. To optimize the park design and improve the park utilization rate in Taiwan, several environmental factors such as *sound* pressure levels and numbers of park visitors were measured, and PET values and SVF values were calculated from primary micrometeorological data such as wind speed and globe temperature. This study proposed equivalent continuous *sound* pressure level (Leq) as a novel indicator to represent park activity intensity and investigated the correlation between Leq and SVF at different PET values. Leq was more appropriate than was attendance density in representing the park activity intensity in Taiwan. In addition, Leq was highly negatively correlated with SVF when visitors felt that the *outdoor thermal* comfort condition was hot or very hot. In other words, a lower degree of shading in the park resulted in lower activity intensity. Park visitors tended to engage in activities in the shaded regions because of more favorable *thermal* comfort conditions (i.e., neutral PET). The established quantitative relationships among Leq, PET, and SVF can serve as a reference for park planning.

Article

132 **Open windows for natural airflow and environmental noise reduction** — Mediastika C.E., Kristanto L.,Anggono J., Suhedi F.,Purwaningsih H. — Architectural Science Review , 61(5) — 2018 — 12

Hide abstract Find It @ UCL View at Publisher Related documents

For buildings in tropical climates, the use of open windows for natural ventilation can not only provide low cost and low energy comfort but also provide *thermal* delight for occupants. However open windows let in environmental noise. The size and location of windows in walls are key but this study set out to determine whether there are any window forms that can effectively reduce the level of *sound* ingress into a building. A top-hung window was chosen for this study looking at the dimensions of the window opening and its orientation in relation to the environmental noise source. The top-hung form was selected for its potential to balance the functions of allowing airflow while potentially blocking and reducing noise levels with its window pane angle. The window pane was tested in a laboratory at three opening angles: 0° (closed), 5°, and 10° to let the *outdoor* air in. The angles were also tested in three different orientations in relation to the noise source position: perpendicular, sideways 60°, and sideways 90°. The test was conducted at 1/3 octave band frequency as specified by ASTM E90-09 to obtain the transmission loss, then ASTM E1332-90 was referred to calculate the *outdoor*-indoor transmission class (OITC) of the specimens. The study revealed that window orientation and extent of the openings and window pane angle have little effect on noise reduction. The paper concludes with a discussion of how higher levels of natural ventilation can be achieved, particularly in noisy urban areas. The top-hung window, once open, barely blocks environmental noise. However, when the window was closed, the perpendicular orientation offered more noise reduction when compared windows placed sideways to the noise source. The adjustable pane-angle of a top-hung window placed perpendicular to the airflow, and thus the noise source, seemed to have the most potential to balance the functions of allowing airflow when opened and reducing significant noise when closed. Nonetheless, an open window that through its design alone can significantly reduce the ingress of ambient noise into a building is still an issue.

	Document title	Authors	Source	Year	Citations
133	Article **Acoustic inversion for Monin-Obukhov similarity parameters from wind noise in a convective boundary layer**	Hart C.R.,Nykaza E.T., White M.J.	Journal of the Acoustical Society of America, 144(3)	2018	3

Hide abstract Find It @ UCL View at Publisher Related documents

The prediction accuracy of *outdoor sound* is in large part limited by uncertainties in the state of the atmosphere. These uncertainties can potentially be reduced by inferring scaling parameters of the atmospheric surface layer from wind noise. Screened microphones sense wind noise as a result of mean atmospheric flow, turbulent eddy interaction with the windscreen, and pressure fluctuations within the turbulent flow. Under conditions of terrain homogeneity and atmospheric quasi-steadiness, the Monin-Obukhov similarity theory (MOST) states that only a handful of parameters governs the dynamics of the atmospheric surface layer. This study explores the relationships of atmospheric similarity parameters to the acoustic spectrum of wind noise in a convective boundary layer. Ambient noise data collected in a high desert during a 2007 long-range *sound* propagation experiment are analyzed for the purposes of establishing a nondimensional empirical relationship between the acoustic power spectrum and MOST parameters. Furthermore, this paper examines the consequences of inferring surface-layer scaling parameters with different parameter priors. This study shows that, for minimizing the variance in the inversion, the most important parameter to constrain is the Obukhov length.

	Document title	Authors	Source	Year	Citations
134	Article • *Open access* **Predicting integrated thermal and acoustic performance in naturally ventilated high-rise buildings using CFD and FEM simulation**	Yu X.,Zhang Q.,Kang J., Cui F.	Building Simulation, 11(3)	2018	21

Hide abstract Find It @ UCL View at Publisher Related documents

The study of ventilation windows for both natural ventilation and noise mitigation has drawn significant attention recently. This paper presents the numerical approaches to analyse the integrated *thermal* and acoustical performance of ventilation windows, for a residential building in tropical climate which employs double-layer noise mitigation window for natural ventilation. Given a set of *outdoor* wind conditions, the distributions of indoor flow and temperature fields are simulated using Computational Fluid Dynamics (CFD) model. The *thermal* comfort is evaluated using statistical Predicted Mean Vote (PMV) method. For the acoustic performance, noise radiation from road traffic is assumed as the noise source, and the *sound* insulation of building façade is simulated using Finite Element Method (FEM). From the simulation results, it is found that the *thermal* satisfaction response is closely related to the inlet wind temperature and speed, and the window opening size greatly affects the ventilation performance. From the case study in Singapore, during certain season, day/night time and with sufficient wind flow, the ventilation window can provide enough fresh air, maintain adequate *thermal* comfort and quiet acoustic environment for the occupants. The numerical approaches presented in this paper are applicable to general window design studies, and the simulation findings can be incorporated into green building planning. The advantages of using simulation approaches are highlighted and their limitations are discussed.

	Document title	Authors	Source	Year	Citations
135	Article • *Open access* **An acoustic thermometer for air refractive index estimation in long distance interferometric measurements**	Pisani M.,Astrua M., Zucco M.	Metrologia, 55(1)	2018	14

Hide abstract Find It @ UCL View at Publisher Related documents

We present a method to measure the temperature along the path of an optical interferometer based on the propagation of acoustic waves. It exploits the high sensitivity of the speed of *sound* to air temperature. In particular, it takes advantage of a technique where the generation of acoustic waves is synchronous with the amplitude modulation of a laser source. A photodetector converts the laser light into an electronic signal used as a reference, while the incoming acoustic waves are focused on a microphone and generate the measuring signal. Under this condition, the phase difference between the two signals substantially depends on the temperature of the air volume interposed between the sources and the receivers. A comparison with traditional temperature sensors highlighted the

	Document title	Authors	Source	Year	Citations

limit of the latter in the case of fast temperature variations and the advantage of a measurement integrated along the optical path instead of a sampling measurement. The capability of the acoustic method to compensate for the interferometric distance measurements due to air temperature variations has been demonstrated to the level of 0.1 °C corresponding to 10^{-7} on the refractive index of air. We applied the method indoor for distances up to 27 m, *outdoor* at 78 m and finally tested the acoustic *thermometer* over a distance of 182 m.

Conference Paper

136 **Outdoor comfort simulation of complex architectural designs a review of simulation tools from the designer perspective** — Naboni E.,Silvia C., Meloni M.,Scartezzini J.-L. — ASHRAE and Ibpsa Usa Building Simulation Conference — 2018 — 4

Hide abstract Find It @ UCL View at Publisher Related documents

The expanding design interest in *outdoor* comfort design is today supported by a few software tools. Given their different user-friendliness, modelling environments and simulation engines, a rational inquiry is how they are integrated into the digital design flow of architects and urban planners, which may include complex forms developments. Preliminary work is conducted to select architect-friendly tools that support the analysis of urban microclimate. A minimum set of criteria (amongst them the use of 3D models, the capability of calculating Mean Radiant Temperature and visual graphical outputs) led to the pre-selection of CitySim Pro, ENVI-met, Autodesk *Thermal* CFD, Grasshopper plug-ins Honeybee / Ladybug. The complex modelling experiment is conducted by simulating the *outdoor thermal* comfort of the space underneath the Rolex Learning Centre in Lausanne with each of the tools above. The paper describes and compares the principles, the procedures and the resources needed to prepare *sound* models that reduce the time of computation without compromising the quality of results. The potential applicability of tools in design is finally discussed from a user's point of view. The tools' capability of creating a 3D complex geometric model, or of importing one from a typical architectural tool, such as Rhino, is studied. Furthermore, it is debated how models can be used in the broader digital environment.

Conference Paper • *Open access*

137 **Comparative analysis of thermal and acoustic performance of composites made from wood fibres, recycled rubber and ABS** — Georgescu S.V., Cosereanu C. — Iop Conference Series Materials Science and Engineering, 399(1), 012019 — 2018 — 2

Hide abstract Find It @ UCL View at Publisher Related documents

The paper investigates the *thermal* and acoustic properties of composites made from wood fibres (WF) and recycled rubber (R) crumbs and acrylonitrile butadiene styrene (ABS) shavings resulted in the particleboards' edge banding. Panels with a target density of 300 kg/m^3 were manufactured for testing the *thermal* conductivity coefficient and *sound* absorption coefficient. Mixed panels WF:ABS, R:ABS with participation rates (in %) of 10:90, 20:80, 30:70; 40:60; 50:50 and WF:R:ABS with participation rates (in %) of 5:5:90, 10:10:80, 15:15:70; 20:20:60; 25:25:50 were investigated in this paper. The experiment simulated the indoor and *outdoor* temperature conditions for the winter and summer seasons, namely 20 °C for indoor and -10 °C to 35 °C for the *outdoor* temperatures. The results show that, with the increase of ABS share, the *thermal* performance of the panel increases. The presence of WF in the composition has a good influence on the *thermal* performance of the panels, whilst the presence of rubber brings a better acoustic performance of the composites. The analysis of the results show that the *thermal* conductivity coefficient (λ) experimentally determined depends on the *outdoor* temperature and recorded the best value for the composite WF-ABS (0.0434 $Wm^{-1}K^{-1}$) followed by WF-R-ABS (0.0460 $Wm^{-1}K^{-1}$) and R-ABS (0.0477 $Wm^{-1}K^{-1}$). Maximum values were recorded for R-ABS structure at a temperature of 35 °C (0.0573 $Wm^{-1}K^{-1}$), but this structure recorded the best *sound* absorption coefficient (0.87).

Article • *Open access*

138 **Design and development of a low-cost, portable monitoring device for indoor environment quality** — Tiele A.,Esfahani S., Covington J. — Journal of Sensors, 2018, 5353816 — 2018 — 96

	Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

This article describes the design and development of a low-cost, portable monitoring system for indoor environment quality (IEQ). IEQ is a holistic concept that encompasses elements of indoor air quality (IAQ), indoor lighting quality (ILQ), acoustic comfort, and *thermal* comfort (temperature and relative humidity). The unit is intended for the monitoring of temperature, humidity, PM2.5, PM10, total VOCs (x3), CO_2, CO, illuminance, and *sound* levels. Experiments were conducted in various environments, including a typical indoor working environment and *outdoor* pollution, to evaluate the unit's potential to monitor IEQ parameters. The developed system was successfully able to monitor parameter variations, based on specific events. A custom IEQ index was devised to rate the parameter readings with a simple scoring system to calculate an overall IEQ percentage. The advantages of the proposed system, with respect to commercial units, is associated with better customisation and flexibility to implement a variety of low-cost sensors. Moreover, low-cost sensor modules reduce the overall cost to provide a comprehensive, portable, and real-time monitoring solution. This development facilities researchers and interested enthusiasts to become engaged and proactive in participating in the study, management, and improvement of IEQ.

Conference Paper

☐ 139 **Acoustic performance test assessment on sound generated by rainfall on multilayer-etfe-cushion systems**

Chivante M.R.P.,Toso S., Moritz K.

8th International Conference on Textile Composites and Inflatable Structures Structural Membranes 2017

2018

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Ethylene-tetra-fluor-ethylene (ETFE) membrane structure presents itself as an innovative construction element, taking advantage of its flexibility, lightweight, transparency and *thermal* insulation properties. However, for large enclosed areas, concerns exist related to the *sound* insulation characteristic of the system, especially related to the internal noise levels generated during rainfall conditions. Therefore, aiming to both assess the effect of rain noise on these structural elements, as well as generating a clear picture to this condition to designers and clients, Taiyo Europe promoted a measurement campaign on a predefined set of pneumatic ETFE roof elements. The set of pneumatic cushions investigated was produced from the same 250-micron thick ETFE foil. The number of layers, number of air chambers and the level of the pressure inside the cushion was variable, following a carefully predefined test plan. The elements were tested with and without an additional rain suppressor fabric mesh on top of the specimen. The tested elements were installed on a special *outdoor* structure designed and built for this purpose. This work will describe the tests performed, as well as an assessment of the results achieved and their impact on the design of enclosed buildings cladded with ETFE pneumatic structures.

Conference Paper

☐ 140 **Building entry loss for traditional and thermally efficient houses between 0.4 and 73 GHz**

Towers J.L.,Salous S., Raimundo X.,Cheema A.

Iet Conference Publications, 2018(CP741)

2018

3

Hide abstract Find It @ UCL View at Publisher Related documents

This paper presents results of building entry loss measurements into a *thermally* efficient building and a traditional building between 0.4 and 73 GHz using two custom designed channel *sounders* developed at Durham University. Comparative measurements indicate that opening external windows reduces the penetration loss by 10 dB in the millimeter wave bands between 25-73 GHz. A difference of ~2-16 dB between the two types of build is measured across the entire frequency range.

Article

☐ 141 **The evolution of human sleep: Technological and cultural innovation associated with sleep-wake regulation among Hadza hunter-gatherers**

Samson D.R., Crittenden A.N., Mabulla I.A., Mabulla A.Z.P.

Journal of Human Evolution , 113

2017

36

	Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

Sleep is necessary for the survival of all mammalian life. In humans, recent investigations have generated critical data on the relationship between sleep and ecology in small-scale societies. Here, we report the technological and social strategies used to alter sleep environments and influence sleep duration and quality among a population of hunter-gatherers, the Hadza of Tanzania. Specifically, we investigated the effects that grass huts, *sound* levels, and fire had on sleep. We quantitatively compared *thermal* stress in *outdoor* environments to that found inside grass hut domiciles to test whether the huts function as *thermoregulated* microhabitats during the rainy season. Using physiological equivalent temperature (PET), we found that the grass huts provide sleep sites with less overall variation in *thermal* stress relative to outside baseline environments. We also investigated ambient acoustic measures of nighttime environments and found that *sound* significantly covaried with sleep-wake activity, with greater *sound* levels associating with less sleep. Finally, after controlling for ecological variables previously shown to influence sleep in this population, fire was shown to neither facilitate nor discourage sleep expression. Insofar as data among contemporary sub-tropical foragers can inform our understanding of past lifeways, we interpret our findings as suggesting that after the transition to full time terrestriality, it is likely that early Homo would have had novel opportunities to manipulate its environments in ways that could have significantly improved sleep quality. We further conclude that control over sleep environment would have been essential for migration to higher latitudes away from equatorial Africa.

Article

☐ 142 **Frame effects on outdoor–indoor transmission class of fixed and open glass windows** Mediastika C.,Kristanto L., Anggono J.,Suhedi F., Purwaningsih H. Advanced Science Letters, 23(7) 2017 5

Hide abstract Find It @ UCL View at Publisher Related documents

The effect of a frame around glass fixed windows on *outdoor*-indoor transmission class (OITC) was studied here. Three frame materials were selected, i.e., timber, Aluminum and unplasticized Poly Vinyl Chloride (uPVC). The use of real open window (to differentiate it from partially open double layered window) for natural ventilation was also studied. The top hung style was selected due to common usage and the possibility of noise blockage by the shutter. Laboratory test complies with ASTM E-90 was employed. The study showed that weight and density, which play significant roles in *sound* insulation quality of a material was not borne out for the window frame. Here, timber as the heaviest material gave lowest OITC and transmission loss (TL) contour. This was due to the absence of sealant and rubber strips which are unusual for timber caused by large *thermal* expansion coefficient. Top hung openings of 5° and 10° showed similar transmission loss contours, but the OITC of the 5° open window was slightly higher than the 10°.

Article • *Open access*

☐ 143 **Meteorological effects on the noise reducing performance of a low parallel wall structure** Van Renterghem T., Taherzadeh S.,Hornikx M., Attenborough K. Applied Acoustics, 121 2017 5

Hide abstract Find It @ UCL View at Publisher Related documents

Numerical calculations, scale model experiments and real-life implementations have shown that the insertion of a closely spaced array of low parallel walls of finite dimension beside a road is potentially useful for road traffic noise abatement. However, previous studies did not consider atmospheric effects. In this work, numerical techniques have been used to predict the *sound* reduction provided by a low parallel wall structure, subject to wind and temperature related atmospheric effects. Three full-wave prediction schemes show very good agreement when looking at the insertion loss of a low 6 m wide parallel wall structure, consisting of 24 regularly spaced 0.2-m high rigid walls. Meteorological effects are predicted not to deteriorate the insertion loss (relative to rigid flat ground) of the parallel wall array in the low frequency range. However, at high *sound* frequencies the insertion loss is strongly reduced by downward refraction at a distance of 50 m in case of strong wind. Consequently, overall A-weighted road traffic noise insertion loss will be significantly lower during wind episodes. Although weak turbulence does not alter the energy time-averaged insertion losses, strong turbulence reduces the noise shielding in the high frequency range also. As with conventional noise walls, when considering use of low parallel wall structures for noise reduction outdoors, even at short distances, atmospheric effects should be considered.

	Document title	Authors	Source	Year	Citations
	Conference Paper				
144	**Circadian House as a vision for healthy and human-centric building design**	Roy N.,Foldbjerg P., Andersen P.A., Christoffersen J.	Proceedings of 33rd PLEA International Conference Design to Thrive PLEA 2017, 1	2017	0

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Much attention on sustainable buildings has been put on energy aspects. However, health is the most precious resource we have, and energy is only one aspect of sustainability. A primary goal for sustainability should be to sustain good human health and provide healthy living environments. This was the starting point for a series of workshops with international experts. The result is a vision to realize healthy homes that support the different biological needs of their occupants with a focus on the entrainment of circadian rhythms, sleep-wake cycles, light exposure, *thermal* comfort, air quality, and contact to *outdoor* environments. Three key principles were identified: Live in balance with nature, Adaptability and Sensibility. Furthermore, those principles are supported by ten key factors including: Variation, Stimulation/absence of stimulation, *Outdoor*/indoor relation, Light/darkness, Electrical lighting, Cool/warm, Silence/*sounds*, Rest/activity, Control and Flexibility related to seasons. The principles and guidelines presented here can be used to guide and improve the design of residential buildings of all types, including apartment buildings, and are applicable to both new and existing dwellings.

	Article				
145	**Using the ICF's environmental factors framework to develop an item bank measuring built and natural environmental features affecting persons with disabilities**	Heinemann A.W.,Lai J.-S., Wong A.,... Miskovic A., Gray D.B.	Quality of Life Research , 25(11)	2016	23

Hide abstract Find It @ UCL View at Publisher Related documents

Objectives: To develop a measure of natural environment and human-made change features (Chapter 2 of the international classification of functioning, disability, and health) and evaluate the influence of perceived barriers on health-related quality of life. Methods: A sample of 570 adults with stroke, spinal cord injury, and traumatic brain injury residing in community settings reported their functioning in home, *outdoor*, and community settings (mean age = 47.0 years, SD = 16.1). They rated 18 items with a 5-point rating scale to describe the influence of barriers to moving around, seeing objects, hearing *sounds*, hearing conversations, feeling safe, and regulating temperature and indicated whether any difficulties were due to environmental features. We used Rasch analysis to identify misfitting items and evaluate differential item functioning (DIF) across impairment groups. We computed correlations between barriers and patient-reported outcomes measurement information system (PROMIS) social domain measures and community participation indicators (CPI) measures. Results: The 18 items demonstrated person reliability of.70, discriminating nearly three levels of barriers. All items fit the Rasch model; impairment-related DIF was negligible. Ceiling effects were negligible, but 25 % of the respondents were at the floor, indicating that they did not experience barriers that they attributed to the built and natural environment. As anticipated, barriers correlated moderately with PROMIS and CPI variables, suggesting that although this new item bank measures a construct that is related to participation and health-related quality of life, it also captures something unique. Known-groups validity was supported by wheelchair users reporting a higher level of barriers than did ambulatory respondents. Conclusions: Preliminary evidence supports the reliability and validity of this new measure of barriers to the built and natural environment. This measure allows investigators and clinicians to measure perceptions of the natural environment and human-made changes, providing information that can guide interventions to reduce barriers. Moderate relationships between barriers and PROMIS and CPI variables provide support for the measurement and theory of environmental influences on social health and participation.

	Conference Paper • *Open access*				
146	**Experimental study on HVAC sound parameters**	Bujoreanu C.,Benchea M.	Iop Conference Series Materials Science and Engineering, 147(1), 012051	2016	10

Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

HVAC system represent major source of buildings internal noise and therefore they are designed to provide a human acoustic comfort besides the *thermal* and air quality requirements. The paper experimentally investigates three types of commercial air handler units (AHU) with different ducts cross-section sizes and inlet-outlet configuration. The measurements are performed in an anechoic room. The measurements are carried out at different fan's speeds, ranging the power-charge from 30-100% while the duct air flow is slowly adjusted from full open to full closed, between 0-500 Pa. The *sound* pressure levels of the radiant units are rated using NR curves. Also, the supply and the *outdoor* ducts *sound* levels are compared in order to point the frequencies where the noise must be reduced. Third-octave band analysis of random noise of an air handling unit from a HVAC system is realized, using measurement procedures that agrees the requirements of the ISO 3744:2011 and ISO 5136:2010 standards. The comparatively results highlight the effects of the geometry, air flow pressure and power-charging dependencies upon the *sound* level. This is the start for a noise reduction strategy.

Conference Paper

147 **Higher sound classes possible also in areas with traffic noise, combining high façade sound insulation with fresh air supply and efficient thermal control** — Simmons C. — Proceedings of the Inter Noise 2016 45th International Congress and Exposition on Noise Control Engineering Towards A Quieter Future — 2016 — 0

Hide abstract Find It @ UCL View at Publisher Related documents

Sweden has recently introduced a new governmental regulation that allows dwellings close to noisy roads, railways and industries. However, the buildings must then provide at least one so called "quiet side", which is defined as a façade with $L_{pAeq,24h,freefield}$ < 55 dBA, corresponding to a L_{DEN} of about 58 dBA road traffic noise. However, looking close to the results of some questionnaire surveys, it becomes apparent that their results may be interpreted in a new way. The role of the façade *sound* insulation seems to have been systematically underestimated, as well as the effects of poor indoor air quality and over-temperature. If dwellings would provide high *sound* insulation in the façades as well as clean air supply and comfortable temperature at all times, occupants would not be forced to sleep with their windows opened and therefore not need a quiet side to the same extent. Thus, such buildings could provide satisfying indoor noise conditions also in areas with heavy traffic in all directions, where no quiet side can be realized. However, there should still be *outdoor* areas designed for social activities and rest that are easily accessed, e.g. on top of the roof or in a nearby space. These conclusions will be suggested to the working group ISO TC 43/SC 2/WG 29, working on a new international standard on *sound* classification of dwellings (ISO/NP 19488).

Conference Paper • *Open access*

148 **A Maternity Hospital in Shenyang Indoor Environment Performance Influence Factor Analysis** — Yu L.,Chen S.,Jia J.,Ding R.,Wang S. — Procedia Engineering, 146 — 2016 — 2

Hide abstract Find It @ UCL View at Publisher Related documents

In recent years, the city air pollution level of $PM_{2.5}$, ozone, nitrogen oxides and sulfur dioxides in our country occupies first place. Indoor environmental problems caused by *outdoor* air pollution are increasingly significant. Indoor environment of hospital which is a special public building has special requirement. The feature of maternity hospital is crowded, heavy energy consumption, weakness of maternal and neonate, generate the sick and the patients need the better indoor environment quality. Therefore, this paper takes hospital building as research object by using questionnaire method, which analyzes the indoor comprehensive environmental performance of hospital, such as the air quality, the *thermal* environment, the light environment, and the acoustic environment. Results show that most of the patients is satisfied with the indoor comprehensive environmental performance of hospital, and only a small part is dissatisfied with the *sound* environment and the *outdoor* vision. This paper provides a research base for establishing a complete evaluation system and method of comprehensive indoor environmental performance of maternity

	Document title	Authors	Source	Year	Citations

hospital, which will help in building environmental monitoring systems and improving the level of public participation in the construction of hospital management.

Article • *Open access*

149 **A survey on annual change of window opening in detached houses a method to monitor open-width of sliding window and the annual characteristics of open-width**

Hayashi M.,Sugawara M., Honma Y.,Kurabuchi T.

Journal of Environmental Engineering Japan , 81(719)

2016

1

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The aim of this study is to estimate the influence of indoor air quality upon the dweller's health in detached houses. The indoor air quality influences the dwellers health continuously for log times. Especially the long-term exposure of chemical compounds increase the risk of multiple chemical sensitivity. Therefore it is very important to know the annual characteristics of indoor air quality. The annual change of indoor quality depends on the annual change of emission and that of ventilation. The ventilation rate depends on not only the ventilation system but also the dweller's habit of operating windows. The window operation is influenced by not only indoor conditions but also the dwellers living style, the environment of circumstance and the other factors. The sliding windows were used in common detached houses. The dwellers can adjust the open wide of sliding window considering any factors, for example indoor *thermal* condition, air quality *sound*, and these *outdoor* conditions. Therefore the state of the open width is thought to be varied and the prediction is not easy. In order to know the annual characteristics of indoor air quality and to predict the long term exposure of pollutant, the following studies are carried out. An instrument to measure the open-width of sliding window was developed and the open-widths were measured in six detached houses through a year. The results are the followings. 1. An instrument using mechanical system can measure open width of sliding window continuously through a year. 2. The open-width depends on not only indoor temperature but also the daily dweller's schedule ant the other factors. 3. The annual reaction of open-width to the indoor temperature is behind when the room is not cooled normally. 4. When the indoor temperature is high in summer, the room is cooled normally and the open-width is large when the room is not cooled. 5. When the indoor temperature is not high in summer, the room is not cooled normally and the windows open for long time and open-width is operated from middle to large when the room is not cooled. In this study the basic annual characteristics of operating windows became clear. The survey on the window opening in more houses, the modeling of window opening and the prediction of annual indoor air quality are necessary in future.

Conference Paper

150 **Big brother for bees (3B) - Energy neutral platform for remote monitoring of beehive imagery and sound**

Edwards-Murphy F., Magno M.,O'Leary L., ... Whelan P.,Popovici E.M.

Proceedings 2015 6th IEEE International Workshop on Advances in Sensors and Interfaces Iwasi 2015, 7184943

2015

33

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Honey bees have played a major role in the history and development of humankind, in particular for nutrition and agriculture. The most important role of the western honey bee (Apis mellifera) is that of pollination. A large amount of crops consumed throughout the world today are pollinated by the activity of the honey bee. It is estimated that the total value of these crops stands at 155 billion euro annually. The goal of the work outlined in this paper was to use wireless sensor network technology to monitor a colony within the beehive with the aim of collecting image and audio data. These data allows the beekeeper to obtain a much more comprehensive view of the in-hive conditions, an indication of flight direction, as well as monitoring the hive outside of the traditional beekeeping times, i.e. during the night, poor weather, and winter months. This paper outlines the design of a fully autonomous beehive monitoring system which provided image and *sound* monitoring of the internal chambers of the hive, as well as a warning system for emergency events such as possible piping, dramatically increased hive activity, or physical damage to the hive. The final design included three wireless nodes: a digital infrared camera with processing capabilities for collecting imagery of the hive interior; an external *thermal* imaging camera node for monitoring the colony status and activity, and an accelerometer and a microphone connected to an off the shelf microcontroller node for processing. The

	Document title	Authors	Source	Year	Citations

system allows complex analysis and sensor fusion. Some scenarios based on *sound* processing, image collection, and accelerometers are presented. Power management was implemented which allowed the system to achieve energy neutrality in an *outdoor* deployment with a 525 x 345 mm solar panel.

Article

151 **Ventilation and condensation water failure in coupled windows - Experimental laboratory- and outdoor measurement about the ventilation and the condensation water failure in den window interspace of a coupled window | Hinterlüftung und Tauwasserausfall bei Verbundfenstern: Experimentelle Labor- und Freilanduntersuchungen zu Hinterlüftung und Tauwasserausfall im Fensterzwischenraum eines Verbundfensters Hinterlüftung und Tauwasserausfall bei Verbundfenstern**

Bauer M.,Baumler A., Bauer M.

Bauphysik, 37(4) 2015 0

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Coupled windows are always used when increased physical construction requirements are placed on the *sound* and heat protection in winter. This is achieved through the double skin window structure, with the additional benefit that a sunblind can be placed in the window space between the outer and inner window sash [1]. In practical use, condensation water occurs on different glazing types. This condensation water occurs especially on the inside of the outer glazing (Fig. 1, pos. 2) where it is least wanted. In the course of detailed studies on a coupled window of the WICONA brand, this transient condensation water occurrence is scientifically evaluated based on laboratory and field measurements. The monitoring program for the experimental determination of the condensation water occurrence is elaborated on the basis of a construction methodological model. For this purpose, the coupled window is divided into its elementary components and the material and energy flows accounted for. Through the combined approach of building physics and *thermodynamics* probes can be attached preferably to the surfaces where there is potential risk for condensation water occurrence. The metrological investigations show that the ventilation and the risk of condensation water on pos. 2 can be influenced mainly by placing opening sections at the base and head of the coupled window. Trough small opening cross sections, very little condensation water shall form on pos. 2. An increased air exchange between the window interspace and the environment can be detected with larger aperture cross-sections, which also increases the occurrence of condensation water.

Article

152 **Algorithms for crowd surveillance using passive acoustic sensors over a multimodal sensor network**

Agarwal R.,Kumar S., Hegde R.M.

IEEE Sensors Journal, 15(3), 6953090 2015 28

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Crowd detection and monitoring is an active area of research because of its significance in many areas, especially law enforcement. Various sensor modalities, such as infrared imaging, video feed, received signal strength indicator, Radio-frequency identification, GPS signals, and audio tones through mobiles have been used in earlier work. In this paper, a method that uses passive acoustic sensors in a multimodal sensor network for crowd monitoring is described. This multimodal system uses three modalities, namely, carbon dioxide level, *sound* intensity level, and received signal strength for crowd detection and monitoring. The first two modalities are sensed using passive sensors, whereas the last one is an active sensor. This combination makes the proposed algorithms energy efficient and computationally less complex. The proposed multimodal crowd monitoring algorithm requires an effective clustering method. Hence, three clustering algorithms that utilize temporal, spatial, and spatio-temporal information are also proposed. Subsequently, an algorithm that fuses the information in different modalities is also proposed for efficient crowd monitoring. Additional contributions of this paper are the development of attenuation, reverberation, and additivity models, using real sensor deployments. Both simulation and real field experiments are conducted to evaluate the performance of the proposed algorithms in indoor and *outdoor* spaces. The results of crowd detection

	Document title	Authors	Source	Year	Citations

and monitoring obtained from these methods are motivating enough to use the proposed method in real small-scale deployment scenarios.

Article • *Open access*

153 **Sound propagation over soft ground without and with crops and potential for surface transport noise attenuation**

Bashir I.,Taherzadeh S., Shin H.-C., Attenborough K.

Journal of the Acoustical Society of America, 137(1)

2015

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Growing demand on transportation, road, and railway networks has resulted in increased levels of annoyance from road traffic. Optimized use of green surfaces in combination with vegetation may be desirable as a method for reducing the noise impact of road traffic in urban and rural environments. *Sound* propagation over soft ground and through crops has been studied through *outdoor* measurements at short and medium ranges and through predictions. At lower frequencies, ground effect is dominant, and there is little or no attenuation due to crops. At higher frequencies above 3-4 kHz, the attenuation in crops is dominant. It was also found that the ground effects and the influence of crops can be treated independently and can be added to obtain the total effect. *Sound* attenuation by crops is the result of multiple scattering between the stems and leaves, loss of coherence, and viscous and *thermal* losses due to foliage. The major contribution is associated with viscous and *thermal* losses. A model for *sound* attenuation by vegetation is proposed. Insertion losses for a typical road traffic noise source have been calculated that result either by replacing hard ground with different types of acoustically soft ground or by growing crops along the road sides.

Article • *Open access*

154 **Modern housing retrofit: Assessment of upgrade packages to enerphit standard for 1940-1960 state houses in Auckland**

Leardini P.,Manfredini M.

Buildings, 5(1)

2015

9

Hide abstract Find It @ UCL View at Publisher Related documents

New Zealand state housing includes a significant portion of problematic buildings constructed after the public housing scheme launched in 1936. Most of these houses are still uninsulated, thus, cold, draughty, mouldy, and progressively decaying; however, as they are fundamental elements of the country's culture, society, and environment, and are built with good quality materials and *sound* construction, they are suitable candidates for effective energy upgrades. This paper presents findings of a study on problems and opportunities of retrofitting the state houses built between 1940 and 1960 in the Auckland region. It advocates strategic national policies and initiatives for retrofitting, based on more challenging performance thresholds. The research defines and virtually implements an incremental intervention strategy including different retrofit packages for a typical 1950s stand-alone house. Indoor and *outdoor* environmental parameters were monitored over a year, and data used to establish a base case for *thermal* simulation. The upgrade packages were then modelled to assess their impact on the house's *thermal* performance, comparing heating requirements and comfort of various insulation and ventilation options. The paper reports on effective ways of preserving the integrity of such a house, while improving its *thermal* performance to the EnerPHit standard, and discusses the benefits of introducing this holistic approach into New Zealand retrofit practice.

Conference Paper • *Open access*

155 **Marble ageing characterization by acoustic waves**

Boudani M.E., Wilkie-Chancellier N., Martinez L., ... Verges-Belmin V., Serfaty S.

Physics Procedia, 70

2015

6

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In cultural heritage, statue marble characterization by acoustic waves is a well-known non-destructive method. Such investigations through the statues by time of flight method (TOF) point out *sound* speeds decrease with ageing. However for *outdoor* stored statues as the ones in the gardens of Chateau de Versailles, ageing affects mainly the surface of the Carrara marble. The present paper proposes an experimental study of the marble acoustic properties

Document title	Authors	Source	Year	Citations

variations during accelerated laboratory ageing. The surface degradation of the marble is reproduced in laboratory for 29 mm thick marble samples by using heating/cooling *thermal* cycles on one face of a marble plate. Acoustic waves are generated by 1 MHz central frequency contact transducers excited by a voltage pulse placed on both sides of the plate. During the ageing and by using ad hoc transducers, the marble samples are characterized in transmission, along their volume by shear, compressional TOF measurements and along their surface by Rayleigh waves measurements. For Rayleigh waves, both TOF by transducers and laser vibrometry methods are used to detect the Rayleigh wave. The transmission measurements point out a deep decrease of the waves speeds in conjunction with a dramatic decrease of the maximum frequency transmitted. The marble acts as a low pass filter whose characteristic frequency cut decreases with ageing. This pattern occurs also for the Rayleigh wave surface measurements. The speed change in conjunction with the bandwidth translation is shown to be correlated to the material de-structuration during ageing. With a similar behavior but reversed in time, the same king of phenomena have been observed trough sol-gel materials during their structuration from liquid to solid state (Martinez, L. et all (2004). "Chirp-Z analysis for sol-gel transition monitoring". Ultrasonics, 42(1), 507-510.). A model is proposed to interpret the acoustical measurements.

Article

☐ 156 **PROMETHEUS: heterogeneous sensor database in support of research on human behavioral patterns in unrestricted environments** — Ntalampiras S.,Arsic D., Hofmann M., Andersson M.,Ganchev T. — Signal Image and Video Processing , 8(7) — 2014 — 6

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

The multi-modal multi-sensor PROMETHEUS database was created in support of research and development activities [PROMETHEUS (FP7-ICT-214901): http://www.prometheus-FP7.eu] aiming at the creation of a framework for monitoring and interpretation of human behaviors in unrestricted indoor and *outdoor* environments. The distinctiveness of the PROMETHEUS database comes from the unique sensor sets, used in the various recording scenarios, but also from the database design, which covers a range of real-world applications, correlated to smart-home automation and indoors/outdoors surveillance of public areas. Numerous single-person and multi-person scenarios, but also scenarios with interactions between groups of people, motivated by these applications were implemented with the help of skilled actors and supernumerary personnel. In these scenarios, the actors and personnel were instructed to implement a range of typical and atypical behaviors, and simulations of emergency and crisis situations. In summary, the database contains more than 4 h of synchronized recordings from heterogeneous sensors (an infrared motion detection sensor, *thermal* imaging cameras, overview/surveillance video cameras, close-view video cameras, a 3D camera, a stereoscopic camera, a general-purpose camcoder, microphone arrays, and motion capture equipment) collected in common setups, simulating smart-home environment, airport, and ATM security environment. Selected scenes of the database were annotated for the needs of human detection and tracking. The entire audio part of the database was annotated for the needs of *sound* event detection, *sound* source enumeration, emotion recognition, etc.

Article

☐ 157 **Description and quantification of uncertainty in outdoor sound propagation calculations** — Keith Wilson D.,Pettit C.L., Ostashev V.E., Vecherin S.N. — Journal of the Acoustical Society of America , 136(3) — 2014 — 29

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

The accuracy of *outdoor sound* propagation predictions is often limited by imperfect knowledge of the atmospheric and ground properties, and random environmental variations such as turbulence. This article describes the impact of such uncertainties, and how they can be efficiently addressed and quantified with stochastic sampling techniques such as Monte Carlo and Latin hypercube sampling (LHS). Extensions to these techniques, such as importance sampling based on simpler, more efficient propagation models, and adaptive importance sampling, are described. A relatively simple example problem involving the Lloyd's mirror effect for an elevated *sound* source in a homogeneous atmosphere is considered first, followed by a more complicated example involving near-ground *sound* propagation with refraction and scattering by turbulence. When uncertainties in the atmospheric and ground properties dominate, LHS with importance sampling is found to converge to an accurate estimate with the fewest samples.

	Document title	Authors	Source	Year	Citations

When random turbulent scattering dominates, the sampling method has little impact. A comprehensive computational approach is demonstrated that is both efficient and accurate, while simultaneously incorporating broadband sources, turbulent scattering, and uncertainty in the environmental properties.

Conference Review

158 **3rd International Conference on Civil Engineering and Transportation, ICCET 2013** | [No Authors Found] | Applied Mechanics and Materials, 507 | 2014 | 0

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The proceedings contain 175 papers. The special focus in this conference is on Civil Engineering and Transportation. The topics include: Enclosure exploration of tianjin wudadao district; study on internal layout of UK's day care unit for older people; study on site design of UK's day care unit for older people; study on the construction mode and quality monitoring of security housing in henan province; on the abstract metaphorical design in the transformation of traditional houses in Guanzhong region; analysis on modern residence design; study of architectural space efficacy based on the modern theory of validity; the application of digital technology in the international design competition of large sports center; an inquiry of teaching methods based on the psychological needs of comfort in small architectural space; architectural space evolution of railway passenger station; Himalayan water tower; London city hall and the values it promotes; on application of virtual space in architecture internal space; a brief discussion about effects of Chinese traditional culture on contemporary Chinese architectural design; interpretation of concerning comfort of office building design in cold regions; modern residential design; new changes in current architectural art teaching and corrective measures; strategies of three-dimensional greening design in building sectors; the acoustical design of conference room based on speech acoustic; the acoustic test report of three *outdoor* courtyard in Shenyang palace museum; the building energy conservation design responses of urban underground space; research on the light design for the enclosed target lane; research on indoor *thermal* environment of dai nationality wood dwellings; slits' effects on *sound* insulation for lightweight partitions; an investment-benefit assessing approach for intelligent buildings; comparing the core evaluation system for green building at home and broad; study on the green construction of current constraints and promotion measures in china; study on natural lighting modification of existing steel-structure plants in cold regions; a middleware of IoT-based smart home based on service; the development for internet of things smart home service operation support platform; a reproducible study on the recycling project of old industrial buildings; advances in damage of concrete due to freeze-thaw circles; experimental research on seismic performance of pre-damaged RC frame T-beam strengthened with sprayed BFRP; experimental research on compressive strength of adobe with cement; effect of curing time and $CaSO_4$ whisker content on properties of gypsum board; development and evaluation for microdamage of concrete under uniaxial compressive load; a mixed-mode fracture criterion for brittle materials; experimental study on *sound* absorption property of porous concrete pavement layer; chloride ion penetrability of HPFRCC after loading; study on optimal ratio of compound corrosion inhibitors for reinforcing steel; experimental study on the strength of basalt fiber confined concrete under impact loading; experimental study on frost resistance of concrete in seawater; research on prediction of carbon steel in SO_2 atmospheric corrosion using grey model; the contrast of concrete Brazilian and flattened Brazilian disc specimen under quasi static diametral compression tests; different replacement rate of recycled aggregate concrete research; various replacement rate experimental study on recycled concrete; mechanical properties of stone masonry mechanical properties; a measurement method of tensile strength for irregular small stones; corrosion resistance performance of ethylamines in fly ash blended cement concrete; dynamic response test of reinforced concrete slab under blast loading; hydration properties of steel slag under different curing temperatures; research on sand gradation to ultra strength grouting material properties; experimental study on flexural behavior of damaged reinforced concrete beams strengthened with CFRP; experimental study on flexural capacity of prestressed RC hollow slabs strengthened by CFRP; research on deformation index limits of RC shear walls based on material damage; experimental study for CFRP sheets reinforcing high-titaniferous blast furnace slag concrete beams; experimental research on fireproofing measure of slabs strengthened by carbon fiber sheet bonded with inorganic adhesive; influence of high-temperature curing on the properties of the concrete containing ground iron and steel slag; state-of-the-art report of recycled aggregate concrete and rapid repair concrete; research on the cement matrix of super toughness fiber concrete; laboratory investigation on the strength characteristics of cement-treated base; frost scaling resistance and improvement of concrete; dry-mixed antifreeze concretes design of mixing and performance of antifreeze; study the strength regularity of recycled masonry mortar produced with recycled ceramic tiles; experimental study on the mechanical performance of pumice powder cement soil; fabrication and

Document title	Authors	Source	Year	Citations

properties of new building materials by reutilization refractory materials; study on curing agent of chemical modification on dredged soil; compensation effect of expansive agent on shrinkage of self-compacting concrete; experiment study on the compressive strength of combined aggregate concrete; evaluation of green degree of concrete based on the gray relation; study on the mechanical properties of the latex-mycelium composite; the research on the numerical modeling of foam material; effect of different grain grading of manufactured-sands on properties of dry-mixed mortar; study on the performance of glazed hollow bead aerated concrete and its influence factors; preparation and properties of lime-based curing material for soil; finite element analysis for the crack self-repairing behavior of the concrete beam embedded shape memory alloy; design and study on integrated photovoltaic steel-structure industrial plants in cold regions; a study on the energy-saving evaluation index system of rural residential buildings; a study on the energy-saving evaluation index system of rural residential buildings; optimization of an invention based on a force multiplier mechanism for wave power generation; research on solar water heating system design of the high-rise residential buildings based on sunlight simulation; study on application of the integrated solar hot-water system to high-rise residential buildings in china; a study on integration design of solar hot-water systems in high-rise residential buildings; analyses of the suitability of ground-source systems in typical architecture in northern region; energy-saving technical measures of German low-energy buildings; study on energy efficiency design of self-insulation system with autoclaved aerated concrete block in cold regions; building energy audit and energy-saving analysis for a five-star hotel in Beijing; analysis to urban traffic jam from the perspective of city planning; about architectural aesthetics in the three gorges reservoir area planning; a study of protective exploitation of zhu des former residences tourism environment; technical system and application strategy of digital urban planning management; extraction of ningxia region new rural living environment to repair again; development study on cultural creative industrial parks of Suzhou central city; analysis of spatial evolution characteristics in songyuan main urban area based on space syntax; study on evaluation of rural human settlement in Changchun; discussion about the planning of settlements supporting education facilities in Jinzhou city; historical architectures renovation strategy under the guidance of the ecologic concept; on the protection and utilization of the famous historic and cultural cities, towns and villages in central plain economic region; study on low-carbon settlements spatial form design strategy - the case of northern china; study of fractal structure characteristics of urban system in northern Shaanxi; a strategy research on the development of Tianjin tuanbo new town west district; regenerative protection of urban historic conservation area amid cultural prosperity; thinking on redevelopment of old urban housing area in the viewing angle of peoples livelihood happiness; concept renewal about koguryo cultural heritage protection with "cultural routes" the study on ancient waterway of yuanshui river and its culture from the perspective of cultural routes; humanization design of landscape for commercial pedestrian street; discussion about the landscape planning in contemporary campuses from the traditional Chinese landscape architecture; study on the community environment design for the aged; thinking of village construction in central region under the context of labor migration; research on the experimental device of combined two-phase ABR and SBR; resent advances in phosphorus removal from waste water; design of the reclaimed water reuse project for recirculating cooling water in a *thermal* power plant; emerging technologies for phosphorus removal and recovery; kinetic and *thermodynamic* studies of adsorption of phosphate on steel-slag filter material; synthesis of spherical CuS photocatalyst and the research on photo-degradation activity of methylene blue; effect of hydraulic retention time on pollutant removal performance of biological contact oxidation process treating hospital wastewater; research on flocculation of protein-bearing wastewater by alga flocculant; evaluation of aquatic plants for water purification in the xixi wetland in Hangzhou, china; chlorine decay models and influencing factors; organics and its control in reclaimed water.

Conference Review

☐ 159 **2nd International Conference on Mechanical Engineering, Materials Science and Civil Engineering, ICMEMSCE 2013** | [No Authors Found] | Applied Mechanics and Materials, 470 | 2014 | 0

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The proceedings contain 231 papers. The special focus in this conference is on Mechanical Engineering, Materials Science and Civil Engineering. The topics include: A study on wear properties of GCV material with DLC coating; adsorption of cefradine onto powder and activated carbon prepared from orange peel; bright line defects of hot rolled plate for 2205 duplex stainless steel; catalytic ozonation with activated carbon for treatment of humic substances in water; effect of TiB_2 on damping capacity and dynamic Young's modulus of 2219 alloy; first-principles and experimental study on Cu doped SnO_2 structure; highly solution-processible red light-emitting conjugated

Document title	Authors	Source	Year	Citations

polymers based on benzothiadiazole; luminescence mechanism of $ZnWO_4$ nanopowder synthesized by microwave-assisted heating; preparation and characterization of aqueous polyurethane polyol used in waterbore two-component polyurethane coating; preparation and performance research of the PBC / PLA biodegradable film; rate-dependent constitutive model coupled with temperature softening for open-cell aluminum foam; separation test of ilmenite in Guizhou; separation test of oxide copper ore in Yunnan; structural formability of ABO3-type perovskite compounds; study on the growth quartz ampoule of $AgGaS_2$; study on a novel structure of terbium complex, $C_6H_{14}O_{19}Tb_2$; study on microstructural evolution of binary alloy crystal growth in directional solidification process; the research status of self-compensation lubricating composites at high temperature; utilization of pineapple leaf fiber as technical fibers; effect of temperature on friction and wear properties of the modified PTFE three-layer composites; internal defect measurement of bonding structure based on infrared *thermal* image; micro-experimentation research on compression sensitivity of carbon fiber concrete; the effect of ultrasound on oxidation desulfurization of diesel; tribological characteristics of Al 356 alloy reinforced with red mud; influence of ultrasonic vibration on metal plastic forming; a computable model for PCB assembly optimization based on polychromatic sets; numerical solution of temperature field for stress analysis of plate structures; DEM simulation study of crosscutting speed for the cutting header; numerical simulation analysis on stability of coal pillar of empty mine goaf in north of Shanxi province; test-bench design for fiber paying-off simulating; the finite element analysis of batching machine spindle based on ANSYS; dynamics simulation of *sounding* rocket water landing recovery with fluid-structure interaction method; nonlinear finite element analysis for galloping of iced quad-bundled conductor; time series modeling of earthquake ground motions using ARMA-GARCH models; simulation of influence of friction coefficient on extrusion of TC_4 alloy; *thermal* fluid-solid coupling numerical simulation in ultra-supercritical steam trap; analysis of *thermal* radiation models for large open pool fire on ship deck; the numerical simulation of blasting demolition on the solid pier; simulation and topological optimization of the lifting mechanism of a dump truck; study of vehicle longitudinal initiative avoidance collision system; computational models for functionally graded composites using the SMA as the metallic phase; application of three-parameter model in settlement calculation of immersed tube tunnel; a SFMEA method based on modeling and simulation by rhapsody; comparative analysis of static strength of traditional and bionic semi-trailer frame; coupled *thermo*-flowing 3D steady state model of friction stir welding process; study of the effects of transient load on envelope structure; temperature distribution in the molding process of the stator of PCP; study on the full state function of structural parameters for materials; visual design and computational analysis of the hydraulic support based on MATLAB; *thermal* analysis of lithium ion batteries using in HEV; nonlinear analysis of rotary sealing performance of rubber O-ring; the optimization of the beam of high-speed computerized embroidery machine; design method of the skull restoration based on user-centered; design of automatic production line for electrode defects inspection of Li-ion power battery; design on cleaning device for slag of narrow gap submerged arc welding; design of the disc cam profile with negative radius translating roller follower; evaluation of roundness error based on the position of the centers; parameters optimization of axial vibration drilling based on the exit burr; parametric design of six typical semi-*outdoor* reticulated domes; shield lining segments design research based on characteristic parameters; spiral roll forming machine design for metal rubber; flexible rope transmission device and preliminary study on its design; lightweight design for lower beam of NC isothermal hydraulic press; novel design of manual mechanical rotary table; study on dedicated test bed for the sealing performance of metal rubber seal; the application of reverse engineering in innovation design; design and analysis of a magnetic QZS vibration isolator; aerodynamic force of a cantilevered cylinder in uniform flow and turbulent boundary layer; analysis and application of the impact absorption behavior of the composite tube; experimental investigation of influence of normal pressure on rotational friction behavior; mechanical analysis and structural optimization to the training device for squatting; numerical investigation of static strength for tubular joints reinforced by inner plate; study on engine idling torque and speed ripple cancellation of HEV; vertical vibration characteristics analysis of stand rolls system in six-roll cold tandem mill; experimental study on the surface roughness of 40Cr by ultrasonic finishing machining; life prediction of two rotor steels under creep-fatigue interaction at elevated temperature; the analysis of processing factors for electro-arc machining; study on application of the mine anti-impact and energy-absorption device; a power optimization algorithm for ad hoc networks with variable rate control; a algorithm of path planning based on multiple mobile robots; forward kinematics of parallel robot based on Neuro-fuzzy system; improvement of production index for cell production by mobile robots; new structural design of coal mine rescue robot; time optimal path planning of palletizing robot; nonlinear model-based control for vacuum altitude simulator; research on the technology of two-axis synchronization control based on fuzzy PID; PLC-based stamp puncher; study on CCT curve and microstructure evolution of HRB500E; the monitoring system on energy consumption of device based on wireless network; the methods for increasing of the efficiency in the intelligent assembly cell; research on automatic winding

Document title	Authors	Source	Year	Citations

system of coreless package; leakage detection method for piping network based on BP neural network; performance analysis of LDPC coded SFH/BPSK in LFM jamming; research on customized indoor route recommendation; the model of load-shedding determined in power market; dynamical variation of Weierstrass-Mandelbrot function in higher dimensional space; vibration measurements of steel foot-bridges using mobile-phone; immune-based algorithm for power economic emission dispatch; application of the minimum cut sets of system fault based on OMA; mechanical properties of fiber reinforced self-compacting concrete; cheap mix design of the rock-filled concrete materials; research on separation of mixed copper-lead concentrate; fine aggregate interference on *thermal* stability of asphalt mixture; research on the mechanical properties and micro-morphology of the cement stabilized gravel with rubber particles; research on bonding strength of steel and concrete with different bonding interfaces; the dynamic response simulation analysis of U-shaped metro tunnel; predicting model of safe distance between tunnel and karst cave; stress analysis and strength checking for high pressure gas pipeline in the earthquake rupture area; seismic behavior of RC columns with corner focus longitudinal bars; fire resistance performance of high-strength concrete columns reinforced with pre-stressed wire ropes; reliability index study of asphalt pavement construction based on analytical hierarchy process; tunnel strength and excavation effect under different stress paths; review on aquifer structure theory in aquifer layered distribution area; nonlinear analysis of bearing capacity of reinforced concrete structures using strip method; analysis on consolidation and settlement characteristics of composite foundation with T-shaped bidirectional soil-cement deep mixing columns; collapse cause analysis and treatment research of west slope tunnel; analysis of stress deformation characteristics of composite geomembrane rock-fill dam with core wall; analytical study on seismic performance of switchboard with hybrid spring supports; supporting design and stability analysis of a subway station's main body tunneling; test research on strengthening in-service damaged truss with prestressed cable; the investigation of the hall loudness evaluation; *thermal* technical assessment of selected constructions of wooden houses; trenchless repair technology and application of urban sewer system; wind resistant design of low-rise steel buildings; nonlinear static pushover analysis for shear wall structures in SAP2000 program; advance of research of unbonded flexible risers.

Conference Paper

☐ 160 **Adaptive comfort and control protocols for natural ventilation** — Philip S.,Shameson A., Brown N.,Loisos G., Ubbelohde S. — 30th International PLEA Conference Sustainable Habitat for Developing Societies Choosing the Way Forward Proceedings, 1 — 2014 — 0

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Recently there has been a shift in international standards towards adaptive *thermal* comfort. ASHRAE had formally incorporated adaptive comfort into ASHRAE Standard 55-2010. In adaptive comfort, the range of indoor temperatures perceived as comfortable drifts upwards in warm weather and downwards in cooler weather. This is especially true where the occupants have control over their *thermal* environment (for example, with clothing or operable windows) and when occupants are aware of the variable *outdoor* conditions. This paper presents the ventilation control strategies used to provide adaptive comfort without air conditioning (compressive cooling) in a major retrofit for an office and classroom building on the University of Hawaii campus. The building retrofit is designed for cross-ventilation while taking care to acoustically isolate the spaces from each other and to dampen *sound* from outside. A supplementary fan system is designed and configured to complement the cross-ventilation strategy. The fan system is controlled to actively draw just enough air to provide adaptive comfort conditions when wind is calm. If the wind is sufficient, the controls sense the speed and direction and shut down the fans allowing cross ventilation to do the cooling. The fan controls sense the conditions inside, conditions outside the building, and weather conditions in the recent past.

Article • *Open access*

☐ 161 **On antenna design objectives and the channel capacity of MIMO handsets** — Nielsen J.O.,Yanakiev B., Caporal Del Barrio S., Pedersen G.F. — IEEE Transactions on Antennas and Propagation , 62(6), 6756957 — 2014 — 13

	Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

The branch correlation coefficient (BCC), the branch power ratio (BPR), and the total mean power (TMP) are often used to characterize the mobile multiple-input multiple-output (MIMO) channel. This work investigates to which degree these parameters are useful for maximizing the channel capacity of multiple-input multiple-output (MIMO) handheld devices used in data mode. A statistical point of view is applied, using about 2800 *outdoor* to indoor channel *sounder* measurements obtained with combinations of ten different handsets, four to eight test users, and a variety of different use cases (UCs). All measurements were made in an urban environment in a setup with two different base stations (BSs) and with the users inside a single building. For each measurement combination, the mean capacity (MC) and associated values of BCC, BPR, and TMP are obtained. From the data it is found that the MC is only weakly correlated with both the BCC and the BPR, while the MC is highly correlated with the TMP. © 2014 IEEE.

Conference Paper

☐ 162 **Optimization of the building envelope in hot and humid climates** — Nehme B.A.,Annan G.R. — 2014 International Conference on Efficient Building Design Materials and HVAC Equipment Technologies Icebd Met 2014 — 2014 — 0

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This paper aims to promote building sustainable designs to reduce energy consumption by implementing optimized building envelope for high performance buildings. Several building envelope case studies for office, airport, retail and residential buildings, located in hot and humid climates, are analyzed in WUFI, IES VE and CadnaA software in order to assess, simulate and optimize building envelope performance. A detailed building *thermal* energy performance simulation, assessing the impact of solar radiation and external shading devices, using IES VE, has been conducted to evaluate the effectiveness of the proposed building envelope designs and weigh the impact of sustainability measures on enhanced indoor environmental quality and building loads reduction. Also, a condensation analysis for external wall assemblies is conducted via WUFI software to study moisture influence on the *thermal* performance of the construction and long-term system behavior. Additionally, *sound* control software such as CadnaA assists in validating noise mitigation techniques from *outdoor* equipment to promote indoor occupant comfort. The simulation results also suggest building envelope composition optimized for hygrothermal and acoustic performance, window to wall ratios and appropriate shading strategies that promote reductions in building annual cooling energy for the different case studies to a range of savings varying from 5 % to 35% with respect to ASHRAE standard 90.1-2007 for hot and humid climates.

Article

☐ 163 **Experimental analysis on arc step voltage of fuse with rectangular notch** — Xiao Y.,Zhuang J.,Wang C.,Chen B.,Yuan Z. — Gaodianya Jishu High Voltage Engineering, 39(4) — 2013 — 4

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Step voltages of a fuse dramatically influence the current-limiting action and electrical insulation of the fuse. For quantitatively describing the fuse electric characteristic during initial arcing stage, we experimentally studied step voltages produced in short circuit interruption with fuses containing rectangular notch. An approximately linear relationship between arc-starting current density, which ranged within 10~45 kA/mm^2, and the initial arc column voltage gradient was found, when pre-arcing energy remained the same as the packing density of silica sand. Meanwhile the effect of silica sand on the step voltage was discussed. Two more sets of comparative tests, in which fuse was linked with different notch lengths and notch numbers in series, showed the *sound* fitness of the linear relationship. The results can be applied in further investigations of small scale or full scale experiments on multiple notches in series for high voltage fuses.

Article

	Document title	Authors	Source	Year	Citations
164	**Assessment of indoor environmental quality in open-plan offices**	Hwang T.,Kim J.T.	Indoor and Built Environment , 22(1)	2013	39

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This study aims to analyze the factors of indoor environment on occupants' health and comfort by post-occupancy evaluation (POE) and to accumulate the fundamental data of occupants' behaviour in relation to indoor environment, in an effort to ascertain the optimal building energy management system (BEMS). For this purpose, indoor temperature, relative humidity, vertical temperature distribution, air velocity, predicted mean vote (PMV), radiant temperature, *outdoor* temperature, and humidity were measured for evaluating *thermal* environment for nearly 2 years. Indoor air contaminants, such as PM_{10}, CO_2, formaldehyde, and volatile organic compounds (VOCs) were measured in the building and at *outdoor* locations. Working plane illuminance, window luminance, and the equivalent continuous a-weighted *sound* pressure level (Leq) were also monitored. In addition, the occupants in the intelligent office building were surveyed regarding the indoor environmental quality via the intranet. Results showed that all indoor environmental factors satisfied the recommendation, except the indoor relative humidity and PMV. The physical factors of *thermal* environment had been stable since two months after it was built. However, it should be necessary to improve the *sound* privacy caused by low background noise, seasonal air-conditioning for appropriate humidity, air freshness, and Video Display Terminal (VDT) glare in order to reduce the level of occupants' annoyance. © 2012 The Author(s).

Review

	Document title	Authors	Source	Year	Citations
165	**Towards sustainability index for healthy buildings - Via intrinsic thermodynamics, green accounting and harmony**	Kim J.T.,Todorovic M.S.	Energy and Buildings , 62	2013	36

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As, fossil energy resources are closer to their exhaustion, global warming in raise, and more catastrophic weather extremes are occurring worldwide, there are more and more warnings that the risks to the Earth humanity survival are also in growth. The Earth/humanity as a whole is becoming more complex system than ever. Trying to find a way to make a turn from the current one-way irreversibility to sustainability, it is necessary to find universal schemes, quantities, indicators and criteria relevant for the Earth and humanity's resilience and sustainability. Presented is a review of the multi-criteria sustainability analysis methods (intrinsic *thermodynamic* based on energy, exergy, sustainability index, analytic hierarchy process, etc.). It has been shown that crucial sensitivity of all methods is related to the selected sets of energy criteria (economical, social and environmental) and to the mathematical algorithms for the determination of the weighted factor and sub-indicators agglomeration. Also, shown is that concerning buildings sustainability is missing indicator relevant to the indoor comfort and people health, accompanied with a comprehensive, physically *sound* assessment methodology for building's synergetic environmental performance - twofold metrics with the reference to *outdoor* and indoor built environment concerning people health and comfort. A scheme of a structure and spatio-temporal vision of the Global sustainable development is presented - showing that the approach to the global sustainable development can be reliable only if it is based on a system of real human and ethical values applicable to every social, cultural and economic situation. Finally paper summarizes needs for future multidisciplinary, inter- and cross-disciplinary study on healthy sustainable buildings, interwoven with harmony and holistic people health and well-being research. That research is to be physically *sound* - intrinsic *thermodynamic*, harmonious green, and relevant metrics is to be free of the short-term politically shaped and quasi-economically distorted approaches, because the world is to get what is painfully missing - a standard metrics of the sustainability index, universally acceptable. © 2013 Elsevier B.V. All rights reserved.

Conference Paper

	Document title	Authors	Source	Year	Citations
166	**Numerical predictions for sustainable development of cities: Acoustic propagation in presence of urban vegetation**	Guillaume G.,Gauvreau B., Fortin N.	41st International Congress and Exposition on Noise Control Engineering 2012 Inter Noise 2012, 7	2012	2

	Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

Outdoor sound predictions in urban media are both a societal stake and a scientific issue, including sustainable development of cities. Thus SPL forecasts in such built-up areas will soon have to take into account particular boundary conditions, e.g. acoustic impedance of facades and/or roof surfaces in expectation of buildings greening or other use of vegetation on urban surfaces. In this context and in the framework of the "VegDUD" project, the Transmission Line Matrix (TLM) method has been used to model acoustic propagation in such complex media. This research project from the french research agency (ANR) involves different laboratories (Ifsttar, Météo-France, INRA, CSTB, ECN, etc.), various disciplinaries (acoustics, meterolorology, *thermics*, hydrologics, etc.), diverse approaches (physical, chemical and human sciences) and different meanings (experimental results and model predictions). Regarding acoustics, it aims at quantifying the best vegetation effort with respect to *soundscape* indicators, e.g. SPL, EDT, etc. This paper presents TLM predictions in (and out of) a canyon street for several configurations of roofs and/or facades vegetalization: ratio and location of vegetation covers, receivers positions, impedance characteristics of surface cover, etc.

Conference Paper

☐ 167 **A Sonic boom propagation model including mean flow atmospheric effects** Salamone J.,Sparrowa V.W. Aip Conference Proceedings , 1474(1) 2012 4

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This paper presents a time domain formulation of nonlinear lossy propagation in onedimension that also includes the effects of non-collinear mean flow in the acoustic medium. The model equation utilized is an augmented Burgers equation that includes the effects of nonlinearity, geometric spreading, atmospheric stratification, and also absorption and dispersion due to *thermoviscous* and molecular relaxation effects. All elements of the propagation are implemented in the time domain and the effects of non-collinear mean flow are accounted for in each term of the model equation. Previous authors have presented methods limited to showing the effects ofwind on ray tracing and/or using an effective speed of *sound* in their model equation. The present work includes the effects of mean flow for all terms included in the augmented Burgers equation with all of the calculations performed in the time-domain. The capability to include the effects of mean flow in the acoustic medium allows one to make predictions more representative of real-world atmospheric conditions. Examples are presented for nonlinear propagation of N-waves and shaped sonic booms. [Work supported by Gulfstream Aerospace Corporation]. © 2012 American Institute of Physics.

Article

☐ 168 **A numerical approach to sound levels in near-surface refractive shadows** Cheinet S. Journal of the Acoustical Society of America , 131(3) 2012 21

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The present study formulates a consistent method to simulate the *outdoor*, near-surface *sound* propagation through realistic refractive conditions. The correlated atmospheric stratification and turbulence properties are derived from standard meteorological quantities through flux-profile similarity relationships. The propagation of a monochromatic *sound* field is simulated in presence of the turbulence and stratification effects and an impedance ground. The propagation model uses a numerical solution of a second-order moment parabolic equation, which is introduced and evaluated. The so-formed coupled atmospheric-acoustic model is used to systematically investigate the *sound* levels in near-surface refractive shadows. In an illustrative propagation scenario, the shadow zone *sound* levels are predicted to show significant variations with the meteorological conditions. Specifically, the *sound* levels decrease with the adverse wind, as a consequence of enhanced mean upward refraction. Conversely, they increase with the absolute value of the surface heat flux, as a consequence of enhanced turbulence scattering. Implications for the assessment of the *sound* levels in shadow zones are discussed. © 2012 Acoustical Society of America.

Conference Paper

	Document title	Authors	Source	Year	Citations
169	**How DFV process can improve split AC installation process**	Lopes C.A.	51st Annual Conference of SAVE International 2011	2011	1

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The Split Air Conditioning product has become very popular in Brazil in recent years. The system consists of two units physically separated - the indoor unit contains the evaporator and the *outdoor* unit contains the condenser. Copper tubes connect both units which transfers the refrigerant fluid from the indoor unit to the *outdoor* unit. The main benefit of this system over a traditional single unit system is the lower *sound* level combined with the improved *thermal* comfort. The customer concern about purchasing this product is the installation process. It requires a specialist to install it and the cost can reach 30% to 40% of the product's price. Consumers are dissatisfied with the installation cost and the need to depend on a technical specialist. Most customers wish they could install it by themselves if the product would allow it. A Value Engineering Workshop was held in February, 2010 with a multifunctional team of fourteen people in Joinville, Brazil, to discuss the installation main issues and generate solutions to improve this complex process using the Value Methodology. This Function Analysis approach generated 64 ideas, where at least 27 were considered easy to implement with high saving or increased value. Seven Business Cases were developed by the team requiring $147K in investment with an annual risk weighted estimated $21.5M of savings for the company and consumers. This paper will demonstrate the success of using the Value Engineering approach to improving the design of a product for simpler installation.

	Article				
170	**Impact of portable HEPA filtration units on fungal exposure in a post transplant unit during | Einfluss dezentraler HEPA filtrationsgeräte auf die schimmelpilz-exposition in einer transplantations-nachsorgeeinheit während des abbruchs eines benachbarten bettenhauses**	Engelhart S.,Simon A., Hirschberg S.,Exner M.	Hygiene Medizin, 36(5)	2011	0

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Background: Building construction, renovation, or demolition works are commonly associated with a significant release of dust and fungal spores, including an increased risk of invasive aspergillosis in immunocompromized patients. Method: During the demolition of an old patient care building, we conducted an infection control risk assessment and initiated measures to reduce dust and fungal emissions and to minimize exposure of patientsat-risk. The nearby post-transplant-unit was equipped with portable HEPA-filtration units (MedicCleanAir® Pro). During the demolition period, we monitored air counts of *thermotolerant* fungi in filtered and unfiltered air using the Merck MAS-100, and conducted an active surveillance for invasive aspergillosis in patients-at-risk. Results: *Outdoor* fungal emissions were only moderately elevated during demolition, indicating the effectiveness of emission *thermoreducing* measures. Moreover, by means of the HEPA filtration units, indoor fungal exposure could be reduced by about one order of magnitude and did not exceed the mean indoor reference level. No cases of invasive aspergillosis occurred. Conclusion: We conclude that a *sound* infection control risk assessment and appropriate infection control measures can successfully minimize the excess risk of invasive aspergillosis in patients-at-risk. Portable HEPA-filtration units can be useful for limited periods of increased exposure, but noise level, indoor climate, and patient compliance restrict the use of these devices.

	Article				
171	**A simple outdoor criterion for assessment of low frequency noise emission**	Broner N.	Acoustics Australia, 39(1)	2011	24

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A criterion for examining low frequency noise (LFN) emission is presented. The modulation effect is clearly observable and can be seen to begin once the screens start operating. The modulation period is approximately 60 seconds. Transportation noise sources such as aircraft and diesel trains also are sources of LFN. Based on empirical and laboratory studies, it can be shown that the primary effect due to LFN appears to be annoyance and that this affect is greater than would be expected based on the A-weighted level alone. One obvious reason for this is that the A-

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weighting network significantly decreases the contribution of low frequency energy in a *sound* due to the reduced loudness sensitivity of a person's hearing at low frequencies. It has been known for many decades that gas turbines, boilers, forced draft fans and other sources can produce low frequency noise which can cause feelings of annoyance due to nausea, headache and uneasiness and vibration induced rattle.

Article • *Open access*

172 **The most problematic variable in the course of human-biometeorological comfort assessment - The mean radiant temperature** — Kantor N.,Unger J. — Central European Journal of Geosciences, 3(1) — 2011 — 335

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This paper gives a review on the topic of the mean radiant temperature T_{mrt}, the most important parameter influencing *outdoor thermal* comfort during sunny conditions. T_{mrt} summarizes all short wave and long wave radiation fluxes reaching the human body, which can be very complex (variable in spatial and also in temporal manner) in urban settings. *Thermal* comfort researchers and urban planners need easy and *sound* methodological approaches to assess T_{mrt} . After the basics of the T_{mrt} calculation some of the methods suitable for obtaining T_{mrt} also in urban environments will be presented.. Two of the discussed methods are based on instruments which measure the radiation fluxes integral (globe *thermometer*, pyranometer-pyrgeometer combination), and three of the methods are based on modelling the radiation environment with PC software (RayMan, ENVI-met and SOLWEIG). © 2011 Versita Warsaw and Springer-Verlag Wien.

Conference Paper

173 **Acoustical indicators and index for urban quality evaluation** — Magrini A.,Alfano F.R.D. — 20th International Congress on Acoustics 2010 Ica 2010 Incorporating Proceedings of the 2010 Annual Conference of the Australian Acoustical Society, 3 — 2010 — 4

Hide abstract Find It @ UCL View at Publisher Related documents

In the urban health evaluation, the environmental comfort (*thermo*-hygrometrical, acoustical and lighting comfort and IAQ) represents a fundamental aspect to quantify the influence of the climate and of the human activities on the men's health. The correlations among *outdoor* comfort, urban landscape and architectural features offer wide perspectives, as they could represent very useful means to give a qualitative and quantitative judgment on the existing real estate or to choose the major actions for the restoration of the urban environment. To obtain urban health global judgment, all the playing factors must be correctly weighted and associated. An analysis based on homogeneous quantities is needed, avoiding multiplying judgment scales and using, for example, one scale for each involved parameter comparable with the others. To quantify and correlate the various environmental elements, a first attempt was performed aimed at a global assessment on the basis of a mutual judgment system. In this paper a proposal of schematisation based on indicators and index has been developed. With the definition of them, relating to the acoustical *outdoor* comfort evaluation, it is possible to obtain a global quality environment evaluation considering also other aspects like, for example, *thermo*-hygrometrical and atmospheric pollution effects. First analyses to validate the method on the basis of experimental and simulated *sound* pressure levels have been developed. Moreover, noise level data, traffic and population density were considered as basic parameters on which it is possible to develop single indicators and a global acoustic quality index. In the meantime a subjective investigation was performed, to correlate the values of the acoustical index to the individual sensation of the people living in the zones of the investigation.

Conference Paper

	Document title	Authors	Source	Year	Citations
174	**Fusion of acoustic and optical sensor data for automatic fight detection in urban environments**	Andersson M., Ntalampiras S., Ganchev T., ... Ahlberg J., Fakotakis N.	13th Conference on Information Fusion Fusion 2010, 5712105	2010	34

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We propose a two-stage method for detection of abnormal behaviours, such as aggression and fights in urban environment, which is applicable to operator support in surveillance applications. The proposed method is based on fusion of evidence from audio and optical sensors. In the first stage, a number of modality-specific detectors perform recognition of low-level events. Their outputs act as input to the second stage, which performs fusion and disambiguation of the first-stage detections. Experimental evaluation on scenes from the *outdoor* part of the PROMETHEUS database demonstrated the practical viability of the proposed approach. We report a fight detection rate of 81% when both audio and optical information are used. Reduced performance is observed when evidence from audio data is excluded from the fusion process. Finally, in the case when only evidence from one camera is used for detecting the fights, the recognition performance is poor.

Conference Paper

175	**Outdoor elements providing urban comfort: The role of fountains in the soundscape**	Semidor C., Venot-Gbedji F.	PLEA 2009 Architecture Energy and the Occupant S Perspective Proceedings of the 26th International Conference on Passive and Low Energy Architecture	2009	5

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Urban planners and other town designers need information to improve the environmental quality of cities. It is well known that water constitutes an important element for mitigating *outdoor thermal* comfort, especially in hot climates. The use of water takes different forms: the evapotranspiration effect from vegetation, river currents or the water jets from a fountain. It also concerns the urban acoustic comfort. As the matter of fact knowing how to provide pleasant and quiet outside areas in order to ensure pleasant and quiet inside areas is one of the challenges for a better urban environment. So, it is necessary to know what type of urban design enables such areas to exist and how to characterise their *sound* quality. This paper deals with the data analysis from different case studies (Barcelona, Bristol and Genoa) depending on the type, the place, and the surroundings of the fountains. In each case, some *soundwalks* were carried out in order to highlight the quality of the urban *soundscape*.

Conference Paper

176	**Optimization of the acoustic performances of lightweight sandwich roof elements**	Dijckmans A., Vermeir G., Niggebrugge J.W.	38th International Congress and Exposition on Noise Control Engineering 2009 Inter Noise 2009, 3	2009	5

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Lightweight roof elements are commonly used in buildings because of the ease and speed of installing and the potential for very high and reliable *thermal* insulation. The *sound* insulation of the sandwich panels however may be too low. In fact, the combination of low mass and high stiffness typically results in a lower *sound* transmission loss over a large frequency band. But regularly the presence of higher *outdoor* levels is reflected in stricter requirements, so making improvements on the *sound* insulation of these roof elements necessary. The *sound* transmission loss of different lightweight sandwich panels with a core of expanded polystyrene has been measured in the lab. Also the structure-borne *sound* sensitivity has been measured with a mini-tapping machine. These measurements are compared with an analytical model based on the transfer matrix method and a numerical, wave based model.

	Document title	Authors	Source	Year	Citations

Primarily the influence of geometrical size of the panel and of the composition of the interlayer is discussed in view of optimizing the results.

Book Chapter

177 **Numerical methods to model infrasonic propagation through realistic specifications of the atmosphere** — Norris D.,Gibson R., Bongiovanni K. — Infrasound Monitoring for Atmospheric Studies — 2009 — 45

Hide abstract | Find It @ UCL | View at Publisher | Related documents

To model infrasonic propagation, two areas must be adequately addressed. First, the environment must encompass a domain spanning from the ground to the lower reaches of the *thermosphere*, and its properties must be resolved at a scale comparable to the acoustic wavelengths of interest. Second, the relevant fundamental physics that influence the wave propagation must be captured in the numerical models that are applied. To address these challenges, a large amount of work has been completed by researchers in fields spanning from traditional acoustics to upper atmospheric physics. In this chapter, this body of work is reviewed with an emphasis on its applicability to the modeling and interpretation of infrasonic observations. Global climatological models have largely been replaced in current infrasound modeling practice by atmospheric specifications that combine output from numerical weather prediction models for the lower atmosphere with empirical models for the upper atmosphere. Recently developed specifications incorporate higher-resolution, regional or mesoscale weather analysis products in order to improve fidelity below altitudes of 50 km. Specifications that utilize a terrain-following coordinate system, including high-resolution topography, enable the incorporation of additional physical effects. These specifications are unable to resolve all fine-scale stochastic phenomena, e.g., atmospheric irregularities smaller than the model resolution, fine-scale structures at altitudes above 50 km, and gravity wave fluctuations. In particular, gravity waves are of interest because their spatial scales are of the same order as infrasonic wavelengths. Because atmospheric fine-scale structure is inherently turbulent, spectral formulations are used to capture the energy distribution across space-time scales relevant to infrasound and to generate representative realizations of variable fields. The fundamental physical processes that affect infrasound, much like those of higher frequency acoustics, include refraction, diffraction, scattering, absorption, dispersion, and terrain. In addition, nonlinear effects can become significant near the source and at high altitudes. Several numerical approaches have been used for modeling infrasonic propagation often based on analogous models for underwater and *outdoor sound* propagation. They can be loosely classified under the headings of geometric (ray tracing, Gaussian beam, tau-p), continuous wave (parabolic equation or PE), full-wave (normal mode, finite-difference time domain, time-domain PE), and nonlinear (e.g., NPE). A good introductory source to the underlying computational aspects is (Jensen F, Kuperman W, Porter M, Schmidt H (1994) Computational ocean acoustics. AIP Press, New York). (Although this reference is primarily intended for ocean acoustics, many of the models carry over to the atmosphere and the presentation is superb.) A general overview of the propagation models is presented here with their corresponding strengths and weaknesses highlighted in the context of infrasonic studies. Also included are examples of modeling studies using high-resolution atmospheric specifications. © 2009 Springer Science+Business Media B.V.

Conference Paper

178 **Development of a micro-satellite compatible FTS sounder for sun-occultation measurements** — Giaccari P.,Moreau L.M., Giroux J.G.,Soucy M.-A. — Proceedings of SPIE the International Society for Optical Engineering , 7474, 74741N — 2009 — 0

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The SciSat/ACE mission provided, and still provides, high quality and high spectral resolution measurements of the atmosphere with a FTS *sounder* in sun-occultation configuration. Based on the comprehensive results and models of SciSat/ACE it is foreseen that most of the desired information can also be retrieved from lower spectral resolution measurements with higher signal-to-noise ratio (SNR) and appropriate data treatment. With the Canadian Space Agency under the Space Technologies Development Program, ABB Analytical developed a small size sun-occultation *sounder* compatible with a micro-satellite platform that has identical throughput, spectral bandwidth and vertical resolution as ACE. The spectral resolution is decreased by a factor 25 (0.6 cm^{-1} instead of 0.024 cm^{-1} for ACE) whereas

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the SNR performance is highly increased with an equal factor (target of 2500 instead of 100 for ACE over most of the spectral bandwidth between 750 and 4000 cm $^{-1}$).A prototype of the sun-occultation *sounder* was built, tested under various *thermal* conditions and subjected to vibrations similar to those expected at launch. An *outdoor* experiment was also conducted to test the instrument in sun-occultation conditions. The good behavior of the instrument indicates interesting opportunities for such small footprint *sounder* on a low-cost micro-satellite mission and potentially good earth coverage if several of such instruments are used in coordination. Depending on the scientific needs, it is possible to adapt the proposed instrument to increase the vertical resolution and/or to extend the measurements on lower altitudes due to the higher SNR performances. © 2009 SPIE.

Article

☐ 179 **Effects of soil water potential and moisture content on sound speed** — Lu Z.,Sabatier J.M. — Soil Science Society of America Journal, 73(5) — 2009 — 73

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To monitor and understand the temporal variations of *sound* speed due to the changes of soil physical properties such as soil water potential, moisture content, and soil temperature under natural conditions, a long-term field soil survey has been conducted. In this study, transmitter and receiver acoustic probes, consisting of ten (five pairs) and five transducers respectively, were inserted into a prepared trench containing a soil-sand mixture in an *outdoor* test site. Five time domain reflectometers, five tensiometers, and five *thermocouples* were buried during the trench-filling at the same depths as the acoustic transducers. Measurements of *sound* speed, soil temperature, soil moisture content, and water potential were performed continuously, along with the measurements of surface temperature and precipitation over a period of 2 yr. Analysis of the data shows that there is a power law relationship between the *sound* speed and water potential. It is also found that the water potential is the dominant influence on the *sound* speed whereas the moisture content and temperature have relatively minor impacts. Brutsaert and Luthins theory was employed to calculate the *sound* speed as a function of the effective stress. The theoretical predictions were compared with the experimental data and they are in good agreement. The study suggests that *sound* speed measurement could be used as a new and effective tool for water potential measurement. © Soil Science Society of America.

Conference Paper

☐ 180 **The variability of acoustical turbulence in the atmospheric boundary layer** — Cheinet S. — Proceedings European Conference on Noise Control — 2008 — 0

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Outdoor sound propagation is affected by small-scale turbulence, quantified through the refractive index structure parameter C_n^2. In this study, a Large Eddy Simulation (LES) is used to reveal the spatial and temporal variability of C_n^2 in the atmospheric convective boundary layer. A considerable variability is predicted, of almost two orders of magnitude at all heights. The regions of high C_n^2 match the convective plumes that drive the boundary layer dynamics. The impact of this variability on line-of-sight propagation and on *sound* scattering is illustrated. It is found to result in fluctuations of the considered acoustical diagnostics, which can in some cases induce major changes in the estimation of sensing performances.

Article • *Open access*

☐ 181 **Modeling growth of Matthiola incana in response to saline wastewaters differing in nitrogen level** — Grieve C.M.,Poss J.A., Shouse P.J.,Carter C.T. — Hortscience, 43(6) — 2008 — 11

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The capture and reuse of nutrient-rich greenhouse effluents may be an environmentally *sound* option for floriculture production, which would conserve fresh water resources and reduce off-site pollution of surface and groundwaters. This study was initiated in 24 *outdoor* lysimeters to determine effects of salinity and varying concentrations of nitrogen on the growth, yield, and ion relations of stock [Matthiola incana (L.) R. Br.] cultivar Cheerful White. The experiment was a 4 x 4 factorial, partially replicated design with four irrigation water salinities (2,5,8, and 11 dS·m^{-1})

	Document title	Authors	Source	Year	Citations

and four nitrate concentrations (2.5, 3.6, 5.4, and 7.1 mmol·L^{-1}; N = 35, 50, 75, and 100 ppm). Ammonium nitrogen was included in the nutrient solutions. Stem lengths were measured three times weekly. Measurements at final harvest were stem and inflorescence lengths, stem and floret diameters, number of axillary buds and florets, and shoot and root fresh weights. Time course of stem elongation was quantified as a function of *thermal* time with a phasic growth model. Salinity significantly delayed initiation of the exponential growth phase, shortened its duration, and reduced the rate of plant development. The overall effect was to delay time to harvest of marketable stems. Although length of the flowering stems decreased with increasing salinity, marketable stems (≈60 cm) were produced in all treatments. Mineral ion relations in the plant tissues were influenced significantly, but independently, by both salinity and nitrogen. Leaf sodium, magnesium, and chlorine concentrations increased with increasing salinity; calcium and potassium decreased. In response to increasing external nitrogen, both potassium and chlorine decreased; sodium increased, whereas calcium and magnesium were unaffected. We conclude that in closed-loop irrigation systems, the nitrogen requirements for stock are low and that growers could minimize costs and limit off-site pollution by reducing nitrogen inputs.

Conference Paper

182 **Delamination - A state-of-the-art review** Jana D. International Cement Microscopy Association 29th International Conference on Cement Microscopy 2007 2007 8

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One of the increasing problems in concrete slabs is delamination, which is a plane of "separation," usually forms within the top 1/4 to 1 in. (6.25 to 25 mm) of the finished surface, and orients parallel to the surface. Delamination can occur by a variety of reasons, such as: (a) the use of air entrainment in a concrete slab receiving a hard trowel finish; (b) machine-trowel finishing a lightweight aggregate concrete slab; (c) premature finishing of a slab prior to the cessation of bleeding; (d) top-down stiffening, or surface crusting of a concrete slab in a hot, windy, or dry weather, especially in a concrete undergoing slow and prolonged bleeding; (e) prolonged finishing operations on an *outdoor* air-entrained concrete slab, or, on a slab receiving a mineral or metallic surface hardener; (f) corrosion of reinforcing steel in concrete in the presence of chlorides and/or atmospheric carbon dioxide; and (g) cyclic freezing and thawing of a non-air-entrained, or poorly air-entrained concrete slab at critically saturated conditions. Methods of detecting the delaminated areas in a slab include: (a) metal tapping, (b) manual or automated chain dragging, (c) electro-mechanical *sounding*, (d) impact echo, (d) infrared *thermography*, and (e) ground penetrating radar. Among the techniques available for diagnosing the causes of delamination, petrographic examinations, along with detailed air void analysis of concrete cores from the delaminated areas are determined to be the most effective. After explaining factors responsible for delamination, and the techniques available for quantitative measurements of delaminated areas, the present paper provides case-study-based examples of detailed investigations of various mechanisms of delamination by diagnosing the evidence present in the delaminated concrete slabs. Finally, a list of common industry recommendations is provided to avoid such an increasing nuisance in the modern concrete slabs. Increasing occurrences of delamination in both indoor and *outdoor* concrete slabs are, unfortunately, the results of modern fast track construction schedules, where inadequate design or, failure to follow a design causes the failure, such as either by accidentally incorporating air in a slab that will not be exposed to freezing during its service and will receive a machine trowel, or not incorporating air in an *outdoor* slab that will be exposed to freezing.

Conference Paper

183 **Analyzing thermal environment efficiency of raised-floor Ondol with ventilation** Sung J.-H.,Han S.-M.,Kim Y.-M.,Sohn J.-Y. Iaqvec 2007 Proceedings 6th International Conference on Indoor Air Quality Ventilation and Energy Conservation 2007 0

	Document title	Authors	Source	Year	Citations
			in Buildings Sustainable Built Environment, 1		

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A floor heating, so called 'Ondol', is widely used in korea, and it has been known that Ondol provids good *thermal* environment. [1] However, the impact *sound* through floor is great and ventilation rate depends on infiltration in rooms where the Ondol is applied.[2] It is reported that impact *sound* is reduced on raised-floor Ondol system.[3] This study aims at analyzing *thermal* environment efficiency of raised-floor Ondol with ventilation system which is an alternative plan about Ondol with insufficient ventilation. The study is performed at a residential house located on Yang-Pyoung, Kyoung-gi do in Korea. A size of sample house is 4.45m (L) x 3.35m (D) x 2.85 (H). Experimentation compared factors of *thermal* environment of each case; Ondol-heating only (Case A), introduction of *outdoor*-air (Case B), using raised-floor Ondol with ventilation system. Case A was the highest in every factor of *thermal* environment, and then case C, case B was. The case C is likely to effectively contribute to keep required indoor environment with better indoor air quality.

Article

☐	184	**Investigation and analysis on indoor environment of urban residential buildings in Chongqing**	Zheng J.,Shun J.,Zhou Y.-L., Yang C.-J.	Journal of Harbin Institute of Technology New Series , 14(SUPPL.)	2007	0

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It is purposed to look into the actual conditions of the residential indoor environment in Chongqing during the autumn. Based on buildings' age and placement, twenty typically households were chosen and field measure was conducted in November 2005, these data were collected in living-room, kitchen, bedroom, toilet, etc. Finally, the characters of the indoor environment were obtained. Investigation revealed that, in this period, residential indoor *thermal* comfort level is better, but the *sound* environment is poor, with 42.68% of households dissatisfied with the noise. The indoor air is most polluted by formaldehyde seriously; and in the season *outdoor* concentration of certain pollutants (such as formaldehyde, etc.) are higher than indoor generally, therefore it is believed that residents should be strengthened ventilation to improve indoor air quality after that the *outdoor* air should be handled.

Conference Paper

☐	185	**Microphone noise when measuring in wind**	Van Den Berg G.P.	Euronoise 2006 the 6th European Conference on Noise Control Advanced Solutions for Noise Control	2006	0

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Microphone noise induced by wind is the result of turbulence in the air. *Outdoor* air turbulence is created by friction and vertical temperature differences and can be quantified with a model containing expressions for surface roughness (friction) and atmospheric stability (*thermal*). The resulting pressure level as observed with a microphone can be calculated when the microphone is in wind over a relatively flat, constant roughness area. Microphone wind noise is important when doing measurements in relation to wind turbine noise and ambient background *sound*; both are measured at wind velocities up to 20 m/s. It can be shown that measurement results may be contaminated by wind-induced microphone noise. The level of wind-induced noise can be determined when the wind velocity is measured at two different heights.

Article

	Document title	Authors	Source	Year	Citations
186	**Wind-induced noise in a screened microphone**	Van Den Berg G.P.	Journal of the Acoustical Society of America , 119(2)	2006	63

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To reduce the effect of wind on a microphone, wind screens are used, usually made of foam or cloth. Although this measure to reduce wind noise is well known and widely used, it lacks a theoretical explanation. One possible explanation was the turbulent wake in the air flow behind the wind screen, another the pressure variations due to turbulence inherent in *outdoor* wind. In the present paper it is shown that atmospheric turbulence is indeed the cause of *outdoor* microphone wind noise, and the knowledge that is available in atmospheric boundary layer physics can be used to model the effects of atmospheric turbulence on a bare or screened microphone. A wind screen can be seen as a first order low pass filter for pressure variations due to atmospheric turbulence. Induced wind noise is thus determined by (average) wind speed and wind screen diameter, but also by factors related to frictional and *thermal* turbulence (roughness length and atmospheric stability). This approach leads to predicted spectral values of the induced "*sound*" pressure that match experimental data. The result can be used to obtain proper *outdoor* ambient wind related *sound* levels by correcting measured *sound* levels with calculated wind-induced pressure levels. © 2006 Acoustical Society of America.

Conference Paper

	Document title	Authors	Source	Year	Citations
187	**A coupling method for SPL predictions in complex situations using acoustic and micrometeorological numerical codes**	Gauvreau B.,Lihoreau B., Berengier M., Blanc-Benon P.,Calmet I.	Forum Acusticum Budapest 2005 4th European Congress on Acustics	2005	2

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The modelling of *outdoor* long-range *sound* propagation implies to take into account complex propagation situations including mixed influence of ground characteristics (topography, obstacles, impedance, etc.) and atmospheric conditions (refraction and turbulence). We have developed a new PE code coupled with a micrometeorological code in order to calculate the *sound* propagation above a non flat ground for such realistic *outdoor* situations. Both codes are detailed in the present paper. Our acoustic code (MW-WAPE) can deal with different boundary conditions like the introduction of impedance jumps, thin screens or complex topography. Therefore, predicting long-range *sound* propagation over a non-urban site with complex propagation media requires the knowledge of micrometeorological fields in the lower part of the atmospheric boundary layer. Thus our PE code is coupled with a dedicated micrometeorological code (SUBMESO) which simulates wind and temperature fields over moderately complex terrain with high resolution. Its output data are used as input data for our PE code through a suited routine. This paper also presents some comparisons between numerical predictions and experimental data issued from the LCPC experimental (monitoring) site located at Saint-Berthevin (F). They show a very good agreement for both acoustical and micrometeorological predictions.

Conference Paper

	Document title	Authors	Source	Year	Citations
188	**BEM-studies of sound propagation in viscous fluids**	Langer S.	Eccomas 2004 European Congress on Computational Methods in Applied Sciences and Engineering	2004	2

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Sound propagation in real media is involved with loss of energy. The loss of energy is caused by various damping mechanism, especially internal friction, *thermal* conduction, and molecular acoustical absorption. Each of these effects can be described by a decay coefficient ß. Considering this coefficient leads to a modified complex Helmholtz-equation (1), $p(x)_{,ii} + \bar{k}^2 p(x) = 0$ (1) describing the *sound* pressure p in a viscous fluid dependant on the frequency LO, speed of *sound* c, and the complex wave-number $\bar{k} = k + i\mu = \omega/c + i\mu$. Here, using the Boundary Element Method, a numerical study of the influences of viscosity on *outdoor sound* propagation is presented. With exemplary calculations

	Document title	Authors	Source	Year	Citations

the relevance of considering energy losses in air for *outdoor sound* propagation is discussed. Using the modified Helmholtz-equation a secondary effect appears: The uniqueness of the integral representation is positively influenced. This is confirmed in numerical studies.

Conference Paper

189 **Adapative delay system (ADS) for sound reinforcement** — Smith E.A.,Detsky J.S. — Canadian Acoustics Acoustique Canadienne, 32(4) — 2004 — 0

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At concerts and presentations, the *sound* system must be carefully calibrated to ensure the entire audience can hear the presenters clearly. For both indoor and *outdoor* venues, this is done by positioning speakers throughout the audience to reinforce the *sound* produced on stage. This technique introduces an added complexity, whereby the electrical signal to the speakers in the crowd travels much faster than the *sound* wave coming from the stage. The Adaptive Delay System (ADS) for *Sound* Reinforcement is a new method for synchronizing the *sound* throughout the audience. Unlike existing methods, it does not require complex calculations when initially configuring the *sound* system. Furthermore, it is also capable of accounting for time-variant conditions such as wind, which are neglected in the current methods of speaker synchronization. Maximum length sequences, a special type of pseudo-random noise, are injected into a speaker for several seconds. A specially placed microphone picks up this *sound* which is then cross correlated with the original noise to determine the propagation delay. As this *sound* is barely audible it can be used during a concert to adaptively correct for changing conditions.

Article

190 **Neural network modelling of outdoor noise levels in a pilot area** — Avsar Y.,Saral A., Gonullu M.T., Arslankaya E.,Kurt U. — Turkish Journal of Engineering and Environmental Sciences, 28(3) — 2004 — 18

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Noise measurements were made for 16 relevant *outdoor* points in the central campus area of Yildiz Technical University. All measurements were performed in accordance with Turkish Standards using an HD 9019 *sound* level meter. A new and sophisticated modelling technique called artificial neural networks was used to model the variation of noise levels from traffic around the campus area. The model inputs were the position of the measurement station, the geographical situation between the noise source and the measurement station, wind speed and direction, air temperature and relative humidity, and time of day. All parameters were measured throughout the study for 5 months. © Tubitak.

Conference Paper

191 **Corrosion study of stainless steel bars in cracked concrete** — Yamaji T.,Hirasaki T., Takahashi R.,Mizuma S., Yamakawa M. — American Concrete Institute ACI Special Publication, SP-222 — 2004 — 0

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Corrosion of stainless steel bars in concrete was investigated using *sound* and pre-cracked concrete specimens. Three types of stainless steel were investigated, such as 18Cr, 18Cr-8Ni and 18Cr-12Ni-2.5Mo. Concrete specimens were exposed to two environments, where wetting and drying alternately repeated. One was in the *outdoor* with atmospheric temperature, and the other was in a controlled chamber, where the temperature was 60°C during wetting and 15°C during drying. The detail investigation was carried out after two years. No corrosion was observed on stainless steel bar in both *sound* and pre-cracked concrete exposed to *outdoor*. The maximum chloride ion concentration was 7.0 kg/m^3 for 18Cr-8Ni, 8.0 kg/m^3 for 18Cr-12Ni-2.5Mo and 6.0 kg/m^3 for 18Cr at the crack region of concrete. This result indicated that the chloride ion threshold level for stainless steel was larger than these values under marine environment with atmospheric temperature. No corrosion was observed on both 18Cr-12Ni-2.5Mo and 18Cr in both *sound* and pre-cracked concrete exposed to controlled chamber. However, corrosion was observed only

	Document title	Authors	Source	Year	Citations

for 18Cr-8Ni at the crack region, even when the chloride ion concentration at the crack region was 6.0 kg/m^3. This was considered to be due to the effect of a high temperature.

Article

192 **A note on the effect of gestation housing environment on approach test measures in gilts**

Marchant J.N., Bradshaw R.H., Marchant-Forde R.M., Broom D.M.

Applied Animal Behaviour Science, 80(4)

2003

21

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Systems' welfare evaluation, including behavioural testing, is becoming increasingly popular in farm animal assurance schemes. The aims of this study were to investigate whether fairly short-term exposure to gestation housing systems, which varied in physical, environmental and human-input factors, influenced behavioural and physiological measures during a human approach test - often used to identify problems in human-animal interactions. Twenty-four Large White x Landrace gilts were initially subjected to identical human contact and daily husbandry. Forty-two days after service, the gilts were randomly assigned to either an indoor housing system (n = 16) or an *outdoor* housing system (n = 8), which differed physically and in the amount of human contact and daily husbandry. The indoor system used an electronic sow feeder (ESF), was more space-limited and *thermally*-controlled and had human contact centered on cleaning out. The *outdoor* system was more extensive, had much greater space accessible, was not *thermally*-controlled and had human contact that centered around feeding. The human approach test was carried out on all gilts 30-44 days after entry to the gestation system. At testing, each individual was fitted with heart rate monitor and then moved into a test arena. After 2 min an unfamiliar human entered the pen and stood motionless for 3 min against one wall and then approached the gilt and touched her snout. Throughout the experimental period, behaviour and *sound* within the test arena were recorded continuously. During the 2 min familiar-isation period, *outdoor* gilts had lower heart rates (108.2 bpm versus 123.7 bpm, $P < 0.05$) and tended to perform fewer short vocalisations (0.5 calls per min versus 3.4 calls per min, $P < 0.1$). *Outdoor*-housed gilts also carried out less locomotor behaviour (2.2 sections crossed versus 4.0 sections crossed, $P < 0.05$) and tended to perform fewer short (1.4 calls per min versus 5.0 calls per min, $P < 0.1$) and long vocalisations (0.2 calls per min versus 1.8 calls per min, $P < 0.1$) over the 3 min test period. *Outdoor* gilts tended to be slower to approach within 0.5 m of the human (69.9 s versus 19.3 s, $P < 0.1$) but they then took less extra time to make physical contact (3.3 s versus 52.7 s, $P < 0.1$). Mean heart rate was significantly lower in *outdoor* sows over the whole 3 min period (99.5 bpm versus 115.5 bpm, $P < 0.05$). The results demonstrate that short-term exposure to different housing systems did influence behavioural and physiological measures during a standard human approach test and thus, systems differences should be taken into account before making judgements about the human-animal relationship on any commercial farm, based on results of behavioural tests of this type. © 2002 Elsevier Science B.V. All rights reserved.

Article

193 **A linearized Eulerian sound propagation model for studies of complex meteorological effects**

Blumrich R.,Heimann D.

Journal of the Acoustical Society of America, 112(2)

2002

110

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Outdoor sound propagation is significantly affected by the topography (including ground characteristics) and the state of the atmosphere. The atmosphere on its part is also influenced by the topography. A *sound* propagation model and a flow model based on a numerical integration of the linearized Euler equations have been developed to take these interactions into account. The output of the flow model enables the calculation of the *sound* propagation in a three-dimensionally inhomogeneous atmosphere. Rigid, partly reflective, or fully absorptive ground can be considered. The linearized Eulerian (LE) *sound* propagation model has been validated by means of four different scenarios. Calculations of *sound* fields above rigid and grass-covered ground including a homogeneous atmosphere deviate from analytic solutions by ≤1 dB in most parts of the computed domain. Calculations of *sound* propagation including wind and temperature gradients above rigid ground agree well with measured scale model data. Calculations of *sound* propagation over a screen including ground of finite impedance show little deviations to measured scale model data which are probably caused by an insufficient representation of the complex ground

	Document title	Authors	Source	Year	Citations

impedance. Further calculations included the effect of wind on shading by a screen. The results agree well with the measured scale model data. © 2002 Acoustical Society of America.

Conference Paper

194 **Overview of Integrating Dedicated Outdoor Air Systems with Parallel Terminal Systems** — Mumma S.A. — ASHRAE Winter Meeting Cd Technical and Symposium Papers — 2001 — 1

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Integration of dedicated *outdoor* air (OA) systems with parallel terminal systems represents a huge shift in the standard engineering design paradigm. This paper will present some of the issues that guided the migration path toward separate dedicated OA ventilation systems and away from delivering the ventilation via single all-air variable air volume systems. The paper will then attempt to show that once a shift in the design paradigm in favor of a separate dedicated OA ventilation system has occurred, questions concerning the *thermodynamic* state of the delivered ventilation arise. Some of the boundary conditions that impact the best state of the delivered ventilation air will be addressed. Specifically, the first variable of interest is supply air temperature and its impact on the size and, hence, the cost of the terminal equipment and the terminal reheat energy. The second specific variable is the dew-point temperature (DPT) and the resultant latent cooling capability of the delivered ventilation air and its impact on the types of terminal equipment that may be practical. If in the paradigm shift the supply air DPT is dropped low enough to remove all of the space latent loads, then sensible-only terminal equipment, such as radiant ceiling panels, may become viable choices in some situations and eliminate the production of microbial problems in the spaces. Finally, this line of reasoning is extended further to note that when the space dew points are controlled with the dedicated OA, the terminal sensible equipment may use uninsulated sprinkler piping as the energy transport conduit. Companion papers (Mumma and Shank 2001; Shank and Mumma 2001; Janus 2001; Conroy and Mumma 2001) will address specific issues related to the overall integration of these concepts into a *sound* working design.

Article

195 **The attenuation of aircraft noise by wood stud walls** — Bradley J.S.,Birta J.A. — Canadian Acoustics Acoustique Canadienne , 29(3) — 2001 — 0

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Measurements were made according to the ASTM E90 procedure but with the frequency range of measurements extended from 50 to 5 kHz. The overall performance of the walls was rated using the ASTM *Outdoor*-Indoor Transmission Class (OITC). In general, it was found that the overall *sound* insulation of wood stud exterior walls is limited by poor performance at the low frequencies due to one of 2 types of low frequency resonance.

Conference Paper

196 **Overview of integrating dedicated outdoor air systems with parallel terminal systems** — Mumma S.A. — ASHRAE Transactions , 107 PART. 1 — 2001 — 60

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Integration of dedicated *outdoor* air (OA) systems with parallel terminal systems represents a huge shift in the standard engineering design paradigm. This paper will present some of the issues that guided the migration path toward separate dedicated OA ventilation systems and away from delivering the ventilation via single all-air variable air volume systems. The paper will then attempt to show that once a shift in the design paradigm in favor of a separate dedicated OA ventilation system has occurred, questions concerning the *thermodynamic* state of the delivered ventilation arise. Some of the boundary conditions that impact the best state of the delivered ventilation air will be addressed. Specifically, the first variable of interest is supply air temperature and its impact on the size and, hence, the cost of the terminal equipment and the terminal reheat energy. The second specific variable is the dew-point temperature (DPT) and the resultant latent cooling capability of the delivered ventilation air and its impact on the types of terminal equipment that may be practical. If in the paradigm shift the supply air DPT is dropped low enough to remove all of the space latent loads, then sensible-only terminal equipment, such as radiant ceiling panels,

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Article

☐ 197 **A study of energy use and satisfactory zone ventilation of different outdoor air ventilation strategies for terminal reheat variable air volume systems** — Reddy T.A., Liu M., Claridge D.E. — Energy and Buildings, 29(1) — 1998 — 9

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Increased building indoor air quality (IAQ) complaints due to reduced *outdoor* air ventilation rates led to ASHRAE Standard 62-1989. Even though the stipulated standard total *outdoor* ventilation flow rate may be drawn into the HVAC system, *thermal* imbalances in the various zones of the building can lead to certain zones being starved of the specified ventilation flow rate, thereby creating localized IAQ problems. The objective of this paper is to compare the differences in energy use and ventilation air flow rates supplied to different zones in the building for three different practical *outdoor* air ventilation strategies all of which are identical in performance at design conditions but which differ under part-load operation. A simplified simulation methodology (which past studies have demonstrated to be useful for field evaluation of actual buildings) has been used to predict the heating and cooling energy use of a two-zone terminal reheat variable air volume (TRVAV) system during part-load operation specified by varying *outdoor* temperature and humidity conditions. The trade-off between *outdoor* air intake and energy use are studied for the following ventilation strategies for a typical 10,000 m^2 commercial building: (i) constant outside air intake based on a value 20% higher than the ASHRAE minimum ventilation rate, (ii) constant ventilation air intake fraction, and (iii) ventilation air intake based on the unfavorable zone requirements (even though the other zone may be over-ventilated). How this trade-off is impacted by building size has also been investigated. Finally, we use bin data for Dallas, TX (a moderately hot and humid location) and Seattle, WA (a mild location) in order to study the differences in energy use and zone ventilation flows of different ventilation strategies due to building location. The effect of economizer cycles and of varying ventilation strategies depending upon diurnal building schedules have not been considered in this study. The results of this study which are based on a simplified HVAC simulation approach are consistent with conclusions reached by other researchers Using more detailed simulation models. This suggests that *sound* and meaningful diagnostic insights of actual building performance and operating strategies can be obtained from such simplified simulations. © 1998 Elsevier Science S.A. All rights reserved.

Article

☐ 198 **Some practical considerations for predicting outdoor sound propagation in the presence of wind and temperature gradients** — Taherzadeh S., Li K.M., Attenborough K. — Applied Acoustics, 54(1) — 1998 — 11

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The propagation of low-frequency *sound* above an impedance plane in a vertically stratified medium is considered. The combined effects due to the vector wind and temperature gradients on *sound* propagation are studied. Recently, a WKB-type approximation has been combined with turning point theory and applied to the efficient computation of the height-dependent Green's function, which, in turn, has been used in the fast field program (FFP). Although this method is applied to a temperature-stratified medium, it leads to a significant reduction in the computational time for predicting *sound* propagation outdoors. The present paper extends such a WKB-type approximation to include vector wind. Attention is focused on the development of efficient algorithms for numerical evaluation of the inverse Fourier transform and on considerations of numerical stability. The limitations of the approximation are examined for logarithmic profiles in particular. © 7997 Elsevier Science Ltd.

Review

	Document title	Authors	Source	Year	Citations
199	**IEA ANNEX 27: Evaluation and demonstration of domestic ventilation systems. Assessments on noise**	Op't Veld P., Passlack-Zwaans C.	Energy and Buildings, 27(3)	1998	7

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The acceptance and appreciation of ventilation systems is mainly determined by the perceived indoor air quality, *thermal* comfort and noise. Noise in relation to ventilation systems can be divided into three categories: (1) *outdoor* noise (entering the dwelling through ventilation openings, cracks, mechanical supply and exhaust openings, etc.); (2) noise generated by components of the ventilation system and: (3) the impact of ventilation systems on *sound* reduction of partitions (between dwellings, rooms, etc.). Depending on the type of ventilation system, one or more of these aspects are of concern. Noise related to the ventilation system and components, can result in turning off the ventilation system or closing vents, etc. This can have a negative influence on ventilation and indoor air quality. In the framework of IEA ANNEX 27, several noise aspects of domestic ventilation systems have heen evaluated. *Outdoor* noise: In noise loaded areas, the selection and the applicability of different types of ventilation systems are determined hy the noise level on the facades. A simplified tool is developed to select ventilation systems as a function of the required noise reduction of the facade, room dimensions and design and construction of the facade. System noise: Controlling noise levels caused by ventilation systems is in practice one of the most important factors to contribute to the satisfaction with a ventilation system. Air duct systems in dwellings transport noise generated by fans and aerodynamic noise generated by bends, control valves, grilles, etc. ANNEX 27 gives basic equations to estimate *sound* power levels of fans and grilles, indication of sixes for silencers and guidelines for design. Impact on noise reduction of partitions: The composite *sound* reduction is the result of different *sound* channels from one room to the other. One of this *sound* channels may be the ventilation system (cross-talk). Cross-talk can be brought about through the air duct system, overflow grilles and ventilation openings in partitions, duct transitions, etc. ANNEX 27 provides guidelines for *sound* reductions of partitions and insertion losses for ducts to eliminate the influence of cross-talk. @ 1998 Published by Elsevier Science S.A. All rights reserved.

	Conference Paper				
200	**Effects of refraction on short range outdoor propagation over asphalt surfaces**	Hartwig Troy, Bolton J.Stuart	Proceedings National Conference on Noise Control Engineering, 1	1997	0

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The effects of wind and temperature gradients on short distance propagation measurements over asphalt surfaces were studied in the context of two-microphone method. In this paper, the fundamental equations governing *outdoor sound* propagation are presented, together with the methodology that was used to model refraction. The experimental verification of the methodology are also presented.

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